

HYPERMAGMAS AND COLORED OPERADS: HEADS, PHASES, AND THETA ROLES

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ABSTRACT. We show that head functions on syntactic objects extend the magma structure to a hypermagma, with the c-command relation compatible with the magma operation and the m-command relation with the hypermagma. We then show that the structure of head and complement and specifier, additional modifier positions, and the structure of phases in the Extended Projection can be formulated as a bud generating system of a colored operad, in a form similar to the structure of theta roles. We also show that, due to the special form of the colored operad generators, the filtering of freely generated syntactic objects by these coloring rules can be equivalently formulated as a filtering in the course of structure formation via a colored Merge, which can in turn be related to the hypermagma structure. The rules on movement by Internal Merge with respect to phases, the Extended Projection Principle, Empty Category Principle, and Phase Impenetrability Condition are all subsumed into the form of the colored operad generators. Movement compatibilities between the phase structure and the theta roles assignments can then be formulated in terms of the respective colored operads and a transduction of colored operads.

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1. INTRODUCTION

This paper, in combination with [20], is aimed at providing a mathematical model of the filtering of the freely formed syntactic objects generated by the action of the free symmetric Merge on workspaces, by viability for interpretation at the interfaces (Syntax-Semantic interface and Externalization). While [20] focused on filtering on the bases of correct assignment of theta roles and the theta criterion, as well as the dichotomy between External and Internal Merge in their roles with respect to theta theory, in this paper we focus on filtering by well formed head and complement structure and phases. We also analyze the relation between the filtering by phases and the filtering by theta theory. As in the case of theta theory, we show that the filtering can equivalently be interpreted as simultaneous to structure formation, in the form of a “colored Merge” that performs structure building and filtering at each step rather than first unconstrained structure formation followed by filtering. Since these two processes are in fact equivalent, adopting one or the other viewpoint is more a matter of convenience (decoupling formation and filtering for deriving formal arguments in abstract modeling, combining them for algorithmic optimization avoiding combinatorial explosions). As in the case of theta theory, the filtering for phase structure is governed by the same algebraic formalism involving colored operads, bud generating systems, and algebras over operads. To show some specific examples of non-trivial interaction between the filtering by theta theory and the filtering by phases we analyze the cases of passivization, of exceptional case marking, and of double object constructions. We propose a formalism, based on a notion of transduction (or correspondence) of colored operads, for the combined implementation of filtering by theta roles and by phase structure.

We start by considering the algebraic properties describing the structure of head and complement, and phases, starting from the formulation discussed in Chapter 1 of [19]. We show that, when one incorporates the structure of head, the magma of syntactic objects needs to be extended to a hypermagma that captures the behavior of the syntactic head. This shows, for example, that while the c-command relation is fully describable in terms of the magma of syntactic objects, the m-command relation requires the hypermagma structure.

We then show that the structure of head and complement, and phases, can be formulated in terms of coloring rules, and a bud generating system for a colored operad, in much the same way as we have done for theta roles assignments in [20]. This determines a coloring algorithm for the full abstract binary rooted trees of syntactic objects that assigns the phase structure.

We then compare the coloring by phases with the coloring by theta roles. The compatibility between the two imposes restrictions on certain kinds of movement explanations for theta role configurations, and generally for the behavior of theta roles in relation to movement.

A note for the readers: the initial part, in §2, on the hypermagma structure clarifies and provides further details about the properties of the head function discussed in §1.13 of [19]. It is used again

in this paper only in §5. Thus, §2, the implication on command structures in §3, and the last part of §5 can be skipped by the reader interested in the linguistics applications to phases and labeling algorithms, more than in the details of the mathematical structures involved. (However, note that Definition 2.4 in §2 is used in the rest of the paper.)

2. THE ALGEBRAIC STRUCTURE OF SYNTACTIC OBJECTS AND HEAD FUNCTIONS

In this section we recall some notions of hyperstructures (in particular the notion of hypermagma) associated to algebraic structure (in this case the notion of magma). We recall how syntactic objects are obtained as the free non-associative commutative magma over the set of lexical items and syntactic features. We then show that, when one takes into consideration also the presence of a syntactic head, whose properties are encoded in the abstract notion of a head function, this naturally leads from the magma to a hypermagma structure.

2.1. Magmas and hypermagmas. Hyperstructures are generalizations of algebraic structures where the operations are set valued, [24]. In particular, we are interested here in the notion of hypermagma, [21], [23], [25].

Definition 2.1. For a given set X , let $\mathcal{P}(X) = 2^X$ be the powerset of X . Then a hyperoperation on X is a function

$$(2.1) \quad \star : X \times X \rightarrow \mathcal{P}(X).$$

Any such hyperoperation extends to an operation on the powerset

$$\star : \mathcal{P}(X) \times \mathcal{P}(X) \rightarrow \mathcal{P}(X)$$

by setting, for $A, B \subset X$,

$$A \star B = \bigcup_{(a,b) \in A \times B} a \star b.$$

A hyperoperation is *total* if (2.1) takes values in $\mathcal{P}(X)^o \subset \mathcal{P}(X)$, the set of non-empty subsets of X . It then extends to an operation on $\mathcal{P}(X)^o$. A pair $(X, \star)$ of a set X together with a hyperoperation as in (2.1) is called a *hypermagma*. A *magma* is a *hypermagma* where $a \star b$ is always a singleton, for all $a, b \in X$, namely (2.1) is an actual binary operation on X ,

$$\star : X \times X \rightarrow X.$$

The magma operation and the hypermagma hyperoperation are in general not associative: when associativity holds the resulting structure is referred to as a monoid, or hypermonoid.

Remark 2.2. Hypermagmas form a category $\mathcal{HM}$ with morphisms $\phi : (X, \star_X) \rightarrow (Y, \star_Y)$ given by functions $\phi : X \rightarrow Y$ satisfying the *colax* condition

$$\phi(a \star_X b) \subseteq \phi(a) \star_Y \phi(b).$$

A strict morphism is one that also satisfies the *lax* condition $\phi(a \star_X b) \supseteq \phi(a) \star_Y \phi(b)$, so that $\phi(a \star_X b) = \phi(a) \star_Y \phi(b)$.

A hypermagma $(X, \star)$ is unital if there is a (unique) element $1 \in X$ such that $1 \star x = x \star 1 = \{x\}$. Morphisms of unital hypermagmas also map the unit to the unit.

We also recall some definitions from [24] that will be useful in the following.

Definition 2.3. A morphism $\phi : (X, \star_X) \rightarrow (Y, \star_Y)$ of hypermagmas is said to be *short* if it is surjective and

$$\phi(\phi^{-1}(a) \star \phi^{-1}(b)) = a \star b, \quad \forall a, b \in Y,$$

and is said to be *coshort* if it is injective and

$$u \star v = \phi^{-1}(\phi(u) \star \phi(v)), \quad \forall u, v \in X.$$

Given a hypermagma $(X, \star_X)$, a *strict subhypermagma* $(Y, \star_X)$ is a subset $Y \subset X$ such that $a \star_X b \subseteq Y$, for all $a, b \in Y$. A *weak subhypermagma* $(Y, \star_Y)$ of $(X, \star_X)$ is a subset $Y \subset X$ with a hyperoperation $\star_Y$ satisfying

$$(2.2) \quad a \star_Y b = (a \star_X b) \cap Y.$$

A coshort morphism is the same as the inclusion of its image as a weak subhypermagma. Any subset $Y \subset X$ has an induced hyperoperation defined by the right-hand-side of (2.2) that makes is a weak subhypermagma.

Note that, even if the hypermagma $(X, \star_X)$ is total, weak subhypermagmas need not be total, as the intersections $(a \star_X b) \cap Y$ can be empty even if $a \star_X b$ is never empty.

2.2. The magma of syntactic objects. As shown in [19], syntactic objects can be described as the elements of the free non-associative commutative magma

$$\mathcal{SO} = \text{Magma}_{c,na}(\mathcal{SO}_0, \mathfrak{M})$$

generated by a finite set $\mathcal{SO}_0$ of lexical items and syntactic features. This means that syntactic objects are obtained, starting with the elements of $\mathcal{SO}_0$ by repeated applications of a single non-associative, commutative binary operation $\mathfrak{M}$. Thus, one recursively obtains increasingly complex syntactic objects, such as $\mathfrak{M}(\alpha, \beta)$, $\mathfrak{M}(\mathfrak{M}(\alpha, \beta), \gamma)$, $\mathfrak{M}(\delta, \mathfrak{M}(\alpha, \mathfrak{M}(\beta, \gamma)))$, etc. The magma operation $\mathfrak{M} : \mathcal{SO} \times \mathcal{SO} \rightarrow \mathcal{SO}$ acts as $(T, T') \mapsto \mathfrak{M}(T, T')$.

It is a well known fact that the set obtained as the free non-associative commutative magma on a given finite set S is canonically isomorphic to the set $\mathfrak{T}_S$ of non-planar binary rooted trees with leaves labelled by elements of the set S . As in [19] when we say binary rooted trees we always mean *full* binary rooted trees (meaning without non-branching vertices: every non-leaf vertex has exactly two descendant vertices). The fact that the trees are non-planar (without an assigned planar embedding) corresponds to the commutativity of the magma operation, while the non-associativity is responsible for the non-trivial tree structures. Thus, here as in [19], we will also use the notation $\mathfrak{T}_{\mathcal{SO}_0}$ to denote the set of syntactic objects, endowed with its magma structure. The magma operation, in this form is given by

$$(T, T') \mapsto \mathfrak{M}(T, T') = \widehat{T} \widehat{T'},$$

where one should keep in mind that these trees are non-planar, namely

$$\widehat{T} \widehat{T'} = \mathfrak{M}(T, T') = \mathfrak{M}(T', T) = \widehat{T'} \widehat{T}.$$

We will always assume that a binary rooted tree T is oriented downward from the root, so each non-leaf vertex v has two edges with source v (the two edges just below). Thus, we will call successor and preceding vertices or edges, or above and below, with respect to this orientation, with the root at the top.

2.3. Head functions and hypermagmas. A head function on an abstract binary rooted tree is defined in the following way (Definition 1.13.3 of [19]).

Definition 2.4. A *head function* on an abstract binary rooted tree T is a function $h_T : V^o(T) \rightarrow L(T)$ from the set $V^o(T)$ of non-leaf vertices of T to the set $L(T)$ of leaves of T , with the property that if $T_v \subseteq T_w$ and $h_T(w) \in L(T_v) \subseteq L(T_w)$, then $h_T(w) = h_T(v)$.

It is proved in Lemma 1.13.7 of [19] that the description of head functions given in Definition 2.4 is equivalent to the description given in [3]. As shown in Lemmas 1.13.4 and 1.13.5 of [19] there are exactly 2^{n-1} possible head functions on an abstract binary rooted tree T with n leaves. They correspond to the 2^{n-1} different possible planar embeddings of the tree. This means that the characterization given in Definition 2.4 has $2^{\#L(T)-1}$ possible solutions.

Among all the $2^{\#L(T)-1}$ possible abstract head function, one is interested in particular in the head function that corresponds to the actual syntactic head, which depends on the specific lexical items and syntactic features assigned to the leaves of T . This, however, is not always well defined, as the structure T may contain exocentric constructions that are not label-able, hence do not have a well determined syntactic head. The existence of exocentric constructions is a manifestation of the fact that the datum of a head function is not directly compatible with the magma operation on syntactic objects. This incompatibility was observed in [19], where in order to model the syntactic head, one introduces an assignment of head functions $h : T \mapsto h_T$ that is in general only defined on a domain $\text{Dom}(h) \subset \mathfrak{T}_{\mathcal{SO}_0}$ that is *not* a sub-magma. We discuss here more in detail the relation between head functions and the magma structure.

Lemma 2.5. *Let $\mathfrak{H}_{\mathcal{SO}_0}$ denote the set of pairs (T, h_T) of a syntactic object $T \in \mathfrak{T}_{\mathcal{SO}_0}$ and a head function h_T on T as in Definition 2.4. The magma operation $\mathfrak{M}$ on $\mathfrak{T}_{\mathcal{SO}_0}$ does not induce a compatible magma operation on $\mathfrak{H}_{\mathcal{SO}_0}$, but it induces a compatible commutative hypermagma structure defined by the hypermagma operation*

$$(2.3) \quad \mathfrak{M}^{\mathfrak{H}}((T, h_T), (T', h_{T'})) = (\mathfrak{M}(T, T'), \{h_T, h_{T'}\}).$$

Proof. Given two pairs (T, h_T) and $(T', h_{T'})$ of a syntactic object and a head function, there is no unique way of assigning a head function $h_{\mathfrak{M}(T, T')}$ to $\mathfrak{M}(T, T')$ determined by h_T and $h_{T'}$. Indeed, there are two possible choices of how to extend h_T and $h_{T'}$ to a head function $h_{\mathfrak{M}(T, T')}$, by assigning to the new root vertex v of $\mathfrak{M}(T, T')$ either the value $h_{\mathfrak{M}(T, T')}(v) = h_T(v_T)$ or $h_{\mathfrak{M}(T, T')}(v) = h_{T'}(v_{T'})$, where v_T and $v_{T'}$ are, respectively, the root vertex of T and of T' . The data of (T, h_T) and $(T', h_{T'})$ do not suffice in general to determine one or the other choice at the new root v . However, there is a hypermagma structure on the set $\mathfrak{H}_{\mathcal{SO}_0}$, which we denote by $\mathfrak{M}^{\mathfrak{H}}$, that is defined as in (2.3), where the notation $(\mathfrak{M}(T, T'), \{h_T, h_{T'}\})$ stands for the set

$$\{(\mathfrak{M}(T, T'), h_T), (\mathfrak{M}(T, T'), h_{T'})\}$$

where the pair $(\mathfrak{M}(T, T'), h_T)$ (respectively, $(\mathfrak{M}(T, T'), h_{T'})$) consists of $\mathfrak{M}(T, T')$, the syntactic object obtained by multiplication in the magma, endowed with the head function $h_{\mathfrak{M}(T, T')}$ with $h_{\mathfrak{M}(T, T')}(v) = h_T(v_T)$ (respectively, $h_{\mathfrak{M}(T, T')}(v) = h_{T'}(v_{T'})$). The hypermagma is commutative, since $\mathfrak{M}(T, T') = \mathfrak{M}(T', T)$ and the set $\{h_T, h_{T'}\}$ is also invariant under exchanging T and T' . $\square$

The relation between the magma $\mathfrak{T}_{\mathcal{SO}_0}$ and the hypermagma $\mathfrak{H}_{\mathcal{SO}_0}$ is described as follows.

Definition 2.6. Given two hypermagmas $(X, \star_X)$ and $(Y, \star_Y)$, a projection $\pi : (Y, \star_Y) \rightarrow (X, \star_X)$ is a surjection $\pi : Y \rightarrow X$ that is a short morphism of hypermagmas. A section σ of the projection π is a coshort morphism of hypermagmas $\sigma : (X, \star_X) \rightarrow (Y, \star_Y)$, satisfying $\pi \circ \sigma = \text{id}_X$. A partial section is a section σ defined on a subset $D(\sigma) \subset X$, with the induced weak subhypermagma structure $(D(\sigma), \star_\sigma)$.

Using the hypermagma formalism, we can refine our understanding of the properties of the domain $\text{Dom}(h) \subset \mathfrak{T}_{\mathcal{SO}_0}$ of a head function, discussed in [19].

Proposition 2.7. *There is a short morphism of hypermagmas $\pi : \mathfrak{H}_{\mathcal{SO}_0} \rightarrow \mathfrak{T}_{\mathcal{SO}_0}$ that forgets the datum of the head function. The assignment of a head function $h : T \mapsto h_T$ defined on a domain $\text{Dom}(h)$ that is a subset of $\mathfrak{T}_{\mathcal{SO}_0}$ determines a weak subhypermagma of $\mathfrak{H}_{\mathcal{SO}_0}$ and a coshort*

hypermagma morphism $\sigma_h : (\text{Dom}(h), \mathfrak{M}_h) \rightarrow (\mathfrak{H}_{\mathcal{SO}_0}, \mathfrak{M}^{\mathfrak{H}})$ that is a partial section of π , where $\mathfrak{M}_h$ is the induced hypermagma structure on $\text{Dom}(h)$.

Proof. Here we view the magma $(\mathfrak{T}_{\mathcal{SO}_0}, \mathfrak{M})$ as a hypermagma where the hyperoperation is singleton-valued. Note that, even if $(\mathfrak{T}_{\mathcal{SO}_0}, \mathfrak{M})$ is an actual magma, subsets of $\mathfrak{T}_{\mathcal{SO}_0}$ will have an induced structure of weak subhypermagma, where the hyperoperation is set values, because we have to include the possibility that the emptyset is a value. In particular, whenever the chosen subset of $\mathfrak{T}_{\mathcal{SO}_0}$ is not a submagma, the induced structure will include the emptyset as a possible value of the hyperoperation. Only in the case of a submagma the induced weak subhypermagma structure is itself a magma structure. Consider then the case of a subset $\text{Dom}(h) \subset \mathfrak{T}_{\mathcal{SO}_0}$ that is the domain of a head function $h : T \mapsto h_T$. We know from [19], §1.13.2, that $\text{Dom}(h)$ is not a submagma. Let $\mathfrak{M}_h$ denote the induced weak subhypermagma structure on $\text{Dom}(h)$. This has the form (for $T, T' \in \text{Dom}(h)$):

$$\mathfrak{M}_h(T, T') = \begin{cases} \mathfrak{M}(T, T') & \mathfrak{M}(T, T') \in \text{Dom}(h) \\ \emptyset & \mathfrak{M}(T, T') \notin \text{Dom}(h). \end{cases}$$

This is clearly also compatibly describing $\text{Dom}(h)$ as a weak subhypermagma of $\mathfrak{H}_{\mathcal{SO}_0}$ with

$$(2.4) \quad \mathfrak{M}^{\mathfrak{H}}((T, h_T), (T', h_{T'})) = \begin{cases} (\mathfrak{M}(T, T'), h_{\mathfrak{M}(T, T')}) & \mathfrak{M}(T, T') \in \text{Dom}(h) \\ \emptyset & \mathfrak{M}(T, T') \notin \text{Dom}(h). \end{cases}$$

Note that this agrees with the weak subhypermagma relation (2.2), since we know that, if $\mathfrak{M}(T, T') \in \text{Dom}(h)$ then by the properties of head functions we have either $h_{\mathfrak{M}(T, T')} = h_T$ or $h_{\mathfrak{M}(T, T')} = h_{T'}$, in the sense discussed above, of being the unique extension of h_T and $h_{T'}$ to $\mathfrak{M}(T, T')$ with value at the root agreeing with either h_T or $h_{T'}$ at the respective roots of T and T' , so that

$$(\mathfrak{M}(T, T'), h_{\mathfrak{M}(T, T')}) = (\mathfrak{M}(T, T'), \{h_T, h_{T'}\}) \cap \text{Dom}(h).$$

This embedding $\sigma_h : \text{Dom}(h) \hookrightarrow \mathfrak{H}_{\mathcal{SO}_0}$ given by $\sigma_h(T) = (T, h_T)$ is then a coshort morphism and it clearly satisfies $\pi \circ \sigma_h = \text{id}$. $\square$

Remark 2.8. Thus, one can think of a choice of a head function $h : T \mapsto h_T$ as a choice of a weak subhypermagma of $\mathfrak{H}_{\mathcal{SO}_0}$ with the subhypermagma structure of the form (2.4).

We will return to discuss the hypermagma structure in §5.

3. COMMAND RELATIONS

In this section we compare the c-command and m-command relations used on syntax. We show that the c-command relation is compatible with the magma structure of the set of syntactic objects, while the m-command relation, which depends not only on the syntactic object but also on the head function, is not compatible with the magma operation but becomes compatible with the hypermagma structure introduced in the previous section, which accounts for the properties of the head function.

3.1. c-command and m-command. Also, recall from [19] that, for $T \in \mathfrak{T}_{\mathcal{SO}_0}$ a syntactic object, the set $\text{Acc}(T)$ of accessible terms of T consists of all the subtrees T_v of T for v a non-root vertex of T , where T_v denotes the subtree rooted at v , and containing everything that lies below v . We also write $\text{Acc}'(T) = \text{Acc}(T) \cup \{T\}$.

Definition 3.1. For a syntactic object $T \in \mathfrak{T}_{\mathcal{SO}_0}$, two vertices, v_1 and v_2 , of T are sisters if there is a vertex v of T above them (i.e., closer to the root) and connected to both v_1 and v_2 .

- A vertex v of T dominates another vertex w if v is on the unique path from the root of T to w .

- A vertex v in T c-commands another vertex w if neither dominates the other and the lowest vertex that dominates v also dominates w . (Equivalently, $w \in T_{v'}$, the accessible term rooted at the sister vertex v' of v .)
- A vertex v asymmetrically c-commands a vertex w if v c-commands w , and v and w are not sisters. (Equivalently, $T_w \in \text{Acc}(T_{v'})$, the set of accessible terms of $T_{v'}$, with v' the sister vertex of v .)
- For a pair (T, h_T) of a syntactic object T and a head function h_T , a maximal projection is a vertex v with the property that T_v is not strictly contained in any larger $T_w \in \text{Acc}'(T)$ with $h_T(w) = h_T(v)$.
- A vertex v in (T, h_T) m-commands another vertex w if neither dominates the other and the maximal projection of v dominates w .

The c-command relation only depends on T , while the m-command relation depends on both T and h_T , as the maximal projection is determined by the head function.

Remark 3.2. It is shown in [19], Lemma 1.14.1, that a head function h_T on a syntactic object $T \in \mathfrak{T}_{\mathcal{SO}_0}$ determines a partition of the vertices of T into a collection of paths (some of them trivially consisting of a single leaf), $\{\gamma_\ell\}_{\ell \in L(T)}$ for $L(T)$ the set of leaves of T , where $v \in \gamma_\ell$ iff $h_T(v) = \ell$. The internal vertex v_ℓ where the path γ_ℓ (if nontrivial) terminates gives the maximal projection of the head ℓ . The resulting accessible term T_{v_ℓ} will also include the specifier position. In Definition 4.17, we will refer to T_{v_ℓ} as the “narrow phase” of ℓ and we will discuss the setting of projection and phases more precisely.

3.2. c-command and the magma structure. We show that the c-command relation on syntactic objects is compatible with the magma structure of $\mathfrak{T}_{\mathcal{SO}_0}$.

Let Rel denote the category where objects are pairs (S, R) of a set and a relation $R \subset S \times S$, and with morphisms $f : (S, R) \rightarrow (S', R')$ that are functions of sets $f : S \rightarrow S'$ that intertwine the relations, namely for all $(x, y) \in R$, we have $(f(x), f(y)) \in R'$. Let $\mathcal{FS}$ denote the category of finite sets.

Definition 3.3. Suppose given a magma $(X, \star_X)$ and a map $\rho : X \rightarrow \mathcal{FS}$ to finite sets that assigns a set $\rho(x) = S_x$ to each element of the magma. Given a collection $\mathcal{R}$ of relations $R_x \subseteq S_x \times S_x$, let $\mathcal{S}_{X, \mathcal{R}} = \{(S_x, R_x) \in \text{Rel}\}$, and let $\tilde{\rho} : X \rightarrow \text{Rel}$ be given by $\tilde{\rho}(x) = (S_x, R_x)$. The collection $\mathcal{R} = \{R_x \subseteq S_x \times S_x \mid x \in X\}$ is *magmatic* if the operation $\star_X$ determines a magma structure $\star_{X, \rho} : \mathcal{S}_{X, \mathcal{R}} \times \mathcal{S}_{X, \mathcal{R}} \rightarrow \mathcal{S}_{X, \mathcal{R}}$, with $\tilde{\rho}(x) \star_{X, \rho} \tilde{\rho}(y) = \tilde{\rho}(x \star_X y)$. In the case of a hypermagma $(H, \star_H)$, and a map $\rho : H \rightarrow \mathcal{FS}$ with $\rho(x) = S_x$, a collection $\mathcal{R} = \{R_x \subseteq S_x \times S_x \mid x \in H\}$ is *hypermagmatic* if $\star_H$ determines a hypermagma structure $\star_{H, \rho} : \mathcal{S}_{H, \mathcal{R}} \times \mathcal{S}_{H, \mathcal{R}} \rightarrow \mathcal{P}(\mathcal{S}_{H, \mathcal{R}})$ on $\mathcal{S}_{H, \mathcal{R}} = \{(S_x, R_x) \in \text{Rel}\}$, with $\tilde{\rho}(x) \star_{H, \rho} \tilde{\rho}(y) \supseteq \tilde{\rho}(x \star_H y)$, where $\tilde{\rho}(x) = (S_x, R_x)$. It is *strictly hypermagmatic* if $\tilde{\rho}(x) \star_{H, \rho} \tilde{\rho}(y) = \tilde{\rho}(x \star_H y)$.

Lemma 3.4. *The c-command relation is magmatic with the respect to the magma $\mathfrak{T}_{\mathcal{SO}_0}$ of syntactic objects and the map $V : T \mapsto V(T)$, the set of vertices of T .*

Proof. We need to show that the magma operation $\mathfrak{M}$ on syntactic objects induces a magma operation $\mathfrak{M}_V$ on the set $\mathcal{S}_{\mathfrak{T}_{\mathcal{SO}_0}, \mathcal{R}_c}$, where $\mathcal{R}_c$ is the c-command relation $R_{c, T} \subset V(T) \times V(T)$ with $R_{c, T} = \{(v, w) \mid v \text{ c-commands } w\}$. The set $\mathcal{S}_{\mathfrak{T}_{\mathcal{SO}_0}, \mathcal{R}_c}$ consists of pairs $(V(T), R_{c, T})$ in Rel . To see that the magma operation $\mathfrak{M}$ on $\mathfrak{T}_{\mathcal{SO}_0}$ induced a magma operation $\mathfrak{M}_V$ on $\mathcal{S}_{\mathfrak{T}_{\mathcal{SO}_0}, \mathcal{R}_c}$ we need to show that we can define $\mathfrak{M}_V((V(T), R_{c, T}), (V(T'), R_{c, T'}))$ in terms of $V(T)$, $V(T')$ and $R_{c, T}$, $R_{c, T'}$, in such a way that it satisfies

$$(3.1) \quad \mathfrak{M}_V((V(T), R_{c, T}), (V(T'), R_{c, T'})) = (V(\mathfrak{M}(T, T')), R_{c, \mathfrak{M}(T, T')}).$$

This relation shows that the magma operation $\mathfrak{M}_V$ on $\mathcal{S}_{\mathfrak{T}_{SO_0}, \mathcal{R}_c}$ should be of the form

$$\mathfrak{M}_V((V(T), R_{c,T}), (V(T'), R_{c,T'})) = (\mathfrak{M}_V(V(T), V(T')), \mathfrak{M}_V(R_{c,T}, R_{c,T'})),$$

with $\mathfrak{M}_V(V(T), V(T')) = V(\mathfrak{M}(T, T'))$ and $\mathfrak{M}_V(R_{c,T}, R_{c,T'}) = R_{c, \mathfrak{M}(T, T')}$. To express $V(\mathfrak{M}(T, T'))$ in terms of $V(T)$ and $V(T')$ and $R_{c, \mathfrak{M}(T, T')}$ in terms of $R_{c,T}$ and $R_{c,T'}$, note that

$$V(\mathfrak{M}(T, T')) = V(T) \sqcup V(T') \sqcup \{v_{\mathfrak{M}(T, T')}\},$$

with $v_{\mathfrak{M}(T, T')}$ the root vertex of $\mathfrak{M}(T, T')$, and

$$R_{c, \mathfrak{M}(T, T')} = R_{c,T} \sqcup R_{c,T'} \sqcup V(T) \times \{v_{T'}\} \sqcup \{v_{T'}\} \times V(T) \sqcup V(T') \times \{v_T\} \sqcup \{v_T\} \times V(T').$$

This shows that the c-control relation $R_{c, \mathfrak{M}(T, T')}$ on $\mathfrak{M}(T, T')$ is uniquely determined as a function $\mathfrak{M}_V(R_{c,T}, R_{c,T'})$ of $R_{c,T}$ and $R_{c,T'}$, which satisfies (3.1). $\square$

3.3. m-command and hypermagmas. Unlike the c-command relation, the m-command relation is not magmatic with respect to the magma $\mathfrak{T}_{SO_0}$. Indeed, first notice that, since the m-command relation requires a head function, it can only be defined on a pair (T, h_T) . So we write $R_{m, (T, h_T)} \subset V(T) \times V(T)$ for the relation

$$R_{m, (T, h_T)} = \{(v, w) \mid v \text{ m-commands } w\},$$

where v m-commands w with respect to the maximal projection induced by the head function h_T . So we can assume given a choice of a head function $h : T \mapsto h_T$ defined on a domain $\text{Dom}(h) \subset \mathfrak{T}_{SO_0}$, with respect to which the m-command is computed. For $T, T' \in \text{Dom}(h)$, we then have the m-command relations $R_{m, (T, h_T)} \subset V(T) \times V(T)$ and $R_{m, (T', h_{T'})} \subset V(T') \times V(T')$. The syntactic object $\mathfrak{M}(T, T')$ need not be in $\text{Dom}(h)$. When $\mathfrak{M}(T, T') \notin \text{Dom}(h)$, the m-command relation is not defined on $\mathfrak{M}(T, T')$. Thus, we see that m-command is not magmatic for the magma $\mathfrak{T}_{SO_0}$.

Proposition 3.5. *The m-command relation is strictly hypermagmatic for the weak subhyper-magma $\text{Dom}(h)$ of $\mathfrak{H}_{SO_0}$.*

Proof. Given a head function $h : T \mapsto h_T$ with domain $\text{Dom}(h)$ and the hypermagma $(\text{Dom}(h), \mathfrak{M}_h)$ with $\mathfrak{M}_h$ induced by its embedding as weak subhypermagma of $\mathfrak{H}_{SO_0}$ as in Proposition 2.7. To see that m-command is strictly hypermagmatic with respect to $(\text{Dom}(h), \mathfrak{M}_h)$, we take $\rho(T, h_T) = V(T)$ and $\tilde{\rho}(T, h_T) = (V(T), R_{m, (T, h_T)})$. The induced hypermagma structure $\mathfrak{M}_{V, h}$ on $\mathcal{S}_{\text{Dom}(h), \mathcal{R}_m} = \{(V(T), R_{m, (T, h_T)}) \mid T \in \text{Dom}(h)\}$ should satisfy

$$\mathfrak{M}_{V, h}((V(T), R_{m, (T, h_T)}), (V(T'), R_{m, (T', h_{T'})})) = (V(\mathfrak{M}(T, T')), R_{m, (\mathfrak{M}(T, T'), h_{\mathfrak{M}(T, T')})})$$

if $\mathfrak{M}(T, T') \in \text{Dom}(h)$ and $\mathfrak{M}_{V, h}((V(T), R_{m, (T, h_T)}), (V(T'), R_{m, (T', h_{T'})})) = \emptyset$ if $\mathfrak{M}(T, T') \notin \text{Dom}(h)$. This gives

$$\mathfrak{M}_{V, h}((V(T), R_{m, (T, h_T)}), (V(T'), R_{m, (T', h_{T'})})) = (\mathfrak{M}_{V, h}(V(T), V(T')), \mathfrak{M}_{V, h}(R_{m, (T, h_T)}, R_{m, (T', h_{T'})})),$$

with

$$\mathfrak{M}_{V, h}(V(T), V(T')) = V(\mathfrak{M}(T, T')) = V(T) \sqcup V(T') \sqcup \{v_{\mathfrak{M}(T, T')}\},$$

as in the c-command case, with $v_{\mathfrak{M}(T, T')}$ the root vertex of $\mathfrak{M}(T, T')$, and $\mathfrak{M}_{V, h}(R_{m, (T, h_T)}, R_{m, (T', h_{T'})}) = \emptyset$ if $\mathfrak{M}(T, T') \notin \text{Dom}(h)$, while for $\mathfrak{M}(T, T') \in \text{Dom}(h)$, we have

$$\mathfrak{M}_{V, h}(R_{m, (T, h_T)}, R_{m, (T', h_{T'})}) = R_{m, (\mathfrak{M}(T, T'), h_{\mathfrak{M}(T, T')})}$$

where depending on whether $h_{\mathfrak{M}(T, T')} = h_T$ or $h_{\mathfrak{M}(T, T')} = h_{T'}$ (in the sense of extending either h_T or $h_{T'}$ to the root of $\mathfrak{M}(T, T')$), we have, respectively

$$R_{m, (\mathfrak{M}(T, T'), h_{\mathfrak{M}(T, T')})} = R_{m, (T, h_T)} \sqcup R_{m, (T', h_{T'})} \sqcup \gamma_{h(v_T)} \times V(T'),$$

$$R_{m,(\mathfrak{M}(T,T'),h_{\mathfrak{M}(T,T')})} = R_{m,(T,h_T)} \sqcup R_{m,(T',h_{T'})} \sqcup \gamma_{h(v_{T'})} \times V(T),$$

where v_T and $v_{T'}$ are the roots of T and T' , respectively, and γ_ℓ are the paths determined by the head function, as in Lemma 1.14.1 of [19], as recalled above. $\square$

4. COMPLEMENTED HEADS AND COLORING

We now consider, in addition to the presence of a head function h_T on the syntactic object T , in the general sense recalled in Definition 2.4, the finer property that this head function is complemented, in the sense of Definition 1.14.2 of [19].

First observe that the assignment of a head function h_T to a syntactic object is equivalent to a bi-coloring of the edges of T according to the following rule.

Remark 4.1. Assigning a head function h_T to a syntactic object T is the same as assigning a bi-coloring (say B/W) of the edges of T , with the rule that one of the edges below each non-leaf vertex v is colored B and the other is colored W (see Lemma 1.13.4 of [19]), where one of the two colors (say B) marks the direction of the head $h_T(v)$. In particular, the paths γ_ℓ mentioned in Remark 3.2 are the paths of B-colored edges in T , according to this coloring.

We show here that assigning a complemented head, and keeping track of the specifier-head-complement structure also has an interpretation in terms of coloring, with appropriate coloring rules as discussed below.

4.1. Complemented head, extended projection, and coloring. We follow here a procedure analogous to the analysis of theta role carried out in [20]. We first consider a coloring algorithm that only accounts for the basic part of the structure, which in this case consists of the three roles: specifier, head, and complement. We then enrich this fundamental structure by allowing for additional modifier positions. This is analogous to the coloring for theta roles of [20], where we first introduced what we called “bare theta structures”, and then enlarged those to include non-theta positions.

We start here by recalling the notion of “complemented head” introduced in Definition 1.14.2 of [19].

Definition 4.2. A *complemented head function* h_T is a function $h_T : V^\circ(T) \rightarrow L(T) \times (\text{Acc}(T) \cup \{1\})$ that assigns to a non-leaf vertex v the corresponding head $\ell = h_T(v) \in L(T)$, together with an associated accessible term $Z_\ell \in \text{Acc}(T)$ (or possibly empty $Z_\ell = 1$, the formal empty tree), that is the complement of the head ℓ . When nonempty, $Z_\ell \subseteq T_{s_{h_T(v)}}$, is either equal to or contained in the accessible term rooted at the sister vertex $s_{h_T(v)}$ of the leaf $\ell = h_T(v)$.

Note that the last condition, that $Z_\ell \subseteq T_{s_{h_T(v)}}$, will be relaxed, by allowing $Z_\ell \subseteq T_v$ for some other $v \in \gamma_\ell$, when we also introduce the possibility of additional modifier positions.

The complement Z_ℓ of the head ℓ is an accessible term of T , so it is itself a syntactic object, and it is also endowed with a head function, which is the restriction of h_T to Z_ℓ . This head function is complemented, hence Z_ℓ itself recursively decomposes into substructures containing heads and their complements.

We refine the structure of head and complement by also accounting for an additional *specifier* position.

To this purpose, we will need here to distinguish between *maximal projection* and *extended projection*. In our formulation, the maximal projection of a head ℓ is the endpoint v_ℓ of the path γ_ℓ of Remark 3.2. When the head function h_T is the actual syntactic head, this corresponds to the usual notion of maximal projection: the highest level (highest vertex $v = v_\ell$) of the phrase structure that maintains the same (lexical) head ℓ . On the other hand, in the case of the syntactic

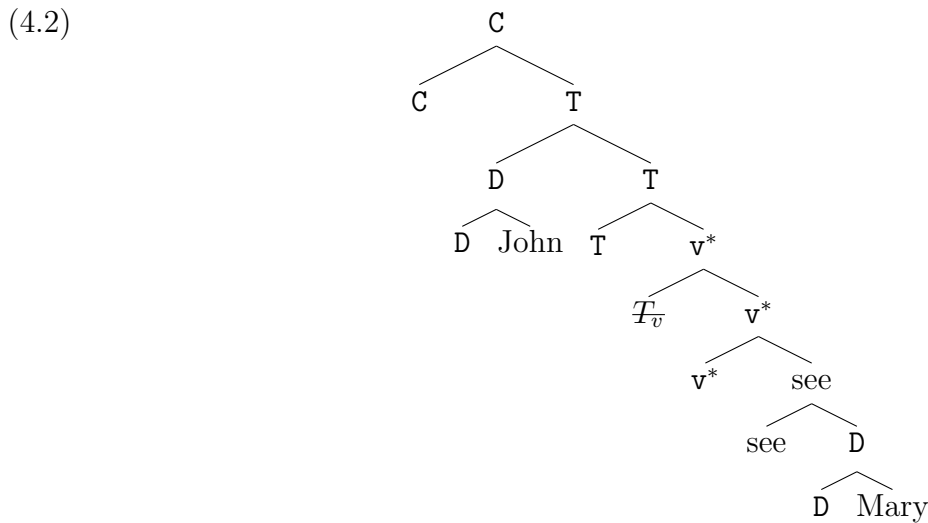
head, the extended projection involves the maximal projection v_ℓ of a *lexical head* and, above that, a certain number of *functional projections*, namely projections of *functional heads* such as tense features (T) or inflection (INFL or I) or complementizer (C), in the extended projection of VPs or determiner (D) in the extended projection of NPs. The Extended Projection Principle states that a viable sentence has a subject located in *specifier position* of an extended projection. We will focus especially on the C, INFL, and v^* functional heads involved in the extended projection. We will also discuss cases involving heads like v , in the extended projection, that are not phase-heads, hence behave differently with respect to IM movement. (In the setting of [6], TP is part of v not of INFL.) The structure of *phases* and the distinction between the interior and the edge of a phase (see the discussion in Section 1.14 of [19]) and the rules about movement by Internal Merge depend on the structure of the extended projection and its functional heads, as we discuss further below.

4.1.1. *Tree structures and extended projection.* We will use two different ways of representing the non-planar binary rooted trees that are the syntactic objects, a simplified form where the leaves are lexical heads, and a full form where the leaves include the functional heads that occur in the extended projections. In the first case, one can remember the role of the functional projections by encoding them in the coloring. Principles of structural economy suggest that a simplified structure, where the extended projection may be accounted for via some kind of coloring rules, may be preferable for both computational and theoretical purposes. However, we will focus here on modeling the full form of the syntactic objects where the extended projection is explicitly included in the resulting binary rooted tree, so that we can analyze more precisely how the usual formulation of the structure of phases can be described in terms of coloring rules and colored operads bud generating systems.

To see a simple example, consider the sentence *John saw Mary*. The simplified form of the binary tree realizing this sentence as a syntactic object is just of the form



where we identify “saw” as the head, “Mary” as the complement of the head, and “John” as the specifier position. In the full form, a syntactic object would instead look like



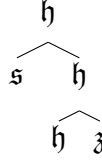
where T_v is the trace of the accessible term

$$T_v = \begin{array}{c} \text{D} \\ \swarrow \searrow \\ \text{D} \quad \text{John} \end{array}$$

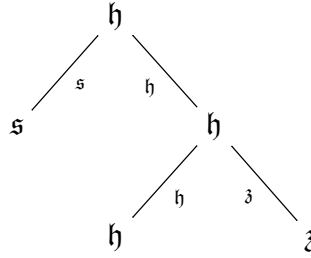
moved by IM, and $\mathbf{C}$, $\mathbf{T}$, $\mathbf{D}$, $\mathbf{v}^*$ are the heads giving rise to functional projections. Note how with this formulation the accessible term T_v is moved to the edge of the phase by Internal Merge, after the functional projections are added on top of the lexical projection. The head structure is represented here by marking the internal vertices along each path γ_ℓ by the head ℓ .

The way this syntactic object is depicted here is as obtained in the Merge derivation generated using Sandiway Fong's Minimalist Machine¹, which can be used for many more examples.

4.1.2. *local coloring moves*. Before describing the general form of the coloring that account for the bare structure of specifier, head, complement, we can see it applied to this example. In the simple form (4.1) we just have



where the three colors $\{\mathbf{s}, \mathbf{h}, \mathbf{z}\}$ mark the specifier $\mathbf{s}$, head $\mathbf{h}$, and complement $\mathbf{z}$ positions. We can also write it in the form



In this case the labels at the edges are redundant, as they simply follow the convention that coloring a certain vertex is equivalent to coloring the unique edge of the tree above that vertex, so the vertex and edge colorings are equivalent. The internal edge, in this case, is also colored $\mathbf{h}$, and so is the root v_ℓ , as in this simplified form they lie on the path γ_ℓ of the head $\ell = \text{saw}$. This example shows a rephrasing in terms of coloring of the form of projections and phases as described in Sections 1.13 and 1.14 of [19].

We can also reinterpret this coloring of the simple form (4.1) in terms of generators of the form

$$(4.3) \quad T_{\mathbf{h},\mathbf{s}}^{\mathbf{c}} = \begin{array}{c} \mathbf{c} \\ \swarrow \quad \searrow \\ (\mathbf{h}, \mathbf{s}^\uparrow) \quad \mathbf{s}^\downarrow \end{array} \quad \text{and} \quad T_{\mathbf{h},\mathbf{z}}^{\mathbf{h}} := \begin{array}{c} (\mathbf{h}, \mathbf{s}^\uparrow) \\ \swarrow \quad \searrow \\ (\mathbf{h}, \mathbf{z}^\uparrow, \mathbf{s}^\uparrow) \quad \mathbf{z}^\downarrow \end{array}$$

as the composition that plugs in the $(\mathbf{h}, \mathbf{s}^\uparrow)$ -marked output at the root of the second generator to the $(\mathbf{h}, \mathbf{s}^\uparrow)$ -marked input at one of the leaves of the first generator. The color $\mathbf{c}$ at the root of the first generator can be either another $\mathbf{s}^\downarrow$ or $\mathbf{z}^\downarrow$ and it depends on what role the whole structure obtained in this way will play if incorporated into a larger structure, either in a specifier position or as the complement of another head in a higher phase, which contributes another $(\mathbf{h}, \mathbf{z}^\uparrow, \mathbf{s}^\uparrow)$. Here one reads the label $\mathbf{h}$ at a vertex as a coloring of the edge immediately above it: so these $\mathbf{h}$ -colored edges are exactly those that form the paths γ_ℓ of the head function. The label $(\mathbf{z}^\uparrow, \mathbf{s}^\uparrow)$ works in the same way as the labels $\theta^\uparrow$ in the case of theta theory in [20]: $(\mathbf{h}, \mathbf{z}^\uparrow)$ means the head is complemented, with a non-empty complement, and the further $\mathbf{s}^\uparrow$ means that it will also carry a specifier. In the generator $T_{\mathbf{h},\mathbf{z}}^{\mathbf{h}}$ the head binds with its complement, which receives the label $\mathbf{z}^\downarrow$ marking it as the complement position, while the output carries the remaining position that the head will bind to, the specifier $\mathbf{s}$. The second generator $T_{\mathbf{h},\mathbf{s}}^{\mathbf{c}}$ then receives the $\mathbf{s}^\uparrow$ through the operad composition from the output of the previous one, and discharges it onto the specifier position that

¹<https://sandiway.arizona.edu/mpp/mm.html>.

gets marked as $\mathfrak{s}^\downarrow$. The output position then remains free to assume a new role, either as a specifier or a complement to another head. This vertex is then the v_ℓ , the maximal projection of the head ℓ carrying the $(\mathfrak{h}, \mathfrak{z}^\uparrow, \mathfrak{s}^\uparrow)$ and the endpoint of the path γ_ℓ which in this case has two edges. The generator $T_{\mathfrak{h}, \mathfrak{s}}^{\mathfrak{c}}$ also accounts for cases where the head $\mathfrak{h}$ has an empty complement so only $(\mathfrak{h}, \mathfrak{s}^\uparrow)$ is injected with the specifier position delivered to the other leaf marked by $\mathfrak{s}^\downarrow$, and an $\mathfrak{c} \in \{\mathfrak{z}, \mathfrak{s}\}$ as output to combine with the head of a larger structure.

This composition of output to input with matching colors is the same type of colored operad structure used in [20] to model theta role assignments.

When we consider the second form (4.2), however, the structure of the γ_ℓ paths is different. Again we want to keep track of the structure of complemented head as we follow the functional projections and encode the structure into local coloring rules.

We are going to interpret this case also as a coloring obtained via a sequence of operad insertions of a certain set of generators. Again the mathematical setting relevant here is that of colored operads and bud generating systems already used in [20], which we will recall more in detail below.

Consider colored structures of the form

$$(4.4) \quad T_{\mathfrak{h}^\omega \mathfrak{z} \mathfrak{s}, \mathfrak{z}}^{\mathfrak{h}^\omega \mathfrak{s}} := \begin{array}{c} (\mathfrak{h}^\omega, \mathfrak{s}^\uparrow) \\ \swarrow \quad \downarrow \quad \searrow \\ (\mathfrak{h}^\omega, \mathfrak{z}^\uparrow, \mathfrak{s}^\uparrow) \quad \mathfrak{z}^\downarrow \end{array}$$

$$(4.5) \quad T_{\mathfrak{h}^\omega \mathfrak{z}, \mathfrak{s} \mathfrak{z}}^{\mathfrak{h}^\omega \mathfrak{s}} := \begin{array}{c} (\mathfrak{h}^\omega, \mathfrak{s}^\uparrow) \\ \swarrow \quad \downarrow \quad \searrow \\ (\mathfrak{h}^\omega, \mathfrak{z}^\uparrow) \quad (\mathfrak{s}^\uparrow, \mathfrak{z}^\downarrow) \end{array}$$

$$(4.6) \quad T_{\mathfrak{h}^\omega \mathfrak{z} \mathfrak{s}, \mathfrak{z}}^{\mathfrak{s} \mathfrak{z}} := \begin{array}{c} (\mathfrak{s}^\uparrow, \mathfrak{z}^\downarrow) \\ \swarrow \quad \downarrow \quad \searrow \\ (\mathfrak{h}^\omega, \mathfrak{z}^\uparrow, \mathfrak{s}^\uparrow) \quad \mathfrak{z}^\downarrow \end{array}$$

$$(4.7) \quad T_{\mathfrak{h}^\omega \mathfrak{z}, \mathfrak{z}}^{\mathfrak{c}} := \begin{array}{c} \mathfrak{c}^\downarrow \\ \swarrow \quad \downarrow \quad \searrow \\ (\mathfrak{h}^\omega, \mathfrak{z}^\uparrow) \quad \mathfrak{z}^\downarrow \end{array} \quad \text{with } \mathfrak{c} \in \{\mathfrak{s}, \mathfrak{z}\}$$

$$(4.8) \quad T_{\mathfrak{h}^\omega \mathfrak{s}, \mathfrak{s}}^{\mathfrak{c}} := \begin{array}{c} \mathfrak{c}^\downarrow \\ \swarrow \quad \downarrow \quad \searrow \\ (\mathfrak{h}^\omega, \mathfrak{s}^\uparrow) \quad \mathfrak{s}^\downarrow \end{array} \quad \text{with } \mathfrak{c} \in \{\mathfrak{s}, \mathfrak{z}\}$$

where we allow a certain set of colors $\{\mathfrak{h}^\omega\}_{\omega \in \Omega_{\mathfrak{h}}}$ that distinguishes the lexical heads ($\omega \in \Omega_{\mathfrak{h}, \text{lex}}$, see Definition 4.4) and all different possible functional heads $\omega \in \Omega_{\mathfrak{h}, f} = \{\mathfrak{v}^*, \mathfrak{T}, \mathfrak{C}, \mathfrak{D}, \dots\}$. Since the trees are not planar, the order of the two lower indices does not matter so we equivalently write $T_{\mathfrak{c}_2, \mathfrak{c}_3}^{\mathfrak{c}_1} = T_{\mathfrak{c}_3, \mathfrak{c}_2}^{\mathfrak{c}_1}$. In all these two structures, we read the labelled half-edge at the top as the output and the two half-edges at the bottom as the inputs.

The way to read the labels on one of these generators $T_{\mathfrak{c}_2, \mathfrak{c}_3}^{\mathfrak{c}_1}$ is by interpreting the label $(\mathfrak{h}^\omega, \mathfrak{c}^\uparrow)$ as a head of type ω that injects colors $\mathfrak{c}$ (either $\mathfrak{z}$ or $\mathfrak{s}$ or a pair $(\mathfrak{s}, \mathfrak{z})$ meaning that this head will have to bind with a received of the role of complement $\mathfrak{z}$ or specifier $\mathfrak{s}$: the arrows $\downarrow, \uparrow$ distinguish whether a leaf (or a whole structure when it occurs at the root vertex) receives the corresponding role, marked as $\mathfrak{c}^\downarrow$, or whether a head is assigning it color marking $\mathfrak{c}^\uparrow$. When a color $\mathfrak{c}^\downarrow$ occurs at

an output it means the entire structure below that vertex receives that role. (The case of theta roles assignments in [20] is analogous.)

In (4.4), (4.5), (4.6), (4.7), (4.8), to simplify notation, we only keep track of the type of head $\mathfrak{h}^\omega$ in the $\mathfrak{h}$ labeling but not explicitly in the associated $\mathfrak{z}$ and $\mathfrak{s}$. To be more precise, as we will need to in §4.6 where we select an appropriate subset of generators for the Extended Projection Principle (EPP) condition, we also write the other labels as $\mathfrak{z}_\omega^\uparrow$, $\mathfrak{z}_\omega^\downarrow$, $\mathfrak{s}_\omega^\uparrow$, $\mathfrak{s}_\omega^\downarrow$, to keep track of the fact that they are associated to a $\mathfrak{h}^\omega$ -labelled head. Thus, for example, we write, more precisely

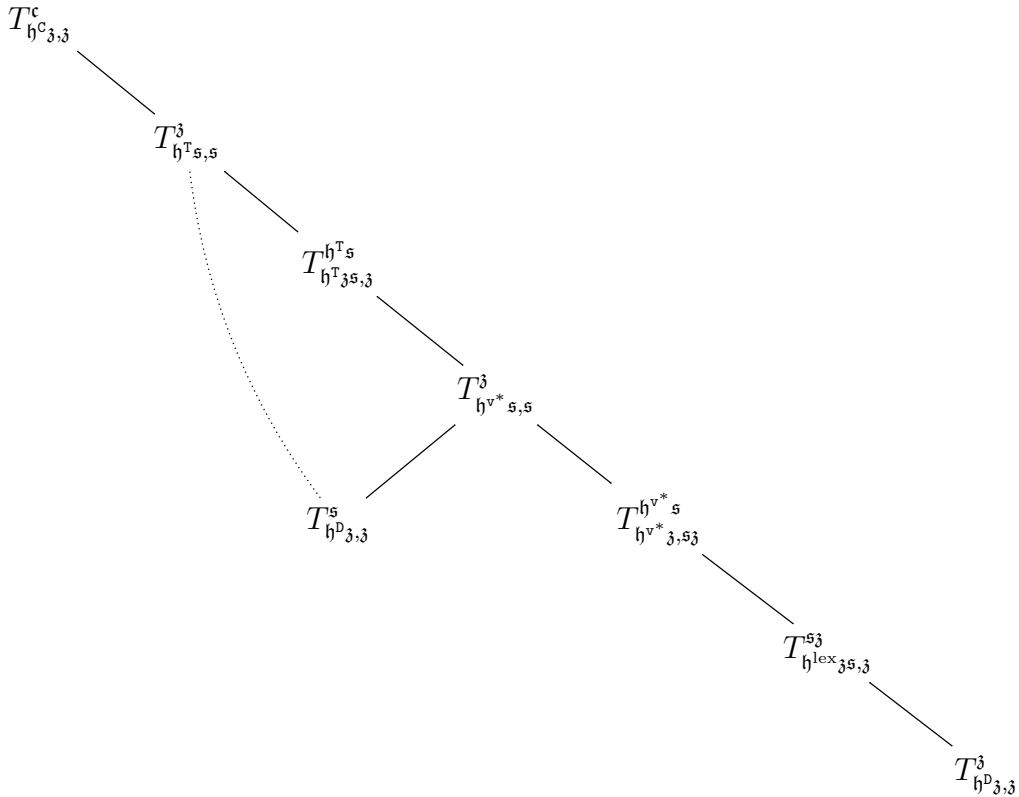
$$(4.9) \quad T_{\mathfrak{h}^\omega \mathfrak{z}^\omega \mathfrak{s}, \mathfrak{z}}^{\mathfrak{s} \mathfrak{z}} := \begin{array}{c} (\mathfrak{s}_\omega^\uparrow, \mathfrak{z}_{\omega'}^\downarrow) \\ \swarrow \quad \downarrow \quad \searrow \\ (\mathfrak{h}^\omega, \mathfrak{z}_\omega^\uparrow, \mathfrak{s}_\omega^\uparrow) \quad \mathfrak{z}_\omega^\downarrow \end{array}$$

$$(4.10) \quad T_{\mathfrak{h}^{\omega'} \mathfrak{z}, \mathfrak{s} \mathfrak{z}}^{\mathfrak{h}^{\omega'} \mathfrak{s}} := \begin{array}{c} (\mathfrak{h}^{\omega'}, \mathfrak{s}_\omega^\uparrow) \\ \swarrow \quad \searrow \\ (\mathfrak{h}^{\omega'}, \mathfrak{z}_{\omega'}^\uparrow) \quad (\mathfrak{s}_\omega^\uparrow, \mathfrak{z}_{\omega'}^\downarrow) \end{array}$$

Here one interprets a label $\mathfrak{z}_{\omega'}^\downarrow$ at the output as indicating that the whole structure below that vertex forms the complement of a head of type $\mathfrak{h}^{\omega'}$ in a larger structure, while a label $\mathfrak{s}_\omega^\uparrow$ in a leaf input indicates that that input can receive an $\mathfrak{s}_\omega$ role coming from a lower head of type $\mathfrak{h}^\omega$.

As will be discussed more in detail in §4.3 and §4.4, the composition rule between these generators corresponds to gluing an output to an input when they carry matching labels.

Then we can see that the example (4.2) can be realized as a chain of colored operad compositions



where each line represent an insertion of output of the lower term to the input of the term above it with matching color (colored operad composition). The dotted line represents the movement by

Internal Merge of the term $T_{\mathfrak{h}^{\mathfrak{D}_{3,3}}}^{\mathfrak{s}}$ from input of $T_{\mathfrak{h}^{\mathfrak{v}^*_{s,s}}}^{\mathfrak{s}}$ to input of $T_{\mathfrak{h}^{\mathfrak{T}_{s,s}}}^{\mathfrak{s}}$. We will describe how this type of movement happens in §5.2 and §6.

4.2. Colored operads and bud generating systems. We refer the reader to Section 2.1 of [20] and Section 3.8 of [19] for a quick summary of colored operads. Operads were originally introduced in [22] and the notion of colored operad is discussed in great detail in [26]. We just recall here that an operad in the category of sets consists of a collection of sets $\mathfrak{D} = \{\mathfrak{D}(n)\}_{n \geq 1}$ whose elements are operations with n inputs and one output. The sets $\mathfrak{D}(n)$ have composition rules that consist exactly of plugging the output of one operation into the input of another,

$$(4.11) \quad \circ_i : \mathfrak{D}(n) \otimes \mathfrak{D}(m) \rightarrow \mathfrak{D}(n + m - 1).$$

where the index i ranges over the set of m inputs of operations $T \in \mathfrak{D}(m)$ and the result $T \circ_i T'$ of combining the output of T' to the i -th input of T has $n + m - 1$ remaining free inputs. The associativity of these compositions are expressed in the form

$$(X \circ_j Y) \circ_i Z = \begin{cases} (X \circ_i Z) \circ_{j+c-1} Y & 1 \leq i < j \\ X \circ_j (Y \circ_{i-j+1} Z) & j \leq i < b+j \\ (X \circ_{i-b+1} Z) \circ_j Y & j+b \leq i \leq a+b-1. \end{cases}$$

The composition operation

$$(4.12) \quad \gamma : \mathfrak{D}(n) \times \mathfrak{D}(k_1) \times \cdots \times \mathfrak{D}(k_n) \rightarrow \mathfrak{D}(k_1 + \cdots + k_n)$$

where all the n inputs of $T \in \mathfrak{D}(n)$ are paired to outputs of operations $T_j \in \mathfrak{D}(k_j)$ is obtained as multiple compositions $\circ_i$,

$$\gamma(X; Y_1, \dots, Y_n) = (\cdots (X \circ_n Y_n) \circ_{n-1} Y_{n-1}) \cdots \circ_1 Y_1).$$

An algebra A over the operad $\mathfrak{D}$ is a set with the property that the operations of $\mathfrak{D}$ can take inputs in A and return outputs in A , namely there is an action

$$(4.13) \quad \gamma_A : \mathfrak{D}(n) \times A^n \rightarrow A$$

satisfying compatibility with composition (4.12) in the operad,

$$(4.14) \quad \begin{aligned} \gamma_A(\gamma(T; T_1, \dots, T_n); a_{1,1}, \dots, a_{1,k_1}, \dots, a_{n,1}, \dots, a_{n,k_n}) = \\ \gamma_A(T; \gamma_A(T_1; a_{1,1}, \dots, a_{1,k_1}), \dots, \gamma_A(T_n; a_{n,1}, \dots, a_{n,k_n})). \end{aligned}$$

A colored operad is a collection of sets

$$\mathfrak{D} = \{\mathfrak{D}(c, \underline{c}) \mid \underline{c} = (c_1, \dots, c_n), \ n \geq 1, \ c, c_i \in \Omega\}$$

with Ω a (usually finite) set of colors. The composition operations are then of the form

$$(4.15) \quad \begin{aligned} \gamma : \mathfrak{D}(c, (c_1, \dots, c_n)) \times \mathfrak{D}(c_1, (c_{1,1}, \dots, c_{1,k_1})) \times \cdots \times \mathfrak{D}(c_n, (c_{n,1}, \dots, c_{n,k_n})) \\ \rightarrow \mathfrak{D}(c, (c_{1,1}, \dots, c_{1,k_1}, \dots, c_{n,1}, \dots, c_{n,k_n})) \end{aligned}$$

subject to the same associativity property, and similarly for the $\circ_i$ compositions. An algebra over a colored operad is a collection of sets $A = \{A_c\}_{c \in \Omega}$ with an action

$$\gamma_A : \mathfrak{D}(c, (c_1, \dots, c_n)) \times A_{c_1} \times \cdots \times A_{c_n} \rightarrow A_c$$

satisfying again compatibility with the composition γ as in the non-colored case.

In [15] colored operads and algebras over colored operads are constructed in terms of *bud generating systems*. This is the formalism we used in [20] to formalize theta theory within the mathematical formulation of Merge and SMT. We recall the setting here briefly.

Given an ordinary operad $\mathfrak{D}$ and a finite set Ω , the bud-operad $\mathbb{B}_\Omega(\mathfrak{D})$ is the colored operad with Ω as the set of colors and with all the possible color assignments

$$(4.16) \quad \mathbb{B}_\Omega(\mathfrak{D})(n) := \Omega \times \mathfrak{D}(n) \times \Omega^n.$$

The local coloring rules, namely the set of generators of the colored operad and algebra to be constructed are chosen among the operations in $\mathbb{B}_\Omega(\mathfrak{D})(n)$.

A bud generating system $\mathbb{B} = (\mathfrak{D}, \Omega, \mathcal{R}, \mathcal{I}, \mathcal{T})$ consists of a (non-colored) operad $\mathfrak{D}$, a finite set of colors Ω , containing a set $\mathcal{I}$ of initial colors and a set $\mathcal{T}$ of terminal colors, and a choice of a finite subset $\mathcal{R} \subset \mathbb{B}_\Omega(\mathfrak{D})$ of operations in the bud-operad (4.16), which gives the set of generators (local coloring rules).

The colored operad $\mathfrak{D}_{\Omega, \mathbb{B}}$ generated by the bud system $\mathbb{B} = (\mathfrak{D}, \Omega, \mathcal{R}, \mathcal{I}, \mathcal{T})$ is

$$(4.17) \quad \mathfrak{D}_{\Omega, \mathbb{B}}(n) := \{x = (c, T, \underline{c}) \in \mathbb{B}_\Omega(\mathfrak{D})(n) \mid \mathbf{1}_c \rightarrow_{\mathbb{B}} x, \ c \in \Omega \setminus \mathcal{T}, \ \underline{c} \in (\Omega \setminus \mathcal{T})^n\},$$

where we write $\mathbf{1}_c \rightarrow_{\mathbb{B}} x$ to mean that x is obtained, starting from $\mathbf{1}_c$ via a sequence of operad compositions $\circ_{i_k} r_k$ of generators $r_1, \dots, r_N \in \mathcal{R}$.

The language $\mathcal{L}(\mathbb{B})$ generated by the bud system $\mathbb{B} = (\mathfrak{D}, \Omega, \mathcal{R}, \mathcal{I}, \mathcal{T})$ is an algebra over the colored operad $\mathfrak{D}_{\Omega, \mathbb{B}}$ given by

$$(4.18) \quad \mathcal{L}(\mathbb{B}) = \{x = (c, T, \underline{c}) \in \mathbb{B}_\Omega(\mathfrak{D}) \mid \mathbf{1}_c \rightarrow_{\mathbb{B}} x, \ c \in \mathcal{I}, \ \underline{c} \in \mathcal{T}^n\}.$$

4.3. The bare colored operad of narrow complemented heads. As a first step, we construct a colored operad that accounts for the specifier, head, and complement structure in the cases of syntactic objects represented in the simple form, such as in the example of (4.1) where all the leaves have lexical items associated to them.

In this case, we take the set of colors to be of the form

$$\begin{aligned} \Omega_b &= \Omega_{b,f} \sqcup \Omega_{b,\text{lex}} \\ \Omega_{b,f} &= \{(\mathfrak{h}, \mathfrak{z}^\uparrow, \mathfrak{s}^\uparrow), (\mathfrak{h}, \mathfrak{s}^\uparrow), \mathfrak{z}^\downarrow, \mathfrak{s}^\downarrow\} \\ \Omega_{b,\text{lex}} &= \mathcal{SO}_0 \times \Omega_{b,f}. \end{aligned}$$

We take $\mathcal{T}_b = \Omega_{b,\text{lex}}$ and $\mathcal{I}_b = \Omega_{b,f}$. As the local coloring rules we choose the three elements $T_{\mathfrak{h},\mathfrak{s}}^\mathfrak{z}$, $T_{\mathfrak{h},\mathfrak{s}}^\mathfrak{s}$, and $T_{\mathfrak{h},\mathfrak{z}}^\mathfrak{h}$ of (4.3),

$$(4.19) \quad \mathcal{R}_b = \mathcal{R}_{b,f} \sqcup \mathcal{R}_{b,\text{lex}} \quad \text{with} \quad \mathcal{R}_{b,f} = \{T_{\mathfrak{h},\mathfrak{s}}^\mathfrak{z}, T_{\mathfrak{h},\mathfrak{s}}^\mathfrak{s}, T_{\mathfrak{h},\mathfrak{z}}^\mathfrak{h}\},$$

and $\mathcal{R}_{b,\text{lex}}$ consisting of the same generators, where at least one of the leaves marked by elements $(\mathfrak{h}, \mathfrak{z}^\uparrow, \mathfrak{s}^\uparrow)$, $\mathfrak{z}^\downarrow$, $\mathfrak{s}^\downarrow$ of $\Omega_{b,f}$ is replaced by a leaf with a corresponding marking $(\alpha_1, \mathfrak{h}, \mathfrak{z}^\uparrow, \mathfrak{s}^\uparrow)$, $(\alpha_2, \mathfrak{z}^\downarrow)$, $(\alpha_3, \mathfrak{s}^\downarrow)$ in $\Omega_{b,\text{lex}}$, where the α_i are lexical items.

Definition 4.3. *The bare colored operad of narrow complemented heads is the operad $\mathfrak{D}_{\Omega_b, \mathbb{B}_b}$, obtained as in (4.17), generated by the bud generating system $\mathbb{B}_b = (\mathcal{M}_h, \Omega_b, \mathcal{R}_b, \mathcal{I}_b, \mathcal{T}_b)$ described above, with the Merge operad $\mathcal{M}_h$ with head, as described in Section 2.2 of [20] and in Section 3.8 of [19]. The language of bare complemented heads is the language $\mathcal{L}(\mathbb{B}_b)$ obtained as in (4.18).*

4.4. The bare colored operad of extended complemented heads.

Definition 4.4. *Let $\Omega_{\mathfrak{h}}$ denote the set of functional categories ($\mathcal{C}$, INFL, $\mathbf{v}$, $\mathbf{v}^*$, etc.) and lexical categories ($\mathbf{V}$, $\mathbf{N}$, etc.) of heads, with these two subsets respectively denoted by $\Omega_{\mathfrak{h},f}$ and $\Omega_{\mathfrak{h},\text{lex}}$,*

$$\Omega_{\mathfrak{h}} = \Omega_{\mathfrak{h},f} \sqcup \Omega_{\mathfrak{h},\text{lex}}.$$

Now we consider the full structure including both the functional and the lexical heads $\mathfrak{h}^\omega$ with $\omega \in \Omega_{\mathfrak{h}}$ and the extended projection. Consider colors of the form

$$\begin{aligned} \Omega_{b,\text{ext}} &= \Omega_{b,\text{ext},f} \sqcup \Omega_{b,\text{ext},\text{lex}} \\ \Omega_{b,\text{ext},f} &= \tilde{\Omega}|_{\omega \neq \text{lex}} \quad \text{and} \quad \Omega_{b,\text{ext},\text{lex}} = \mathcal{SO}_0 \times \tilde{\Omega}|_{\omega = \text{lex}}, \end{aligned}$$

where

$$(4.20) \quad \tilde{\Omega} := \{(\mathfrak{h}^\omega, \mathfrak{z}^\uparrow, \mathfrak{s}^\uparrow), (\mathfrak{h}^\omega, \mathfrak{z}^\uparrow), (\mathfrak{h}^\omega, \mathfrak{s}^\uparrow), \mathfrak{z}^\downarrow, \mathfrak{s}^\downarrow, (\mathfrak{s}^\uparrow, \mathfrak{z}^\downarrow) \mid \omega \in \Omega_{\mathfrak{h}}\},$$

with $\mathcal{T}_{b,ext} = \Omega_{b,ext,lex}$ and $\mathcal{I}_{b,ext} = \Omega_{b,ext} \setminus \mathcal{T}_{b,ext} = \Omega_{b,ext,f}$.

As the local coloring rules we take $\mathcal{R}_{b,ext} = \mathcal{R}_{b,ext,f} \sqcup \mathcal{R}_{b,ext,lex}$ with

$$(4.21) \quad \mathcal{R}_{b,ext,f} = \{T_{\mathfrak{h}^\omega \mathfrak{z}^\omega \mathfrak{z}^\omega}^c, T_{\mathfrak{h}^\omega \mathfrak{z}^\omega \mathfrak{s}^\omega \mathfrak{z}^\omega}^{\mathfrak{h}^\omega \mathfrak{s}^\omega}, T_{\mathfrak{h}^\omega \mathfrak{z}^\omega \mathfrak{s}^\omega \mathfrak{z}^\omega}^{\mathfrak{s}^\omega \mathfrak{z}^\omega}, T_{\mathfrak{h}^{\omega'} \mathfrak{z}^{\omega'} \mathfrak{s}^\omega \mathfrak{z}^\omega}^{\mathfrak{h}^{\omega'} \mathfrak{s}^\omega}, T_{\mathfrak{h}^\omega \mathfrak{s}^\omega \mathfrak{s}^\omega}^c, T_{\mathfrak{h}^{\omega'} \mathfrak{s}^\omega \mathfrak{s}^\omega}^c\},$$

with the generators

$$(4.22) \quad T_{\mathfrak{h}^\omega \mathfrak{z}^\omega \mathfrak{z}^\omega}^c := \begin{array}{c} \mathfrak{c} \\ \swarrow \quad \downarrow \\ (\mathfrak{h}^\omega, \mathfrak{z}^\uparrow) \quad \mathfrak{z}^\downarrow \end{array} \quad T_{\mathfrak{h}^\omega \mathfrak{z}^\omega \mathfrak{s}^\omega \mathfrak{z}^\omega}^{\mathfrak{h}^\omega \mathfrak{s}^\omega} = \begin{array}{c} (\mathfrak{h}^\omega, \mathfrak{s}^\uparrow) \\ \swarrow \quad \downarrow \\ (\mathfrak{h}^\omega, \mathfrak{z}^\uparrow, \mathfrak{s}^\uparrow) \quad \mathfrak{z}^\downarrow \end{array}$$

$$(4.23) \quad T_{\mathfrak{h}^\omega \mathfrak{z}^\omega \mathfrak{s}^\omega \mathfrak{z}^\omega}^{\mathfrak{s}^\omega \mathfrak{z}^\omega} := \begin{array}{c} (\mathfrak{s}^\uparrow, \mathfrak{z}^\downarrow) \\ \swarrow \quad \downarrow \\ (\mathfrak{h}^\omega, \mathfrak{z}^\uparrow, \mathfrak{s}^\uparrow) \quad \mathfrak{z}^\downarrow \end{array} \quad T_{\mathfrak{h}^{\omega'} \mathfrak{z}^{\omega'} \mathfrak{s}^\omega \mathfrak{z}^\omega}^{\mathfrak{h}^{\omega'} \mathfrak{s}^\omega} := \begin{array}{c} (\mathfrak{h}^{\omega'}, \mathfrak{s}^\uparrow) \\ \swarrow \quad \downarrow \\ (\mathfrak{h}^{\omega'}, \mathfrak{z}^\uparrow) \quad (\mathfrak{s}^\uparrow, \mathfrak{z}^\downarrow) \end{array} \quad \text{with } \omega' \neq \omega$$

$$(4.24) \quad T_{\mathfrak{h}^\omega \mathfrak{s}^\omega \mathfrak{s}^\omega}^c := \begin{array}{c} \mathfrak{c} \\ \swarrow \quad \downarrow \\ (\mathfrak{h}^\omega, \mathfrak{s}^\uparrow) \quad \mathfrak{s}^\downarrow \end{array} \quad T_{\mathfrak{h}^{\omega'} \mathfrak{s}^\omega \mathfrak{s}^\omega}^c := \begin{array}{c} \mathfrak{c} \\ \swarrow \quad \downarrow \\ (\mathfrak{h}^{\omega'}, \mathfrak{s}^\uparrow) \quad \mathfrak{s}^\downarrow \end{array} \quad \text{with } \omega' \neq \omega$$

and with $\mathcal{R}_{b,ext,lex}$ containing the same generators with $\omega = \text{lex}$ and with the label of at least one of the two leaves in $\Omega_{b,ext,lex}$, namely with one or both leaf-labels $(\mathfrak{h}^{\text{lex}}, \mathfrak{z}^\uparrow, \mathfrak{s}^\uparrow)$, $(\mathfrak{h}^{\text{lex}}, \mathfrak{z}^\uparrow)$, $(\mathfrak{h}^{\text{lex}}, \mathfrak{s}^\uparrow)$, $\mathfrak{s}^\downarrow$, $\mathfrak{z}^\downarrow$ replaced by a corresponding $(\alpha_1, \mathfrak{h}^{\text{lex}}, \mathfrak{z}^\uparrow, \mathfrak{s}^\uparrow)$, $(\alpha_2, \mathfrak{h}^{\text{lex}}, \mathfrak{z}^\uparrow)$, $(\alpha_3, \mathfrak{h}^{\text{lex}}, \mathfrak{s}^\uparrow)$, $(\alpha_4, \mathfrak{s}^\downarrow)$, $(\alpha_5, \mathfrak{z}^\downarrow)$ where the α_i are lexical items.

Definition 4.5. *The bare colored operad of complemented heads with extended projection is the operad $\mathfrak{O}_{\Omega_{b,ext}, \mathbb{B}_{b,ext}}$ generated by the bud generating system $\mathbb{B}_{b,ext} = (\mathcal{M}_h, \Omega_{b,ext}, \mathcal{R}_{b,ext}, \mathcal{I}_{b,ext}, \mathcal{T}_{b,ext})$ described above. The language of bare complemented heads with extended projection is the language $\mathcal{L}(\mathbb{B}_{b,ext})$ obtained as in (4.18).*

4.5. Cartographic Models. In the choice of generators of the form (4.9) and (4.10),

$$T_{\mathfrak{h}^\omega \mathfrak{z}^\omega \mathfrak{z}^\omega}^{\mathfrak{s}^\omega \mathfrak{z}^\omega} := \begin{array}{c} (\mathfrak{s}^\uparrow, \mathfrak{z}^\downarrow) \\ \swarrow \quad \downarrow \\ (\mathfrak{h}^\omega, \mathfrak{z}^\uparrow, \mathfrak{s}^\uparrow) \quad \mathfrak{z}^\downarrow \end{array} \quad \text{and} \quad T_{\mathfrak{h}^{\omega'} \mathfrak{z}^{\omega'} \mathfrak{s}^\omega \mathfrak{z}^\omega}^{\mathfrak{h}^{\omega'} \mathfrak{s}^\omega} := \begin{array}{c} (\mathfrak{h}^{\omega'}, \mathfrak{s}^\uparrow) \\ \swarrow \quad \downarrow \\ (\mathfrak{h}^{\omega'}, \mathfrak{z}^\uparrow) \quad (\mathfrak{s}^\uparrow, \mathfrak{z}^\downarrow) \end{array},$$

the pairs (ω, ω') occurring in the color label $(\mathfrak{s}^\uparrow, \mathfrak{z}^\downarrow)$ specify the possible hierarchies of heads in well formed syntactic objects and in particular the hierarchy of functional heads in the Extended Projection.

So far we have left the set of pairs (ω, ω') unconstrained. We will discuss in the following sections empirical constraints that one can impose, that restrict the set of generators (4.9) and (4.10), that limit these possibilities according to the usual structure of the Extended Projection in verbal phrases.

One can, more generally, allow for possible models of the constraints on the pairs (ω, ω') that can occur in the generators (4.9) and (4.10). Linguistically these are the cartographic models, in the sense of [12].

Definition 4.6. Let Ω_h denote the set of head categories as in Definition 4.4, with the subset $\Omega_{h,f}$ of functional categories ($\mathbf{C}$, $\mathbf{INFL}$, $\mathbf{v}$, $\mathbf{v}^*$, etc.) and the subset $\Omega_{h,lex}$ of lexical categories ($\mathbf{V}$, $\mathbf{N}$, etc.) of heads. A cartographic model is a choice of a subset $\mathcal{K} \subset \Omega_h \times \Omega_h$ with the properties that the relation $\omega' \succeq_{\mathcal{K}} \omega$ iff $(\omega, \omega') \in \mathcal{K}$ is a partial order. Generators of the form (4.9) and (4.10) where the coloring $(\mathfrak{s}_{\omega}^{\uparrow}, \mathfrak{z}_{\omega'}^{\downarrow})$ satisfy $\omega' \succeq_{\mathcal{K}} \omega$ are a $\mathcal{K}$ -cartographic coloring.

Definition 4.7. Given a cartographic model $\mathcal{K}$ as above, we can restrict the set of generators $\mathcal{R}_{b,ext}$ in (4.21) to a subset $\mathcal{R}_{\mathcal{K},b,ext} \subset \mathcal{R}_{b,ext}$ with the constraint that the generators of the form (4.9) and (4.10) belong to $\mathcal{R}_{\mathcal{K},b,ext}$ iff $\omega' \succeq_{\mathcal{K}} \omega$, namely they give a $\mathcal{K}$ -cartographic coloring.

Definition 4.8. Given a cartographic model $\mathcal{K}$, in the sense of Definition 4.6, the $\mathcal{K}$ -cartographic bare colored operad of complemented heads with extended projection is the operad $\mathfrak{O}_{\Omega_{b,ext}, \mathbb{B}_{\mathcal{K},b,ext}}$ generated by the bud generating system $\mathbb{B}_{\mathcal{K},b,ext} = (\mathcal{M}_h, \Omega_{b,ext}, \mathcal{R}_{\mathcal{K},b,ext}, \mathcal{I}_{b,ext}, \mathcal{T}_{b,ext})$.

In the following we will make some specific assumptions about the cartographic model $\mathcal{K}$ that reflect the structure of the Extended Projection of VPs. In general, one can consider the operad $\mathfrak{O}_{\Omega_{b,ext}, \mathbb{B}_{b,ext}}$ as in Definition 4.5, without any choice of a cartographic model, and implement specific cartographic models in a successive filter at the interfaces (syntax-semantics and Externalization), or incorporate some cartographic assumptions directly in the construction of the colored operad of phases.

4.6. Extended Projection Principle. The Extended Projection Principle (see [2], [4], [5]) states as a linguistic hypothesis that clauses must necessarily contain a noun phrase or determiner phrase in the “subject position”, namely in the specifier position of an inflectional phrase $\mathbf{INFL}$, or of a verb phrase, when subject does not raise to $\mathbf{TP/IP}$, as in NullSubject languages, where one can have $\{\mathbf{INFL}, \{\mathbf{EA}, \mathbf{v}^*\mathbf{P}\}\}$. Note that, in the setting of [6], $\mathbf{T}$ is not in $\mathbf{INFL}$ but is part of light verb v head: this accounts for cases like “*John arrives every day at noon and met Bill yesterday*”, where each conjunct has its own, different, tense, but there is just a single $\mathbf{INFL}$.

More precisely, one can split this into two properties:

- (1) the functional heads occur as extended projection of the lexical heads (above the part of the structure involving the lexical head and its maximal projection, as in the $\mathbf{CP-INFLP-v}^*\mathbf{P/vP}$ extended projection of $\mathbf{VP}$);
- (2) the specifier position of the $\mathbf{INFL}$ head (or possibly $\mathbf{v}^*$ head, as in NullSubject languages) is occupied by a noun phrase or determiner phrase.

For a discussion of movement by Internal Merge and the structure of the extended projection, see §6 and §8 below.

Lemma 4.9. The language $\mathcal{L}(\mathbb{B}_b)$ of narrow bare complemented heads satisfies a weaker form of Extended Projection Principle (EPP). Namely every phrase (meant as the maximal projection of a head) must contain either a lexical item or a phrase in the specifier position.

Proof. Given the form (4.3) of the generators $T_{h,s}^{\mathfrak{z}}$, $T_{h,s}^{\mathfrak{s}}$, and $T_{h,3}^h$ in $\mathcal{R}_b$, every head ℓ is marked by a label $(\alpha, h, \mathfrak{z}^{\uparrow}, \mathfrak{s}^{\uparrow})$ and has to occur in combination with a complement, in a generator of type $T_{h,3}^h$. The output of this generator is labelled $(h, \mathfrak{s}^{\uparrow})$ and must therefore combine with an input with the same marking, hence with the $(h, \mathfrak{s}^{\uparrow})$ -marked leaf of the generator of the form $T_{h,s}^c$, where c is either $\mathfrak{z}^{\downarrow}$ or $\mathfrak{s}^{\downarrow}$. This generator has the other leaf marked $\mathfrak{s}^{\downarrow}$, which means exactly that the phrase (terminating at the v_{ℓ} vertex) must contain either a lexical item or a phrase at the determiner position. $\square$

In the bare colored operad of complemented heads with extended projection and the associated language $\mathcal{L}(\mathbb{B}_{b,ext})$ it is possible to make the Extended Projection Principle more precise. We write

here the generators in the form as in (4.9) and (4.10), keeping tracks of the different functional heads.

Definition 4.10. Consider the subset $\mathcal{R}_{b,ext,EPP} \subset \mathcal{R}_{b,ext}$ consisting of the same generators in as in (4.4), (4.5), (4.6), (4.7), (4.8) in $\mathcal{R}_{b,ext,f}$ and corresponding generators in $\mathcal{R}_{b,ext,lex}$, but with the following restrictions on the generators:

- (1) the generator $T_{\mathfrak{h}\omega', \mathfrak{s}_\omega, \mathfrak{s}_\omega}^c$ only occurs (in both $\mathcal{R}_{b,ext,f}$ and $\mathcal{R}_{b,ext,lex}$) when $\omega' \in \{\mathbf{v}^*, \text{INFL}, \mathbf{C}\}$ (verb phrase, inflectional phrase, complementizer phrase, respectively), for either $\omega = \omega'$ or $\omega \neq \omega'$
- (2) the generators in both sets $\mathcal{R}_{b,ext,f}$ and $\mathcal{R}_{b,ext,lex}$ with output marked by $\mathfrak{s}_{\omega'}^\downarrow$ carry a head label $\mathfrak{h}^\omega$ with ω marking a determiner or noun phrase;
- (3) Let $\Omega_{\mathfrak{h}}$ be the set of heads. The generators $T_{\mathfrak{h}\omega', \mathfrak{z}_\omega, \mathfrak{z}_\omega}^{\mathfrak{s}_\omega \mathfrak{z}_{\omega'}}, T_{\mathfrak{h}\omega', \mathfrak{z}_{\omega'}, \mathfrak{s}_\omega \mathfrak{z}_{\omega'}}^{\mathfrak{h}\omega'}$ belong to $\mathcal{R}_{b,ext,EPP}$ iff the pair (ω, ω') in the label $(\mathfrak{s}_\omega^\uparrow, \mathfrak{z}_{\omega'}^\downarrow)$ has $\omega' \neq \text{lex} \in \Omega_{\mathfrak{h}}$ (functional heads occur above lexical head in the extended projection). When both $\omega, \omega' \neq \text{lex}$, the pair (ω, ω') follows a partial order structure among the functional head that determines the order in which they occur in the extended projection (e.g., $\mathbf{C} > \text{INFL} > \mathbf{v}^*$).

The partial ordering in the Extended Projection that constraints the choice of pairs (ω, ω') can be seen in terms of choices of cartographic models, as in Definition 4.6 and Definition 4.8.

Proposition 4.11. Let $\mathbb{B}_{b,ext,EPP}$ denote the bud generating system with the same colors as $\mathbb{B}_{b,ext}$ and the restricted set of generators $\mathcal{R}_{b,ext,EPP}$, as in Definition 4.10. The resulting colored operad $\mathfrak{D}_{\Omega_{b,ext}, \mathbb{B}_{b,ext,EPP}}$ and the associated language $\mathcal{L}(\mathbb{B}_{b,ext,EPP})$ satisfy the Extended Projection Principle.

Proof. The third condition constrains the composition rules in the operad $\mathfrak{D}_{\Omega_{b,ext}, \mathbb{B}_{b,ext}}$ (namely, the composition rules between generators in $\mathcal{R}_{b,ext}$) implying that the functional heads must occur above a lexical head, a necessary part of the EPP principle. The first and second constraints imply that the discharging of a $\mathfrak{s}_\omega^\downarrow$ can only occur in the Spec position of a head of type $\omega' \in \{\mathbf{v}^*, \text{INFL}, \mathbf{C}\}$. These happen at the top of the extended projection, since the generators involving a leaf labelled $\mathfrak{s}_\omega^\downarrow$ have a single $\mathfrak{c}^\downarrow$ at the output, which means that there is no further upward pointing $\mathfrak{s}^\uparrow$ from the head that requires the phase to be completed. Thus, the EPP principle is satisfied in $\mathcal{L}(\mathbb{B}_{b,ext,EPP})$ and $\mathfrak{D}_{\Omega_{b,ext}, \mathbb{B}_{b,ext,EPP}}$. $\square$

Remark 4.12. The restrictions of Definition 4.10 on the generators of $\mathcal{R}_{b,ext}$ give the minimal restriction (largest subset of $\mathcal{R}_{b,ext}$) that guarantees that EPP is satisfied. On top of this condition, one can further restrict the set of generators of $\mathcal{R}_{b,ext}$ by including other constraints, for example on the type of complement that a certain head can take. Namely given $\mathfrak{h}^\omega$ with either $\omega = \text{lex}$ or with ω a functional category, we can further distinguish the associated $\mathfrak{z}_\omega$ and $\mathfrak{s}_\omega$ into types depending on ω . For example, for lexical heads A, N, P, V the complements would be $\mathfrak{z}_A, \mathfrak{z}_N \in \{\text{CP}, \text{PP}\}$, $\mathfrak{z}_P \in \{\text{IP}, \text{DP}, \text{PP}\}$, $\mathfrak{z}_V \in \{\text{CP}, \text{DP}, \text{PP}\}$. In the case of $\omega \in \Omega_{\mathfrak{h}}$ a functional category, if $\omega = \mathbf{C}$ then $\mathfrak{z}_C = \text{IP}$, if $\omega = \mathbf{I}$ then $\mathfrak{z}_I = \text{VP}$, if $\omega = \mathbf{D}$ then $\mathfrak{z}_D = \text{NP}$, etc. These rules can be implemented as further restrictions on the choice of the generators that form $\mathcal{R}_{b,ext,EPP}$. We can consider these restrictions on the generators as being part of the choice of a cartographic model in the sense of Definition 4.6 and Definition 4.8.

Note further that some of the terminology for functional categories of heads is somewhat different in the recent literature on Minimalism, but the same selection of generators can be performed with the updated notation.

4.7. The colored operad of complemented heads with modifiers. In the colored operads constructed in §4.3 and §4.4, we only consider positions that are heads, complements, specifiers. This approach is similar to the construction done in [20] of the colored operad of bare theta structures, where only theta positions are included. One can then refine this construction by including positions that do not participate in a mandatory set of relations: in the case of head and phases, these non-mandatory relations are the *modifiers*, while in the case of theta theory they correspond to non-theta positions, as in [20].

In the setting considered here, one can include modifier positions, which modify the complement, or the head, or the specifier. We introduce an additional marking $\mathbf{m}$ for these modifiers. Again we can consider a simplified form of the syntactic objects, as in §4.3 or the more complete form as in §4.4 with the Extended Projection, and introduce possible additional modifier positions.

Definition 4.13. Consider the set of colors $\Omega = \Omega_f \sqcup \Omega_{\text{lex}}$ with $\Omega_f = \Omega_{b,f} \sqcup \{\mathbf{m}\}$ and $\Omega_{\text{lex}} = \Omega_{b,\text{lex}} \sqcup \mathcal{SO}_0 \times \{\mathbf{m}\}$, and the set of generators

$$\mathcal{R} = \mathcal{R}_f \sqcup \mathcal{R}_{\text{lex}} \quad \text{with} \quad \mathcal{R}_f = \mathcal{R}_{b,f} \sqcup \{T_{\mathbf{c},\mathbf{m}}^{\mathbf{c}}, \mathbf{c} \in \Omega\}$$

with

$$(4.25) \quad T_{\mathbf{c},\mathbf{m}}^{\mathbf{c}} = \begin{array}{c} \mathbf{c} \\ \diagup \quad \diagdown \\ \mathbf{c} \quad \mathbf{m} \end{array}$$

and with $\mathcal{R}_{\text{lex}}$ with the same generators as $\mathcal{R}_f$ but where at least one of the leaves occurs together with a lexical item, $(\alpha, \mathbf{c})$ or $(\alpha, \mathbf{m})$. The colored operad $\mathfrak{O}_{\Omega,\mathbb{B}}$ of complemented heads with modifiers is generated by the bud system $\mathbb{B} = (\mathcal{M}_h, \Omega, \mathcal{R}, \mathcal{I} = \Omega_f, \mathcal{T} = \Omega_{\text{lex}})$, with associated language $\mathcal{L}(\mathbb{B})$.

4.8. The colored operad of extended complemented heads with modifiers. As in §4.4 we passed from the bare colored operad of narrow complemented heads of §4.3 to the bare colored operad of extended complemented heads, which includes the structure of the Extended Projection, here we pass from the colored operad of complemented heads with modifiers of §4.7, to a version that accounts for the Extended Projection and for the effect of movement.

Definition 4.14. Consider the set of colors

$$\Omega_{\text{ext}} = \Omega_{b,\text{ext}} \sqcup \{\mathbf{m}\} \sqcup \mathcal{SO}_0 \times \{\mathbf{m}\} \sqcup \{(1, \mathbf{m})\},$$

with $\Omega_{\text{ext},f} = \Omega_{b,\text{ext},f} \sqcup \{\mathbf{m}\}$ and

$$\Omega_{\text{ext},\text{lex}} = \Omega_{b,\text{ext},\text{lex}} \sqcup \mathcal{SO}_0 \times \{\mathbf{m}\} \sqcup \{(1, \mathbf{m})\}$$

and the subset $\mathcal{R}_{b,\text{ext},\text{EPP}} \subset \mathcal{R}_{b,\text{ext}}$ as in Definition 4.10. Take

$$\mathcal{R}_{\text{ext}} = \mathcal{R}_{\text{ext},f} \sqcup \mathcal{R}_{\text{ext},\text{lex}} \quad \text{with} \quad \mathcal{R}_{\text{ext},f} = \mathcal{R}_{b,\text{ext},\text{EPP},f} \sqcup \{T_{\mathbf{c},\mathbf{m}}^{\mathbf{c}}, \mathbf{c} \in \Omega\}$$

with $T_{\mathbf{c},\mathbf{m}}^{\mathbf{c}}$ as in (4.25) with $\mathbf{h} = \mathbf{h}^\omega$ for some $\omega \in \Omega_{\mathbf{h}}$,

$$(4.26) \quad T_{\mathbf{c},\mathbf{m}}^{\mathbf{c}} = \begin{array}{c} \mathbf{c} \\ \diagup \quad \diagdown \\ \mathbf{c} \quad \mathbf{m} \end{array}$$

representing modifiers (including modifiers of complement, specifier, and head). We also include in the generators $T_{\mathbf{h}^\omega \mathbf{z}, \mathbf{z}}^{\mathbf{c}}$ and $T_{\mathbf{h}^\omega \mathbf{s}, \mathbf{s}}^{\mathbf{c}}$ in $\mathcal{R}_{b,\text{ext},\text{EPP},f}$ the possibility that $\mathbf{c} = \mathbf{m}$ at the root. The set $\mathcal{R}_{\text{ext},\text{lex}}$ contains generators of the same form as those in $\mathcal{R}_{\text{ext},f}$, but where at least one of the leaves labels is accompanied by a lexical item $(\alpha, \mathbf{c})$, for $\mathbf{c}$ occurring at the leaves of the generators

of $\mathcal{R}_{ext,f}$. Additionally, in $\mathcal{R}_{ext,lex}$ also contains generators that involve the color $(1, \mathbf{m})$, and are of the form

$$(4.27) \quad T_{\mathbf{c},(1,\mathbf{m})}^{\mathfrak{s}} := \begin{array}{c} \mathfrak{s}_{\omega'}^{\downarrow} \\ \swarrow \quad \searrow \\ \mathbf{c}_{\omega}^{\downarrow} \quad (1, \mathbf{m}) \end{array} \quad \text{with } \mathbf{c} \in \{\mathfrak{s}, \mathfrak{z}\}.$$

The colored operad $\mathfrak{D}_{\Omega_{ext}, \mathbb{B}_{ext}, \text{EPP}}$ of extended complemented heads with modifiers is generated by

$$\mathbb{B}_{ext, \text{EPP}} = (\mathcal{M}_h, \Omega_{ext}, \mathcal{R}_{ext}, \mathcal{I}_{ext} = \Omega_{ext,f}, \mathcal{T}_{ext} = \Omega_{ext,lex}),$$

with the associated language $\mathcal{L}(\mathbb{B}_{ext, \text{EPP}})$.

Note that we are also formally including here a new color $(1, \mathbf{m})$ where 1 is the unit of the magma of syntactic objects, namely the formal empty tree. The reason for including this additional color is analogous to the case of $(1, \theta_0)$ in [20], and provides the rules for IM movement. In particular, in (4.27) the color $\mathbf{c}_{\omega}^{\downarrow}$ will determine the position that IM moves from and $\mathfrak{s}_{\omega'}^{\downarrow}$ the position it moves to.

Remark 4.15. Here taking $\mathbf{c} \in \{\mathfrak{s}, \mathfrak{z}\}$ in the generator (4.27) implies not moving from a modifier $\mathbf{m}$ position or from a head $\mathfrak{h}^{\omega}$ position. Both of these restrictions could be lifted, by including additional generators of the form

$$(4.28) \quad T_{\mathbf{m},(1,\mathbf{m})}^{\mathfrak{s}} := \begin{array}{c} \mathfrak{s}_{\omega'}^{\downarrow} \\ \swarrow \quad \searrow \\ \mathbf{m} \quad (1, \mathbf{m}) \end{array}$$

or of the form

$$(4.29) \quad T_{\mathbf{c},(1,\mathbf{m})}^{\mathfrak{s}} := \begin{array}{c} \mathfrak{s}_{\omega'}^{\downarrow} \\ \swarrow \quad \searrow \\ \mathbf{c} \quad (1, \mathbf{m}) \end{array} \quad \text{with } \mathbf{c} \in \{(\mathfrak{h}^{\omega}, \mathfrak{z}_{\omega}^{\uparrow}, \mathfrak{s}_{\omega}^{\uparrow}), (\mathfrak{h}^{\omega}, \mathfrak{z}_{\omega}^{\uparrow}), (\mathfrak{h}^{\omega'}, \mathfrak{s}_{\omega}^{\uparrow})\}.$$

Introducing the possibilities (4.29) creates a possible problem with the labeling algorithm, as we will discuss more in detail in §8.3.2.

The formulation in terms of movement is discussed in §6 below and following sections.

The resulting colored operad $\mathfrak{D}_{\Omega_{ext}, \mathbb{B}_{ext}, \text{EPP}}$ and language $\mathcal{L}(\mathbb{B}_{ext, \text{EPP}})$ still satisfy the EPP, since the part handling the specifier-head-complement structure comes from the generators in $\mathcal{R}_{ext,f} = \mathcal{R}_{b,ext, \text{EPP}}$ that build structures satisfying EPP. The additional presence of the $\mathbf{m}$ modifier positions do not alter this fact, as they do not alter the specifier-head-complement structure.

As in the case of the bare colored operad of extended complemented heads, we can restrict the generators according to a cartographic model, as in Definition 4.6 and Definition 4.8.

4.9. Phases. The formulation of the specifier-head-complement structure (including modifiers) in terms of a coloring algorithm and a colored operad bud generating system then provides a way of describing phases in terms of this same colored operad and its associated language.

In the simplified form of the trees, which corresponds to the colored operad of narrow complemented head, the phases are simply determined by the paths γ_{ℓ} and the maximal projection vertices v_{ℓ} of γ_{ℓ} (when the path is nontrivial). We refer to this structure as *narrow phases*.

On the other hand, in the full form that determines the colored operad of extended complemented heads, where the functional heads are accounted for, and the extended projection, the structure of phases is more subtle, because not all the heads that project determine phases, and this complicates the relation between the paths γ_{ℓ} of the head function and the structure of phases,

see [4], [5], [13]. For example, a lexical V head ℓ now does not complete its phase at the v_ℓ vertex, but at the $v_{\ell'}$ of the functional head $\ell' = \mathbf{v}^*$, and while $\mathbf{C}$ and $\mathbf{v}^*$ heads determine phases, INFL , $\mathbf{v}$ heads are not phase heads. A result of the fact that not all functional heads are phase heads is that a Spec-of- INFL or Spec-of-Root position, for example, is in the interior rather than at the edge of the phase. This has consequences on movement via Internal Merge.

Definition 4.16. Let $\Omega_{\mathbf{h}}$ denote, as in Definition 4.4, the set of possible heads categories, with $\Omega_{\mathbf{h},f} \subset \Omega_{\mathbf{h}}$ the set of functional heads $\omega \notin \Omega_{\mathbf{h},\text{lex}}$. One can further identify a subset $\Omega_{\mathbf{h},\phi} \subset \Omega_{\mathbf{h},f}$ of functional heads that are phase heads.

Definition 4.17. Given a syntactic object $T \in \mathfrak{T}_{\mathcal{SO}_0}$ with a complemented head function h_T , the phases of T can be described in the following way.

- In the simple representation $T \in \mathcal{L}(\mathbb{B})$ as in Definition 4.13, a narrow phase of T is an accessible term $T_v \subset T$ with the properties:
 - (1) the leaf $\ell = h_T(v)$ is labelled by $(\alpha, \mathbf{h}, \mathbf{z}^\uparrow, \mathbf{s}^\uparrow)$ or $(\alpha, \mathbf{h}, \mathbf{s}^\uparrow)$, for some lexical $\alpha \in \mathcal{SO}_0$;
 - (2) the root $v = v_\ell$ is the maximal projection of this head and is a bud generator of the form $T_{\mathbf{h},\mathbf{s}}^c$ as in (4.3).
- In the full form with the Extended Projection, $T \in \mathcal{L}(\mathbb{B}_{\text{ext,EPP}})$ as in Definition 4.14, a phase of T is an accessible term $T_v \subset T$ with the properties:
 - (1) T_v contains an accessible term T_w with $w = v_\ell$ for $\ell = h_T(w)$ a leaf labelled by a lexical head $(\alpha, \mathbf{h}^\omega, \mathbf{z}_\omega^\uparrow, \mathbf{s}_\omega^\uparrow)$ or $(\alpha, \mathbf{h}^\omega, \mathbf{s}_\omega^\uparrow)$, for lexical $\alpha \in \mathcal{SO}_0$ and $\omega \in \Omega_{\mathbf{h},\text{lex}}$;
 - (2) all heads ℓ' in $T_v / {}^d T_w$ with $v_{\ell'}$ on the path from v to w are in $\Omega_{\mathbf{h},f}$;
 - (3) the head $\ell' = h_T(v)$ is in $\Omega_{\mathbf{h},\phi}$, as in Definition 4.16;
 - (4) the root v of T_v is a bud generator of the form $T_{\mathbf{h},\mathbf{s}}^c$ or $T_{\mathbf{h}^\omega, \mathbf{z}_\omega^\uparrow}^c$ as in (4.7) and (4.8).
- in both cases the edge of the phase (Spec-of-Phase) is the accessible term T_{v_1} with v_1 the vertex immediately below v_ℓ and marked by $\mathbf{s}$. The interior of the phase is the remaining $T_{v_2} = T / {}^d T_{v_1}$, with v_2 the sister vertex of v_1 below v_ℓ .

There is a subtlety that needs to be taken into account in the assignment of the set $\Omega_{\mathbf{h},\phi} \subset \Omega_{\mathbf{h},f}$ of functional heads that are phase heads. The case of $\omega = \text{INFL}$ is particular, in the sense that, depending on its interaction with a head $\omega' \succ \omega$, $\omega' = \mathbf{C}$, it may inherit a phase-head role or not.

This can be seen, for instance, by comparing the two sentences:

$$(4.30) \quad \text{What did you say } [C[_1 \text{what}[\text{INFL}[\text{what}[\mathbf{v}^*[\text{caused it }]]]]]]$$

$$(4.31) \quad \text{What did you say } [_1 \text{what}[\text{that } [_2 \text{what}[\text{INFL}[\text{what}[\mathbf{v}^*[\text{caused it }]]]]]]]]$$

In the first case, INFL inherits phase features from $\mathbf{C}$, which labels the position at $[_1$, and “what” can then move from the edge-of-the-phase position, while in the second case it is “that” which has to be phasal, while INFL is not: the node at $[_2$ in the Spec-of- INFL position remains just a trace at the point $[_1$ where the phase is completed and so it is invisible to the labeling algorithm. A consequence is ECP (empty category principle) violation: labeling operating at the phase level (here CP level) fails to label $[_2$ by criterial agreement $\langle \varphi, \varphi \rangle$ since its Spec is invisible. Labeling requirement thus unify EPP and ECP (that-trace filter) for non-null subject languages. This behavior of INFL with respect to phases is discussed in [5].

To account for these two different roles of the INFL functional head, we can assume that the set $\Omega_{\mathbf{h}}$ contains both an element INFL and an element INFL_φ , with the second (which pertains to the cases where INFL inherits the role of phase-head) in $\Omega_{\mathbf{h},\phi}$ and the first (INFL that is not a phase-head) in $\Omega_{\mathbf{h},f} \setminus \Omega_{\mathbf{h},\phi}$.

5. COLORINGS, HYPERMAGMAS, AND COLORED MERGE

As observed also in the case of the colored operad of theta role assignments, the language $\mathcal{L}(\mathbb{B})$ generated by the bud generating system is not a submagma of the magma $(\mathcal{SO}, \mathfrak{M})$ of syntactic objects. In other words the subset $\mathcal{L}(\mathbb{B}) \subset \mathcal{SO}$ of those syntactic objects that are well colored according to the local coloring rules $\mathcal{R}$ of $\mathbb{B}$ is not stable under the magma operation $\mathfrak{M}$.

This can be seen easily in the cases of both $\mathcal{L}(\mathbb{B})$ of Definition 4.13 and $\mathcal{L}(\mathbb{B}_{ext,EPP})$ of Definition 4.14.

Lemma 5.1. *The languages $\mathcal{L}(\mathbb{B})$ of Definition 4.13 and $\mathcal{L}(\mathbb{B}_{ext,EPP})$ of Definition 4.14 are not stable under the magma operation $\mathfrak{M}$.*

Proof. It suffices to produce examples of pairs of elements $T_1, T_2 \in \mathcal{L}(\mathbb{B})$ and $T'_1, T'_2 \in \mathcal{L}(\mathbb{B}_{ext,EPP})$ such that $\mathfrak{M}(T_1, T_2) \notin \mathcal{L}(\mathbb{B})$ and $\mathfrak{M}(T'_1, T'_2) \notin \mathcal{L}(\mathbb{B}_{ext,EPP})$. This can be seen already at the level of generators. For example, take two generators of the form $T_1 = T_{h,s}^3$ and $T_2 = T_{h,s}^s$ as in (4.3). Then $\mathfrak{M}(T_{h,s}^3, T_{h,s}^s)$ cannot be in $\mathcal{L}(\mathbb{B})$ because the root vertex has the two inputs colored by $3^\downarrow$ and $s^\downarrow$ but there is no generator in the bud system $\mathbb{B}$ that has the two inputs labelled in this way. The case of $\mathcal{L}(\mathbb{B}_{ext,EPP})$ is similar: it suffices to take, for example, $T'_1 = T_{h^\omega 3,3}^s$ and $T'_2 = T_{h^\omega s,s}^3$. Then by the same argument $\mathfrak{M}(T_{h^\omega 3,3}^s, T_{h^\omega s,s}^3)$ cannot be in $\mathcal{L}(\mathbb{B}_{ext,EPP})$ because no generator of the bud system $\mathbb{B}_{ext,EPP}$ has inputs labelled by $s^\downarrow$ and $3^\downarrow$. $\square$

This is a typical situation that we have already encountered for the case of head functions, where $\text{Dom}(h)$ is compatible with operad compositions (giving rise to the underlying operad $\mathcal{M}_h$ in all the bud generating systems considered here and in [20]), but not with the magma structure. We also encountered the same behavior with respect to theta role assignments, and here for the structure of phases. This means that, while the magma structure is the crucial mechanism of structure formation (alongside movement that requires the additional coproduct structure of the Hopf algebra of workspaces as discussed in Chapter 1 of [19]), the *filtering* that takes care of eliminating ill formed structures produced by the free symmetric Merge follows a different type of algebraic structure, described by operads and coloring, which does not preserve the magma operation.

5.1. Hypermagmas and coloring algorithms. As we discussed in §2 above, this lack of compatibility with the magma operation can be expressed in terms of a hypermagma structure, as we did for the head function in §2.

It is in fact a natural idea that compositional coloring problems (meaning coloring problems where the local coloring rules can be assembled into large structures via certain specific composition operations obeying a specified algebraic structure) may be formulated in terms of hyperstructures, where the multivalued result of the composition operation keeps track of possible choices of how to combine the local coloring moves forming the elements being composed. Well-colorings would then be described as coshort hypermagma morphisms that are sections of the forgetful projection that forgets the coloring (as $\text{Dom}(h)$ in the case of the head function in Proposition 2.7). Surprisingly, it seems that formulations of coloring algorithms in terms of hyperstructures is rare (but see for instance [16]).

For simplicity of notation, we analyze explicitly the case of the languages $\mathcal{L}(\mathbb{B})$ of Definition 4.13. The case of $\mathcal{L}(\mathbb{B}_{ext,EPP})$ of Definition 4.14 can be treated in exactly the same way. Let $\mathbb{B}_\Omega$ be the bud-operad with arbitrary color assignments in the color set Ω at the leaves and the root of trees $T \in \mathcal{M}_h$, as in (4.16). We write here (T, c_T) for an element in $\mathbb{B}_\Omega$, with c_T denoting an arbitrary assignment $c_T : V(T) \rightarrow \Omega$ of colors, with $V(T)$ the set of vertices.

Definition 5.2. The hypermagma $\mathfrak{H}_{\mathbb{B}_\Omega}$ is obtained by setting

$$\mathfrak{M}_{\mathbb{B}_\Omega}((T, c_T), (T', c_{T'})) = (\mathfrak{M}(T, T'), \{c_{\mathfrak{M}(T, T')}\})$$

where the set of all $c_{\mathfrak{M}(T, T')}$ consists of all possible functions $c_{\mathfrak{M}(T, T')} : V(\mathfrak{M}(T, T')) \rightarrow \Omega$ that agree with c_T and $c_{T'}$ on the respective subsets $V(T)$ and $V(T')$ of vertices of $V(\mathfrak{M}(T, T'))$, with an additional arbitrary assignment $c(v_{\mathfrak{M}(T, T')}) \in \Omega$ at the root vertex of $\mathfrak{M}(T, T')$.

Proposition 5.3. Given $(T, c_T), (T', c_{T'}) \in \mathcal{L}(\mathbb{B}) \subset \mathbb{B}_\Omega$, let

$$\mathcal{C}_{\mathfrak{M}(T, T')} = \{c_{\mathfrak{M}(T, T')} \mid \begin{array}{c} c(v_{\mathfrak{M}(T, T')}) \\ \swarrow \quad \searrow \\ c_T(v_T) \quad c_{T'}(v_{T'}) \end{array} \in \mathcal{R}\},$$

where $v_T, v_{T'}$ are the roots of T, T' . The inclusion $\mathcal{L}(\mathbb{B}) \subset \mathbb{B}_\Omega$ induces a weak subhypermagma structure on $\mathcal{L}(\mathbb{B})$ with

$$(5.1) \quad \mathfrak{M}_{\mathcal{L}(\mathbb{B})}((T, c_T), (T', c_{T'})) = \begin{cases} (\mathfrak{M}(T, T'), \{c_{\mathfrak{M}(T, T')} \in \mathcal{C}_{\mathfrak{M}(T, T')}\}) & \mathcal{C}_{\mathfrak{M}(T, T')} \neq \emptyset \\ \emptyset & \text{otherwise} \end{cases}$$

One can give an analogous hypermagma formulation for the language of the colored operad of theta roles assignments described in [20], following the same construction.

5.2. Colored Merge. As discussed in [20], one can think of the coloring algorithm that selects the well-colored syntactic objects $\mathcal{L}(\mathbb{B})$ and $\mathcal{L}(\mathbb{B}_{ext, EPP})$, among all the possible structures in $\mathfrak{T}_{SO_0}$ freely constructed by Merge, as a filter in I-language that eliminates the ill formed structures: the filter described here being the one that checks for the specifier-head-complement structure and phases, and the EPP condition. Another similar filter, functioning according to the same algebraic mechanism, is the one described in [20] that checks for the proper assignment of theta roles and the theta criterion.

This view as filters that select among the freely formed structures produced by the free symmetric Merge may give the impression that there should be a temporal relation between these two mechanism: *first* all structures are freely formed, *then* they are filtered. However, this is not necessarily the case. What the algebraic formalism shows is that one can *decouple* these two computational mechanisms, which makes it possible to separately analyze them in a more precise and efficient way. However, in actual language production, these operations of structure formation and filtering do not need to occur in the time-ordering suggested. In fact, it was shown in [20] that this description in terms of filtering is *equivalent* to a description in which structures are filtered while they are formed, at each stage of the Merge action on workspaces that leads to structure formation. While the decoupled description of structure formation and filtering is best for separately analyzing the algebraic structures involved and their properties, this “combined” view is preferable for algorithmic implementations, as it avoids the combinatorial explosion inherent in the free structure formation. As shown in [20], this combined view can be formulated in terms of a “colored Merge”, that accounts for theta role assignments during structure formation. We show here the analogous construction for the case of head, complement, and phases discussed above. It is important to keep in mind, though, that these two formulations are equivalent, so describing the combined process as a colored Merge does not mean that the additional structures of phases and theta roles impose constraints on the free structure formation of the free symmetric Merge, which is unchanged. It just means that the filtering by coloring can be done at each stage of structure formation. The crucial property that makes this possible is a peculiar characteristic of the colored operads of both phases and theta theory: *all their bud system generators consist of a single vertex with two leaves inputs and one output*. This is by no means typical, as usually colored operad generators can involve larger structures with multiple vertices.

Remark 5.4. The fact that in both of these two different filters (phases and theta roles) all the colored operad generators have a single vertex *implies* that a colored Merge reformulation is possible, hence that the description as filters after structure formation and the description as filters during structure formation are equivalent.

We discuss here also how the formulation in terms of colored Merge is related to the hypermagma description of §5.1.

In the formulation of the Merge action on workspaces in Chapter 1 of [19], one considers the vector space $\mathcal{V}(\mathfrak{F}_{S\mathcal{O}_0})$ spanned by the set $\mathfrak{F}_{S\mathcal{O}_0}$ of forests whose connected components are syntactic objects in $\mathfrak{T}_{S\mathcal{O}_0}$. A product operation $\sqcup$ on $\mathcal{V}(\mathfrak{F}_{S\mathcal{O}_0})$ is given by the disjoint union of forests, extended by linearity to the vector space. Two forms of coproduct are considered, both of the form

$$(5.2) \quad \Delta(T) = \sum_{\underline{v}} F_{\underline{v}} \otimes T/F_{\underline{v}} = \sum_C \pi_C(T) \otimes \rho_C(T),$$

where an admissible cut C is a cut of a number of edges in the tree with the property that no two of them are on the same path from the root to a leaf, and extended to forests by multiplicativity $\Delta(F) = \sqcup_a \Delta(T_a)$ for $F = \sqcup_a T_a$. The difference between the two versions of (5.2) is in the form of the quotient term $T/F_{\underline{v}}$, respectively indicated by $T/^c F_{\underline{v}}$ (each component T_{v_i} of $F_{\underline{v}}$ is contracted to its root vertex) and $T/^d F_{\underline{v}}$ (the components T_{v_i} are removed and the largest full binary tree determined by what is left is obtained via some edge contractions that eliminate non-branching vertices). These two forms of the coproduct have different algebraic properties as discussed in Chapter 1 of [19], and linguistically they correspond to keeping a trace of movement (as needed at the CI interface) or deleting it (as in Externalization). We will not explicitly distinguish between the two forms of the coproduct, except when addressing specific properties of each. The form of the Merge action on workspaces is then given by a Hopf algebra Markov chain (in the sense of [14]) of the form

$$(5.3) \quad \mathcal{K} = \sum_{S, S'} \mathfrak{M}_{S, S'} = \sum_{S, S'} \sqcup \circ (\mathcal{B} \otimes \text{id}) \circ \delta_{S, S'} \circ \Delta,$$

where the sum is over pairs of syntactic objects, $\delta_{S, S'}$ is the indicator function supported on the terms of the coproduct with $S \sqcup S'$ in the left-hand-side, and $\mathcal{B}$ is the grafting operator that grafts a forest to a common root. The sum is apparently an infinite sum, but it becomes a finite sum when applied to any workspace, as $\delta_{S, S'}$ is equal to one on only finitely many terms and zero otherwise.

Definition 5.5. To simplify notation, we write $\mathbb{B}_{\Phi}$ and $\mathcal{R}_{\Phi}$ for the bud generating system $\mathbb{B}_{\text{ext}, \text{EPP}}$ of Definition 4.14 and for its local coloring rules $\mathcal{R}_{\text{ext}}$, with $\mathcal{L}(\mathbb{B}_{\Phi})$ indicating the resulting language $\mathcal{L}(\mathbb{B}_{\text{ext}, \text{EPP}})$. (The subscript Φ is meant to indicate that this is the coloring filter for the structure of phases.) We define Ξ_{Φ} as the set of triples $(c, \{c', c''\})$ such that there is a generator in $\mathcal{R}_{\text{ext}}$ with output labelled by c and inputs labelled by c' and c'' ,

$$(5.4) \quad \Xi_{\Phi} = \{(c, \{c', c''\}) \mid \begin{array}{c} c \\ \swarrow \quad \searrow \\ c' \quad c'' \end{array} \in \mathcal{R}_{\Phi}\}.$$

Thus, Ξ_Φ contains the following triples

$$(5.5) \quad \begin{array}{ll} (\mathbf{c}, \{\mathbf{c}, \mathbf{m}\}), & ((\mathbf{s}_\omega^\uparrow, \mathbf{z}_\omega^\downarrow), \{(\mathbf{h}^\omega, \mathbf{z}_\omega^\uparrow, \mathbf{s}_\omega^\uparrow), \mathbf{z}_\omega^\downarrow\}), \\ ((\mathbf{h}^{\omega'}, \mathbf{s}_\omega^\uparrow), \{(\mathbf{h}^{\omega'}, \mathbf{z}_\omega^\uparrow), (\mathbf{s}_\omega^\uparrow, \mathbf{z}_\omega^\downarrow)\}), & ((\mathbf{h}^\omega, \mathbf{s}_\omega^\uparrow), \{(\mathbf{h}^\omega, \mathbf{z}_\omega^\uparrow, \mathbf{s}_\omega^\uparrow), \mathbf{z}_\omega^\downarrow\}), \\ (\mathbf{z}_\omega^\downarrow, \{(\mathbf{h}^\omega, \mathbf{z}_\omega^\uparrow), \mathbf{z}_\omega^\downarrow\}) & (\mathbf{s}_\omega^\downarrow, \{(\mathbf{h}^\omega, \mathbf{z}_\omega^\uparrow), \mathbf{z}_\omega^\downarrow\}) \\ (\mathbf{z}_\omega^{\downarrow'}, \{(\mathbf{h}^{\omega'}, \mathbf{s}_\omega^\uparrow), \mathbf{s}_\omega^\downarrow\}) & (\mathbf{s}_\omega^{\downarrow'}, \{(\mathbf{h}^{\omega'}, \mathbf{s}_\omega^\uparrow), \mathbf{s}_\omega^\downarrow\}) \\ (\mathbf{m}, \{(\mathbf{h}^\omega, \mathbf{z}_\omega^\uparrow), \mathbf{z}_\omega^\downarrow\}) & (\mathbf{m}, \{(\mathbf{h}^{\omega'}, \mathbf{s}_\omega^\uparrow), \mathbf{s}_\omega^\downarrow\}) \end{array}$$

subject to the constraints of Definition 4.10, and where, in the last cases, in the label $(\mathbf{h}^{\omega'}, \mathbf{s}_\omega^\uparrow)$ the ω and ω' may be either the same or different, with the constraint of Definition 4.10 if different. The set Ξ_Φ also contains the same triples with one or both of the inputs carrying an item $\alpha \in \mathcal{SO}_0$ with $\omega \in \Omega_{\mathbf{h}, \text{lex}}$, and also the additional triples of the form $(\mathbf{s}_\omega^\downarrow, \{\mathbf{c}^\downarrow, (1, \mathbf{m})\})$ with $\mathbf{c} \in \{\mathbf{s}, \mathbf{z}\}$ (with the possibility of additional $(\mathbf{s}_\omega^\downarrow, \{\mathbf{c}, (1, \mathbf{m})\})$ as in (4.28), (4.29) as discussed in Remark 4.15).

As in [20], we define the colored grafting operator $\mathcal{B}^c$ as the grafting that assigns a color c to the root,

$$\mathcal{B}^c(F) := \begin{array}{c} c \\ \swarrow \quad \downarrow \quad \searrow \\ T_1 \quad T_2 \quad \cdots \quad T_N \end{array}$$

which, in the case of two components $F = T \sqcup T'$, can also be written as

$$\mathcal{B}^c(F) = \begin{array}{c} c \\ \swarrow \quad \searrow \\ T \quad T' \end{array} =: \mathfrak{M}^c(T, T').$$

We obtain the following colored Merge operator exactly as in the case of theta coloring in [20].

Proposition 5.6. *Let $\mathcal{V}(\mathfrak{F}(\mathcal{L}(\mathbb{B}_\Phi)))$ be the vector space spanned by the set $\mathfrak{F}(\mathcal{L}(\mathbb{B}_\Phi))$ of forests whose components are in $\mathcal{L}(\mathbb{B}_\Phi)$. The colored Merge is given by*

$$(5.6) \quad \mathfrak{M}_{S, S'}^c = \sqcup \circ (\mathcal{B}^c \otimes \text{id}) \circ \delta_{\{c_S, c_{S'}\}}^c \delta_{S, S'} \circ \Delta,$$

where

$$(5.7) \quad \delta_{\{c_S, c_{S'}\}}^c = \begin{cases} 1 & (c, \{c_S, c_{S'}\}) \in \Xi_\Phi \\ 0 & (c, \{c_S, c_{S'}\}) \notin \Xi_\Phi \end{cases}$$

with c_S and $c_{S'}$ the colors at the root vertices of $S, S' \in \mathcal{L}(\mathbb{B}_\Phi)$. It is a linear map on $\mathcal{V}(\mathfrak{F}(\mathcal{L}(\mathbb{B}_\Phi)))$.

By construction of (5.6), the result of structure building by the operators $\{\mathfrak{M}_{S, S'}^c \mid c \in \Omega_\Phi, S, S' \in \mathcal{L}(\mathbb{B}_\Phi)\}$, starting from pairs $S, S' \in \Omega_{\Phi, \text{lex}}$ (lexical items with labels in $\Omega_{\Phi, f}$) gives syntactic objects in $\mathcal{L}(\mathbb{B}_\Phi)$ and all those can be obtained in this way, *because* all the generators in $\mathcal{R}_\Phi$ have a single vertex so they can all occur in $\mathfrak{M}_{S, S'}^c$. This ensures that all of $\mathcal{L}(\mathbb{B}_\Phi)$ can also be constructed by the colored Merge, not only by the colored operad, hence giving an equivalent reformulation as discussed above.

One can also see the relation between the colored Merge formulation and the hypermagma formulation described above, by considering the colored version of (5.3), of the form

$$(5.8) \quad \mathcal{K}_\Phi = \sum_{S, S' \in \mathcal{L}(\mathbb{B}_\Phi)} \sum_{c \in \Omega_{\Phi, f}} \mathfrak{M}_{S, S'}^c.$$

Proposition 5.7. *For a set A , with $\mathcal{P}(A) = 2^A$ its power set, and $\mathcal{V}(A)$ the vector space spanned by A , we write $\Sigma : \mathcal{P}(A) \rightarrow \mathcal{V}(A)$ with $\Sigma(B) := \sum_{x \in B} x \in \mathcal{V}(A)$. The colored Merge in the form (5.8) can be equivalently written as*

$$(5.9) \quad \mathcal{K}_\Phi = \sqcup \circ \Sigma \circ (\mathfrak{M}_{\mathbb{B}_\Phi} \otimes \text{id}) \circ \Pi_{(2)} \circ \Delta,$$

where $\mathfrak{M}_{\mathcal{L}(\mathbb{B}_\Phi)}$ is the hypermagma operation of (5.1), the projection $\Pi_{(2)}$ is the linear projection onto the subspace of $\mathcal{V}(\mathfrak{F}(\mathcal{L}(\mathbb{B}_\Phi))) \otimes \mathcal{V}(\mathfrak{F}(\mathcal{L}(\mathbb{B}_\Phi)))$ spanned by elements of the form $(T \sqcup T') \otimes F$ with two components in the left channel of the coproduct.

Proof. The equivalence of (5.8) and (5.9) follows from the fact that the set $\mathcal{C}_{\mathfrak{M}(T,T')}$ of (5.1) can be identified with

$$\mathcal{C}_{\mathfrak{M}(T,T')} = \{(\mathfrak{M}(T, T'), c) \mid (c(v_{\mathfrak{M}(T,T')}), \{c_T(v_T), c'_{T'}(v_{T'})\}) \in \Xi_\Phi\}$$

hence the summation

$$\sum_{T, T' \in \mathcal{L}(\mathbb{B}_\Phi)} \sum_{c \in \Omega_{\Phi, f}} \delta_{c_T(v_T), c'_{T'}(v_{T'})}^c \mathfrak{M}^c(T, T')$$

is the same as the summation

$$\sum_{T, T' \in \mathcal{L}(\mathbb{B}_\Phi)} \Sigma(\mathcal{C}_{\mathfrak{M}(T, T')})$$

where the sum over $T, T' \in \mathcal{L}(\mathbb{B}_\Phi)$ is a finite sum when restricted to T, T' in the accessible terms of a given forest $F \in \mathfrak{F}(\mathcal{L}(\mathbb{B}_\Phi))$. $\square$

An important part of the Merge model is the structure of Internal Merge as a composition $\mathfrak{M}_{T_v, T/T_v} \circ \mathfrak{M}_{T_v, 1}$ of two Merge operations, where in the first one applied, one formally takes $S = T_v$, an accessible term of one of the components of the workspace, and $S' = 1$ the unit of the magma of syntactic objects, the formal “empty tree”. The first operation $\mathfrak{M}_{T_v, 1}$ results in extracting the accessible term T_v and depositing it, alongside the remainder term T/T_v in the resulting workspace, as usually described in the linguistics literature, where they are then merged together by the second part $\mathfrak{M}_{T_v, T/T_v}$ of the operation.

As we discussed in the case of theta theory in [20] this particular form of Internal Merge has important consequences for the rules governing movement on colored structure: for example, in the case of theta theory, it is responsible for the dichotomy (also called duality) in semantics between Internal and External Merge. We analyze in §6 how this formulation of Internal Merge relates to rules about movement and the structure of phases.

This discussion in §6, and the following §7 addresses an issue formulated in the last section of [20] about movement in theta theory and its compatibility with phases.

6. PHASES COLORING AND MOVEMENT

In Minimalism there are strong constraints on movement by Internal Merge and the structure of phases. In the SMT formulation, these constraints are not built into the free structure building operation of Merge, hence they need to be realized at the level of the filtering that accounts for well formed phase structure. In view of the formulation in terms of colored Merge discussed above, this means that constraints on movement by Internal Merge that arise from phases have to be implemented in the formulation of Internal Merge $\mathfrak{M}_{T_v, T/T_v} \circ \mathfrak{M}_{T_v, 1}$, where both $\mathfrak{M}_{T_v, 1}$ and $\mathfrak{M}_{T_v, T/T_v}$ have to be realized by the colored Merge of Proposition 5.6.

In order to investigate how this works, we need to fit into the colored Merge of Proposition 5.6 the additional case of a Merge $\mathfrak{M}_{T_v, 1}$, where one of the objects is the empty tree, the unit of the magma. This case is not directly accounted for by (5.9), since it is not in the range of the projector $\Pi_{(2)}$. As in [19], one needs to add the case of the map $\sum_S \mathfrak{M}_{S, 1} = \sqcup \circ \Pi_{(1)} \circ \Delta$, where $\Pi_{(1)}$ consists of the projection onto the terms of the coproduct with a single component in the left channel. In the colored case, one then replaces $\sum_S \mathfrak{M}_{S, 1}$ with

$$\sum_{S \in \mathcal{L}(\mathbb{B}_\Phi)} \sum_c \delta_{c', c''}^c \mathfrak{M}_{S, 1}^c,$$

where $\delta_{c',c''}^c$ represents the local coloring rule for merging with a colored $(1, c'')$ when the coloring of S has $c_S(v_S) = c'$.

Since 1 is the unit of the magma, $\mathfrak{M}(T, 1) = T$, so merging with 1 does not alter the structure T . Thus, the unit 1 does not participate in building the specifier-head-complement structure, which means that the only reasonable coloring option is $(1, \mathfrak{m})$. This argument is the same as in the theta theory case of [20], where for the same structural reason, the unit 1 does not participate in the assignment of theta roles, hence it can only carry the coloring $(1, \theta_0)$ of a non-theta position.

There is, however, a difference in behavior between the case of the coloring that accounts for phases and the coloring for theta roles.

Remark 6.1. The basic rules that are known to govern compatibility of movement by Internal Merge with respect to phases can be summarized in the following way.

- (1) Internal Merge (IM) can move from the interior of the phase to the edge of the phase, Spec-of-Phase.
- (2) IM can also move within the interior of the phase as $\text{IM}(\text{EA}, \text{INFLP})$ and $\text{IM}(\text{XP}, \text{RootP})$, to Spec-of-INFL and Spec-of-Root positions (simple clauses, raising, exceptional case-marking (ECM))
- (3) IM does not move from the interior of lower phases once the phases are completed (for example, movement to Spec-of-Root cannot be further accessed from higher phases because it is not in Spec-of-Phase position).

These restrictions on the action of Internal Merge have to be implemented via the coloring rules of $\mathcal{L}(\mathbb{B}_\Phi)$ and the colored Merge formulation discussed in §5.2.

The key to obtain this is the rules for the colored version of the Merge operation $\mathfrak{M}_{T,1}$ involved in Internal Merge.

Proposition 6.2. *The additional color in Ω_Φ of the form $(1, \mathfrak{m})$ and the corresponding generators in $\mathcal{R}_\Phi$ of the form (4.27) involving this color are the part of the colored operad that accounts in $\mathcal{L}(\mathbb{B}_\Phi)$ for the results of movement by Internal Merge. In order to ensure that this type of movement satisfies the rules listed above, it is necessary to restrict the pairs (ω, ω') that occur in (4.27) accordingly. Given a lexical head $\ell = (\alpha, \text{lex})$ consider the set $\Omega(\ell)$ of functional heads ℓ' that can occur above ℓ , in the extended projection of ℓ . We take $\Omega(\ell)$ to be partially ordered by the order in which they can occur. The generators (4.27) that give IM movement satisfying the conditions listed above are of the form*

$$(6.1) \quad T_{\mathfrak{c},(1,\mathfrak{m})}^{\mathfrak{s}} = \begin{array}{c} \mathfrak{s}_{\text{RootP}}^\downarrow \\ \swarrow \quad \searrow \\ \mathfrak{c}_{(\alpha,\text{lex},V)} \quad (1, \mathfrak{m}) \end{array},$$

which moves to Spec-of-RootP at v_ℓ and

$$(6.2) \quad T_{\mathfrak{c},(1,\mathfrak{m})}^{\mathfrak{s}} = \begin{array}{c} \mathfrak{s}_{\omega'}^\downarrow \\ \swarrow \quad \searrow \\ \mathfrak{c}_\omega \quad (1, \mathfrak{m}) \end{array},$$

where $\omega, \omega' \in \Omega(\ell)$ with $\omega' > \omega$ and either

- both ω and ω' are phase-heads
- $\omega' = \text{INFL}$ or $\omega' = \text{Root}$

This follows directly by seeing that repeated use of generators (6.1) and (6.2) as stated give rise to movement as described in the cases listed above.

The partial ordering on $\Omega(\ell)$ reflects ordering such as $\mathbf{C} > \text{INFL} > \mathbf{v}^*$ or $\mathbf{C} > \text{INFL} > \mathbf{v}$. This partial ordering can be seen as part of the choice of a cartographic model, in the sense of Definition 4.6 and Definition 4.8.

Remark 6.3. In particular, the Phase Impenetrability Condition (PIC) follows from the set of generators of $\mathcal{L}(\mathbb{B}_\Phi)$ if the generators (4.27) are restricted to satisfy the conditions of Proposition 6.2.

6.1. Labeling. In [4] and [5] a labeling algorithm is discussed. Section 1.15 of [19] discussed this labeling algorithm in the framework of the mathematical formulation of Merge. This algorithm assigns labeling to the internal vertices of syntactic objects $T \in \text{Dom}(h) \subset \mathfrak{T}_{\mathcal{SO}_0}$, by labeling vertices on a path γ_ℓ by ℓ . The algorithm also assigns a labeling to *some* syntactic objects $T = \mathfrak{M}(T_1, T_2)$, where $T_1, T_2 \in \text{Dom}(h)$ but T is only in the hypermagma $\mathfrak{H}_{\mathcal{SO}_0}$ and does not, a priori, have a unique assignment of a head function fully determined by those of T_1 and T_2 (exocentric). Cases of this form that can still be labelled by the labeling algorithm (hence for which a head assignment giving $T \in \text{Dom}(h)$ becomes possible) are those for which raising of either T_1 or T_2 gives a structure that can be seen to be in $\text{Dom}(h)$, namely that has a uniquely assigned head, and this in turn induces a choice of head on T itself which resolves the ambiguity in the hypermagma $\mathfrak{H}_{\mathcal{SO}_0}$.

The simplest case of this (mentioned in Section 1.15 of [19]) would be if either $\mathfrak{M}(T_1, T/^cT_1)$ or $\mathfrak{M}(T_2, T/^cT_2)$ is in $\text{Dom}(h)$, which necessarily means $h_{\mathfrak{M}(T_1, T/^cT_1)} = h_{T_2}$ (meaning that it assigns to the root vertex of $\mathfrak{M}(T_1, T/^cT_1)$ the same value that h_{T_2} assigns to the root vertex of T_2), respectively, $h_{\mathfrak{M}(T_2, T/^cT_2)} = h_{T_1}$. This is usually expressed in terms of ECP (empty category principle) but it follows here directly from the definition of head function. In this case one assigns labeling to the root vertex v of $\mathfrak{M}(T_1, T_2)$ as $h_{\mathfrak{M}(T_1, T/^cT_1)}(v)$ (respectively, $h_{\mathfrak{M}(T_2, T/^cT_2)}(v)$). See Definitions 1.15.1 and 1.15.2 of [19].

One should also allow the possibility that raising yields an object in $\text{Dom}(h)$ in a more general form, for instance with some $T' \in \text{Dom}(h)$ and the syntactic object $\tilde{T} = \mathfrak{M}(T_1, \mathfrak{M}(T', T/^cT_1)) \in \text{Dom}(h)$. This implies that $\mathfrak{M}(T', T/^cT_1)$ has a head function $h_{\mathfrak{M}(T', T/^cT_1)}$ induced by $h_{\tilde{T}}$, which agrees either with $h_{T'}$ or with $h_{\mathfrak{M}(T', T/^cT_1)} = h_{T_2}$ (in the same sense as above). This again means that the root vertex of $\mathfrak{M}(T', T/^cT_1)$, which is the same as the root vertex of T , is assigned a label, hence T gets a labeling too. The argument is the same as for the simpler case of $\mathfrak{M}(T_1, T/^cT_1)$ discussed above: the only reason for considering this more general case is to account for the restrictions on movement imposed by the coloring rules (namely movement to the edge of phase position, and the other cases discussed above).

We show here that the labeling of [4] and [5] is also directly implemented by the coloring in terms of the generators of the colored operad of phases. The main point here is that, while just knowing the head functions of T_1 and T_2 may not suffice to resolve the ambiguity of $T = \mathfrak{M}(T_1, T_2) \in \mathfrak{H}_{\mathcal{SO}_0}$ to a unique head assignment, taking into account the form of the generators of $\mathcal{R}_\Phi$, the local coloring rules applied to the structure obtained after raising resolve the ambiguity and select a unique possibility for the head structure and the coloring of $T = \mathfrak{M}(T_1, T_2)$.

Proposition 6.4. *For all syntactic objects $T \in \mathcal{L}(\mathbb{B}_\Phi)$, the decomposition into generators in $\mathcal{R}_\Phi$ uniquely determines a labeling of the non-leaf vertices of T and this labeling matches the one prescribed by the labeling algorithm of [4] and [5].*

Proof. For any $T \in \mathcal{L}(\mathbb{B}_\Phi)$, the labeling assigned by the labeling algorithm of [4] and [5] can be recovered from the coloring of the vertices by the colored operad insertions realization of T .

We make the following labeling assignment at the vertex of these generators:

$$(6.3) \quad T_{\mathfrak{h}^\omega \mathfrak{z}, \mathfrak{z}}^c := \begin{array}{c} \mathfrak{c} \\ \swarrow \quad \searrow \\ (\mathfrak{h}^\omega, \mathfrak{z}^\uparrow) \quad \mathfrak{z}^\downarrow \end{array} \mapsto \mathfrak{h}^\omega, \quad T_{\mathfrak{h}^\omega \mathfrak{s}, \mathfrak{s}}^c := \begin{array}{c} \mathfrak{c} \\ \swarrow \quad \searrow \\ (\mathfrak{h}^\omega, \mathfrak{s}^\uparrow) \quad \mathfrak{s}^\downarrow \end{array} \mapsto \mathfrak{h}^\omega,$$

$$(6.4) \quad T_{\mathfrak{h}^\omega \mathfrak{z} \mathfrak{s}, \mathfrak{z}}^{\mathfrak{s}} = \begin{array}{c} (\mathfrak{s}_\omega^\uparrow, \mathfrak{z}_{\omega'}^\downarrow) \\ \swarrow \quad \downarrow \quad \searrow \\ (\mathfrak{h}^\omega, \mathfrak{z}_\omega^\uparrow, \mathfrak{s}_\omega^\uparrow) \quad \mathfrak{z}_\omega^\downarrow \end{array} \mapsto \mathfrak{h}^\omega, \quad T_{\mathfrak{h}^{\omega'} \mathfrak{z}, \mathfrak{s}}^{\mathfrak{h}^{\omega'} \mathfrak{s}} = \begin{array}{c} (\mathfrak{h}^{\omega'}, \mathfrak{s}_\omega^\uparrow) \\ \swarrow \quad \downarrow \quad \searrow \\ (\mathfrak{h}^{\omega'}, \mathfrak{z}_{\omega'}^\uparrow) \quad (\mathfrak{s}_\omega^\uparrow, \mathfrak{z}_{\omega'}^\downarrow) \end{array} \mapsto \mathfrak{h}^{\omega'}$$

$$(6.5) \quad T_{\mathfrak{h}^\omega \mathfrak{z} \mathfrak{s}, \mathfrak{z}}^{\mathfrak{h}^\omega \mathfrak{s}} = \begin{array}{c} (\mathfrak{h}^\omega, \mathfrak{s}_\omega^\uparrow) \\ \swarrow \quad \downarrow \quad \searrow \\ (\mathfrak{h}^\omega, \mathfrak{z}_\omega^\uparrow, \mathfrak{s}_\omega^\uparrow) \quad \mathfrak{z}_\omega^\downarrow \end{array} \mapsto \mathfrak{h}^\omega, \quad T_{\mathfrak{h}, \mathfrak{m}}^{\mathfrak{h}} = \begin{array}{c} (\mathfrak{h}^\omega, \mathfrak{z}^\uparrow, \mathfrak{s}^\uparrow) \\ \swarrow \quad \downarrow \quad \searrow \\ (\mathfrak{h}^\omega, \mathfrak{z}^\uparrow, \mathfrak{s}^\uparrow) \quad \mathfrak{m} \end{array} \mapsto \mathfrak{h}^\omega,$$

We also set

$$(6.6) \quad T_{\mathfrak{h}, \mathfrak{m}}^{\mathfrak{h}} = \begin{array}{c} \mathfrak{c} \\ \swarrow \quad \searrow \\ \mathfrak{c} \quad \mathfrak{m} \end{array} \mapsto \mathfrak{h}^\omega, \quad \text{whenever the color } \mathfrak{c} = (\mathfrak{h}^\omega, \mathfrak{c}') \text{ contains a head label } \mathfrak{h}^\omega$$

This assignment of labeling to the vertex, for the generators in (6.3) and (6.4), is obvious because they are on the path γ_ℓ for the head ℓ colored by $\mathfrak{h}^\omega$, so we have to assign $\mathfrak{h}^\omega$ (or equivalently ℓ) to the vertex, and this agrees with what prescribed by the labeling algorithm of [4] and [5] and Section 1.15 of [19]. Then we need to assign a labeling to the vertex in the remaining generators

$$(6.7) \quad T_{\mathfrak{z}, \mathfrak{m}}^{\mathfrak{z}} = \begin{array}{c} \mathfrak{z}_\omega^\downarrow \\ \swarrow \quad \searrow \\ \mathfrak{z}_\omega^\downarrow \quad \mathfrak{m} \end{array}, \quad T_{\mathfrak{s}, \mathfrak{m}}^{\mathfrak{s}} = \begin{array}{c} \mathfrak{s}_\omega^\downarrow \\ \swarrow \quad \searrow \\ \mathfrak{s}_\omega^\downarrow \quad \mathfrak{m} \end{array} \quad \text{and} \quad T_{\mathfrak{s}, (1, \mathfrak{m})}^{\mathfrak{s}} := \begin{array}{c} \mathfrak{s}_{\omega'}^\downarrow \\ \swarrow \quad \searrow \\ \mathfrak{c}_\omega^\downarrow \quad (1, \mathfrak{m}) \end{array}.$$

This labeling is assigned to be the same $\mathfrak{h}^\omega$ label carried by the generator of the form (6.3) whose root output is matched in T , via the operad composition, with the non- $\mathfrak{m}$ -labelled leaf of the generator of the form (6.7). In the case of the first two generators in (6.7) this labeling assignment matched the prescription of the labeling algorithm of [4] and [5] and Section 1.15 of [19] because these vertices are on the γ_ℓ path of the head ℓ colored by $\mathfrak{h}^\omega$.

The last case of (6.7) is what also accounts for the cases of the labeling algorithm of [4] and [5] where the labeling is assigned via raising. To see that, suppose we have a syntactic object $T \in \mathfrak{T}_{\mathcal{SO}_0}$ with $T = \mathfrak{M}(T_1, T_2)$, such that $T_1, T_2 \in \text{Dom}(h)$ but for which one only has $\mathfrak{M}(T_1, T_2) \in \mathfrak{H}_{\mathcal{SO}_0}$. This is the typical situation of exocentric structures like $\{XP, YP\}$, which in terms of the coloring and generators in $\mathcal{R}_\Phi$, correspond to the top nodes of T_1 and T_2 being both of the form (6.3), where $\mathfrak{c}_\omega^\downarrow$ is either a $\mathfrak{z}^\downarrow$ or a $\mathfrak{s}^\downarrow$, but there is no generator with leaves input labels $\{\mathfrak{z}_\omega^\downarrow, \mathfrak{s}_{\omega'}^\downarrow\}$ or $\{\mathfrak{z}_\omega^\downarrow, \mathfrak{z}_{\omega'}^\downarrow\}$ or $\{\mathfrak{s}_\omega^\downarrow, \mathfrak{s}_{\omega'}^\downarrow\}$, hence $\mathfrak{M}(T_1, T_2)$ cannot be colored by $\mathcal{R}_\Phi$ local coloring moves, and is not in $\mathcal{L}(\mathbb{B}_\Phi)$.

However, let's consider a situation where $T = \mathfrak{M}(T_1, T_2)$ is not in $\text{Dom}(h)$ but a structure obtained by raising T_1 is. We assume here that both $T_1, T_2 \in \mathcal{L}(\mathbb{B}_\Theta)$, so they both have a decompositions into generators in $\mathcal{L}(\mathbb{B}_\Theta)$. The fact that $T = \mathfrak{M}(T_1, T_2)$ is not in $\text{Dom}(h)$ means that we cannot assign a good coloring to the input and outputs of the root vertex of $T = \mathfrak{M}(T_1, T_2)$.

Because of the form of the generators in $\mathcal{R}_\Phi$, the smallest structure involving raising of T_1 that can be colorable would be of the form $\mathfrak{M}(T_1, \mathfrak{M}(\alpha, T/\mathfrak{c}T_1))$ where α is a (functional) head. This

means that the root vertex of $\mathfrak{M}(T_1, \mathfrak{M}(\alpha, T/^cT_1))$ must correspond to a generator of the form

$$(6.8) \quad \begin{array}{c} \mathfrak{c} \\ \swarrow \quad \searrow \\ \mathfrak{s}_\omega^\downarrow \quad (\mathfrak{h}^{\omega'}, \mathfrak{s}_\omega^\uparrow) \end{array}$$

where $\omega' \succeq \omega$, with $\mathfrak{c}$ either a $\mathfrak{z}^\downarrow$ or an $\mathfrak{s}^\downarrow$ of an $\mathfrak{m}$. The output at the root of T_1 is labelled by some other $\tilde{\mathfrak{c}}^\downarrow$ (which is either a $\mathfrak{z}$ or a $\mathfrak{s}$) and is attached to a generator of the form

$$(6.9) \quad \begin{array}{c} \mathfrak{s}_\omega^\downarrow \\ \swarrow \quad \searrow \\ \tilde{\mathfrak{c}}^\downarrow \quad (1, \mathfrak{m}) \end{array}$$

whose root matched the $\mathfrak{s}_\omega^\downarrow$ -labelled input of the generator (6.8). We can assign to the vertex of (6.9) the same label as the vertex that appears below the $\tilde{\mathfrak{c}}^\downarrow$ -labelled leaf of (6.8). The coloring of inputs and output of the generator (6.8) in turn implies that the root vertex of $\mathfrak{M}(\alpha, T/^cT_1)$ must be colored by a generator that has output $(\mathfrak{h}^{\omega'}, \mathfrak{s}_\omega^\uparrow)$, which has to be of the form

$$(6.10) \quad \begin{array}{c} (\mathfrak{h}^{\omega'}, \mathfrak{s}_{\omega'}^\uparrow) \\ \swarrow \quad \searrow \\ (\mathfrak{h}^{\omega'}, \mathfrak{z}_{\omega'}^\uparrow, \mathfrak{s}_{\omega'}^\uparrow) \quad \mathfrak{z}_{\omega'}^\downarrow \end{array} \quad \text{or} \quad \begin{array}{c} (\mathfrak{h}^{\omega'}, \mathfrak{s}_{\omega'}^\uparrow) \\ \swarrow \quad \searrow \\ (\mathfrak{h}^{\omega'}, \mathfrak{z}_{\omega'}^\uparrow) \quad (\mathfrak{s}_{\omega'}^\uparrow, \mathfrak{z}_{\omega'}^\downarrow) \end{array}$$

with $\mathfrak{h}^{\omega'}$ the functional head α , depending on whether ω and ω' agree or not. In turn this implies that the root vertex of $T/^cT_1$ needs to be labelled by a generator of the form

$$(6.11) \quad \begin{array}{c} \mathfrak{z}_{\omega'}^\downarrow \\ \swarrow \quad \searrow \\ \mathfrak{s}_\omega^\downarrow \quad (\mathfrak{h}_{\omega'}, \mathfrak{s}_\omega^\uparrow) \end{array} \quad \text{or} \quad \begin{array}{c} (\mathfrak{s}_\omega^\uparrow, \mathfrak{z}_{\omega'}^\downarrow) \\ \swarrow \quad \searrow \\ \mathfrak{z}_\omega^\downarrow \quad (\mathfrak{h}^\omega, \mathfrak{z}_\omega^\uparrow, \mathfrak{s}_\omega^\uparrow) \end{array}$$

respectively, according to the two cases of (6.10). In the first case $\mathfrak{h}_{\omega'}$ is the head of T_2 and it labels the root vertex of T , so this would give a consistent coloring and labeling to $T = \mathfrak{M}(T_1, T_2)$ itself, hence providing a resolution of the head choice for $T \in \mathfrak{H}_{\mathcal{SC}_0}$, consistent with the heads of T_1 and T_2 , this choice means that T can in fact fit in the section $\text{Dom}(h)$ of the hypermagma. The object T is then also in $\mathcal{L}(\mathbb{B}_\Phi)$, with $\tilde{\mathfrak{c}}^\downarrow = \mathfrak{s}_\omega^\downarrow$ in (6.9). In the second case, similarly, the root vertex of T gets labelled by $\mathfrak{h}^\omega$ and this gives a consistent coloring and labeling to $T = \mathfrak{M}(T_1, T_2)$ itself which is then in $\text{Dom}(h)$ and in $\mathcal{L}(\mathbb{B}_\Phi)$, with $\tilde{\mathfrak{c}}^\downarrow = \mathfrak{z}_\omega^\downarrow$ in (6.9). This shows that, in fact, the only cases where $\mathfrak{M}(T_1, \mathfrak{M}(\alpha, T/^cT_1)) \in \mathcal{L}(\mathbb{B}_\Phi)$ occur when already $T = \mathfrak{M}(T_1, T_2) \in \mathcal{L}(\mathbb{B}_\Phi)$, since we have shown that, in such cases, assigning a head function and a good coloring to $T = \mathfrak{M}(T_1, T_2)$ can be achieved directly in terms of the head function on T_1 and T_2 , without having to take into account also the additional information on the possible colorings of $\mathfrak{M}(T_1, \mathfrak{M}(\alpha, T/^cT_1))$. $\square$

Remark 6.5. One can then make an observation about the ECP (empty category principle). The trace of movement should be invisible to labeling. On the other hand, the coloring of the generators in $\mathcal{R}_\Phi$ does assign a color $\tilde{\mathfrak{c}}^\downarrow$ to the trace of movement, which can be either a $\mathfrak{s}^\downarrow$ or a $\mathfrak{z}^\downarrow$ according to (6.11) (with the additional possibilities as in (4.28), (4.29), as in Remark 4.15). However, in all cases (except the possible additional generator (4.29)) this color associated to the position from which IM movement is performed does not participate into the determination of labeling, since in both cases in (6.11) (as well as in (4.28)) the labeling of the node is determined only by the $\mathfrak{h}$ color at the other leaf input. Thus, the fact that movement changes the coloring at the root of the accessible term being moved, from $\tilde{\mathfrak{c}}^\downarrow$ to $\mathfrak{s}_\omega^\downarrow$, by effect of the generator (6.9),

does not affect in any way the labeling algorithm. (The case of the possible generator (4.29) is discussed separately in §8.3.2.)

7. PHASES COLORING AND THETA COLORING: TRANSDUCTIONS OF COLORED OPERADS

A well formed parsable syntactic object has both a consistent system of phases and a correct assignment of theta roles. Thus, both filtering have to apply to the same structure. This means that two different colored operads need to be compared and viable structures should be those that meet both coloring requirements, according to both sets of generators simultaneously. Equivalently, in terms of the formulation via a colored Merge, the well-formed objects should have a derivation according to both types of colored Merge, for phases and for theta roles. This in principle may be interpreted in different ways: for example, it is a priori different to require that a single colored Merge simultaneously implements both filters or that there is a derivation according to the colored Merge of phases and a derivation according to the colored Merge for theta roles, but these two derivations need not be the same. In terms of colored operads, one can require that a singled colored operad with single-vertex bud generators implements both filters for phases and theta roles simultaneously, or that the two colored operads that select for these two properties are in an appropriate sense both obtainable as specializations from a single colored operad.

Ultimately, a good model of the relation between these two colored operads and languages, $\mathcal{L}(\mathbb{B}_\Phi)$ and $\mathcal{L}(\mathbb{B}_\Theta)$, for the bud systems $\mathbb{B}_\Phi$ and $\mathbb{B}_\Theta$ of phases and theta roles, should rely on empirical evidence on when phases and theta roles in Merge derivations do or do not move in coordinated ways, for example, in cases like the more complex theta role assignments of languages such as Mandarin, and their relation to the phase structure. We will present here a proposal for how to combine the filters for theta roles and for phases.

The key idea here is that we want viable structures to “pass both filters” for theta roles and for phases. This would mean that such structures have to be in the intersection $\mathcal{L}(\mathbb{B}_\Phi) \cap \mathcal{L}(\mathbb{B}_\Theta)$. However, to make sense of this intersection, both $\mathcal{L}(\mathbb{B}_\Phi)$ and $\mathcal{L}(\mathbb{B}_\Theta)$ have to be realized as subsets of a common larger set. However, we constructed $\mathcal{L}(\mathbb{B}_\Phi)$ and $\mathcal{L}(\mathbb{B}_\Theta)$ as different colored operads, each with its own bud generating system, hence there is no common ambient set where they would intersect. Thus, we need a more careful formulation of what it means that a syntactic object “passes both filters”. It turns out that the right way to express the idea that such syntactic objects are in the intersection $\mathcal{L}(\mathbb{B}_\Phi) \cap \mathcal{L}(\mathbb{B}_\Theta)$ is to “pull back” $\mathcal{L}(\mathbb{B}_\Phi)$ and $\mathcal{L}(\mathbb{B}_\Theta)$ to a correspondence and intersect them there. This idea is frequently used, and is for example the key to the notion of transducers and transduction in the theory of formal languages. We will discuss in this section how to formulate it in the case of colored operads and their bud generating systems.

It is useful to start by reviewing how one compares different colored operads, especially those obtained via bud generating systems. There is the usual notion of morphisms of colored operads.

Definition 7.1. *A morphism $\varphi : \mathfrak{D} \rightarrow \mathfrak{D}'$ of colored operads consists of a map $\varphi_c : \Omega \rightarrow \Omega'$ between the sets of colors and maps $\varphi_n : \mathfrak{D}(a, (a_1, \dots, a_n)) \rightarrow \mathfrak{D}'(\varphi_c(a), (\varphi_c(a_1), \dots, \varphi_c(a_n)))$ between the operad sets, that are compatible with the operad composition operations, through the appropriate commutative diagrams. In particular, a morphism of ordinary operads (single-colored) $\varphi : \mathcal{O} \rightarrow \mathcal{O}'$ is a collection of maps $\varphi_n : \mathfrak{D}(n) \rightarrow \mathfrak{D}'(n)$ compatible with the operad composition operations.*

On the basis of this, for example, one can expect that there may be a colored operad $\tilde{\mathfrak{D}}$ that maps to both $\mathfrak{D}_{\Omega_\Phi, \mathbb{B}_\Phi}$ and $\mathfrak{D}_{\Omega_\Theta, \mathbb{B}_\Theta}$

$$\begin{array}{ccc} & \tilde{\mathfrak{D}}_{\tilde{\Omega}_\Phi, \Theta, \tilde{\mathbb{B}}_\Phi, \Theta} & \\ \swarrow \varphi & & \searrow \varphi' \\ \mathfrak{D}_{\Omega_\Phi, \mathbb{B}_\Phi} & & \mathfrak{D}_{\Omega_\Theta, \mathbb{B}_\Theta} \end{array}$$

where the structures that simultaneously satisfy both filtering constraints for phases and theta roles would be identified with $\varphi^{-1}(\mathcal{L}(\mathbb{B}_\Phi)) \cap \varphi'^{-1}(\mathcal{L}(\mathbb{B}_\Theta))$.

A somewhat more flexible model comes from generalizing morphisms of colored operads to correspondences between colored operads with bud generating systems. These generalize the usual notion of transduction in formal languages.

Definition 7.2. Suppose given two colored operads $\mathfrak{D}_{\Omega, \mathbb{B}}$ and $\mathfrak{D}'_{\Omega', \mathbb{B}'}$ with bud generating systems $\mathbb{B} = (\mathfrak{D}, \Omega, \mathcal{R}, \mathcal{I}, \mathcal{T})$ and $\mathbb{B}' = (\mathfrak{D}', \Omega', \mathcal{R}', \mathcal{I}', \mathcal{T}')$. Let $\hat{\Omega}$ be a set with maps $\pi : \hat{\Omega} \rightarrow \Omega$ and $\pi' : \hat{\Omega} \rightarrow \Omega'$, and let $\tilde{\mathfrak{D}}$ be an ordinary operad with operad homomorphisms $\varphi : \tilde{\mathfrak{D}} \rightarrow \mathfrak{D}$ and $\varphi' : \tilde{\mathfrak{D}} \rightarrow \mathfrak{D}'$. Consider a bud generating system $\tilde{\mathbb{B}} = (\tilde{\mathfrak{D}}, \tilde{\Omega}, \tilde{\mathcal{R}}, \tilde{\mathcal{I}}, \tilde{\mathcal{T}})$, with $\tilde{\mathcal{I}} = \pi^{-1}(\mathcal{I}) \cap \pi'^{-1}(\mathcal{I}')$ and $\tilde{\mathcal{T}} = \pi^{-1}(\mathcal{T}) \cap \pi'^{-1}(\mathcal{T}')$ and $\tilde{\Omega} = \tilde{\mathcal{I}} \cup \tilde{\mathcal{T}} \subset \hat{\Omega}$, with a given finite set of generators $\tilde{\mathcal{R}} \subset \tilde{\mathbb{B}}_{\tilde{\Omega}}(\tilde{\mathfrak{D}})$, with $\tilde{\mathbb{B}}_{\tilde{\Omega}}(\tilde{\mathfrak{D}})(n) = \tilde{\Omega} \times \tilde{\mathfrak{D}}(n) \times \tilde{\Omega}^n$, as in (4.16). Let $\tilde{\mathfrak{D}}_{\tilde{\Omega}, \tilde{\mathbb{B}}} \subset \tilde{\mathbb{B}}_{\tilde{\Omega}}(\tilde{\mathfrak{D}})$ be the colored operad generated by the bud system $\tilde{\mathbb{B}}$, and $\mathcal{L}(\tilde{\mathbb{B}})$ the associated language. The maps π, π' restricted to $\tilde{\Omega}$ extend the operad morphisms $\varphi : \tilde{\mathfrak{D}} \rightarrow \mathfrak{D}$ and $\varphi' : \tilde{\mathfrak{D}} \rightarrow \mathfrak{D}'$ to morphisms of the colored operads

$$\varphi : \tilde{\mathbb{B}}_{\tilde{\Omega}}(\tilde{\mathfrak{D}}) \rightarrow \mathbb{B}_{\Omega}(\mathfrak{D}) \quad \text{and} \quad \varphi' : \tilde{\mathbb{B}}_{\tilde{\Omega}}(\tilde{\mathfrak{D}}) \rightarrow \mathbb{B}'_{\Omega'}(\mathfrak{D}').$$

A transduction of colored operads $\mathbb{T} : \mathfrak{D}_{\Omega, \mathbb{B}} \rightarrow \mathfrak{D}'_{\Omega', \mathbb{B}'}$ consists of data $(\tilde{\mathbb{B}}, \varphi, \varphi')$ such that

$$(7.1) \quad \varphi'(\varphi^{-1}(\mathfrak{D}_{\Omega, \mathbb{B}}) \cap \tilde{\mathfrak{D}}_{\tilde{\Omega}, \tilde{\mathbb{B}}}) \subset \mathfrak{D}'_{\Omega', \mathbb{B}'}.$$

This induced a transduction of languages $\mathbb{T} : \mathcal{L}(\mathbb{B}) \rightarrow \mathcal{L}(\mathbb{B}')$ if (7.1) restricts to

$$\varphi'(\varphi^{-1}(\mathcal{L}(\mathbb{B})) \cap \mathcal{L}(\tilde{\mathbb{B}})) \subset \mathcal{L}(\mathbb{B}').$$

With this more general model of relations between colored operads with bud systems, we can regard structures that simultaneously satisfy both filtering for phases and theta roles as

$$\varphi^{-1}(\mathcal{L}(\mathbb{B}_\Phi)) \cap \varphi'^{-1}(\mathcal{L}(\mathbb{B}_\Theta)) \cap \mathcal{L}(\tilde{\mathbb{B}}_{\Phi, \Theta}),$$

where $\mathcal{L}(\tilde{\mathbb{B}}_{\Phi, \Theta})$ is the language of the transduction that encodes compatibility restrictions between phases and theta roles, namely further restrictions that are not directly captured just by the intersection $\varphi^{-1}(\mathcal{L}(\mathbb{B}_\Phi)) \cap \varphi'^{-1}(\mathcal{L}(\mathbb{B}_\Theta))$.

8. THETA AND PHASE COLORING AND MOVEMENT

In both cases of Definition 7.1 and Definition 7.2, the relation between the two colored operads does constrain the structures that are accepted by both, and the kind of movement that is achievable via the colored Merge. For example, not all the cases of movement discussed in [20] that satisfies the coloring of $\mathcal{L}(\mathbb{B}_\Theta)$ would also satisfy the coloring of $\mathcal{L}(\mathbb{B}_\Phi)$, since some involved movement within the interior of the phase, to positions other than those marked by $\mathfrak{s}^\downarrow$ in the phase-coloring, for positions within the phase as in Remark 6.1.

In this section we analyze some cases that illustrate more explicitly the constraints on compatibility between the $\mathcal{L}(\mathbb{B}_\Phi)$ and $\mathcal{L}(\mathbb{B}_\Theta)$ filtering systems.

8.1. Theta roles and movement. As an example of interaction between the coloring rules for phases and for theta roles, consider the case of passive. Let's take the example of the sentence “*The fox was chased*”, for which the Merge derivation is analyzed in §6.1.2 of [11]. We follow the generators of both $\mathcal{L}(\mathbb{B}_\Theta)$ and $\mathcal{L}(\mathbb{B}_\Phi)$, or equivalently both colored Merge operations.

We can start with forming the object { chase , the fox }. We have, in $\mathcal{L}(\mathbb{B}_\Phi)$ a generator of $\mathcal{R}_\Phi$

$$\begin{array}{c} \mathfrak{z}_{\text{lex},V}^\downarrow \\ \swarrow \quad \searrow \\ (\text{the}, \mathfrak{h}^{\text{lex},D}, \mathfrak{z}_{\text{lex},D}^\uparrow) \quad (\text{fox}, \mathfrak{z}_{\text{lex},D}^\downarrow) \end{array}$$

that composes with a generator

$$(8.1) \quad \begin{array}{c} (\mathfrak{h}^{\text{lex},V}, \mathfrak{s}_{\text{lex},V}^\uparrow) \\ \swarrow \quad \searrow \\ (\text{chase}, \mathfrak{h}^{\text{lex},V}, \mathfrak{z}_{\text{lex},V}^\uparrow, \mathfrak{s}_{\text{lex},V}^\uparrow) \quad \mathfrak{z}_{\text{lex},V}^\downarrow \end{array}$$

This second generator is the interesting one to compare with the theta role assignments, as in $\mathcal{L}(\mathbb{B}_\Theta)$ it would correspond to a generator of $\mathcal{R}_\Theta$ of the form

$$(8.2) \quad \begin{array}{c} \theta_E^\uparrow \\ \swarrow \quad \searrow \\ (\text{chase}, (\theta_E^\uparrow, \theta_I^\uparrow)) \quad \theta_I^\downarrow \end{array}$$

The Merge derivation described in §6.1.2 of [11] then moves the IA “*the fox*” by IM. This produces, in the two coloring bud systems, respective generators

$$(8.3) \quad \begin{array}{c} \mathfrak{s}_{\text{lex},V}^\downarrow \\ \swarrow \quad \searrow \\ \mathfrak{z}_{\text{lex},V}^\downarrow \quad (1, \mathfrak{m}) \end{array} \quad \text{and} \quad \begin{array}{c} \theta_0 \\ \swarrow \quad \searrow \\ \theta_I^\downarrow \quad (1, \theta_0) \end{array}$$

The output of these is then composed, respectively, with one of the inputs of the generators

$$(8.4) \quad \begin{array}{c} \mathfrak{z}_v^\downarrow \\ \swarrow \quad \searrow \\ \mathfrak{s}_{\text{lex},V}^\downarrow \quad (\mathfrak{h}^{\text{lex},V}, \mathfrak{s}_{\text{lex},V}^\uparrow) \end{array} \quad \text{and} \quad \begin{array}{c} \theta_E^\uparrow \\ \swarrow \quad \searrow \\ \theta_0 \quad \theta_E^\uparrow \end{array}$$

where the other input is respectively matched to the outputs of (8.1) and (8.2). These are then matched, at the next step, with generators

$$(8.5) \quad \begin{array}{c} (\mathfrak{h}^v, \mathfrak{s}_v^\uparrow) \\ \swarrow \quad \searrow \\ (\mathfrak{h}^v, \mathfrak{z}_v^\uparrow, \mathfrak{s}_v^\uparrow) \quad \mathfrak{z}_v^\downarrow \end{array} \quad \text{and} \quad \begin{array}{c} \theta_E^\uparrow \\ \swarrow \quad \searrow \\ \theta_0 \quad \theta_E^\uparrow \end{array}$$

Note that, with this formalism, at this point the θ_E role has not been discharged yet. Note also that, while v^* is a phase-head (transitive case), v is a head that is not a phase-head, so IM here does not have to move IA to Spec positions for v , as it can move directly to the edge of the phase

that contains the v head. In the colored operad of phases, this generator is then followed by

$$(8.6) \quad \begin{array}{c} \mathfrak{z}_I^\downarrow \\ | \\ (\text{was}, \mathfrak{s}_v^\downarrow) \quad (\mathfrak{h}^v, \mathfrak{s}_v^\uparrow) \end{array} .$$

The next step is the merging in of an INFL (I) head. At the level of the colored operad of phases, we can associate to this a generator of the form

$$(8.7) \quad \begin{array}{c} (\mathfrak{h}^I, \mathfrak{s}_I^\uparrow) \\ | \\ (\mathfrak{h}^I, \mathfrak{z}_I^\uparrow, \mathfrak{s}_I^\uparrow) \quad \mathfrak{z}_I^\downarrow \end{array}$$

At the next step we can proceed by moving IA at the spec of INFL position within the phase, using generators

$$\begin{array}{c} \mathfrak{s}_I^\downarrow \\ | \\ \mathfrak{s}_{\text{lex},V}^\downarrow \quad (1, \mathfrak{m}) \end{array} \quad \text{and} \quad \begin{array}{c} \theta_0 \\ | \\ \theta_0 \quad (1, \theta_0) \end{array} .$$

In the colored operad of phases this is finally capped with a generator of the form

$$\begin{array}{c} \mathfrak{c}^\downarrow \\ | \\ \mathfrak{s}_I^\downarrow \quad (\mathfrak{h}^I, \mathfrak{s}_I^\uparrow) \end{array} .$$

However, from the side of theta theory, we need to discharge the $\theta_E^\uparrow$, and the IM movement of the IA to Spec-of-INFL position will only move to a θ_0 -marked position that cannot be the receiver. Thus, the only possibility for matching the theta roles in this case is that the $\theta_E^\downarrow$ is discharged into the $\mathfrak{s}_I^\downarrow$ position, by a generator

$$(8.8) \quad \begin{array}{c} \theta_0 \\ | \\ (\text{was}, \theta_E^\downarrow) \quad \theta_E^\uparrow \end{array}$$

which is paired to the generator (8.6). In other words that the “*was*” of the “*was chased*” acts as receiver of the θ_E assignment of “*chase*” when in the passive form. After the θ_E is discharged there are no more theta roles carried, so the remaining generators (8.6), (8.7) of the colored operad of phases are paired to the generator

$$\begin{array}{c} \theta_0 \\ | \\ \theta_0 \quad (1, \theta_0) \end{array}$$

of the colored operad of theta roles.

Note that adding an optional “*by someone*” to the phrase would not change where the θ_E can be discharged, as that would receive an $\mathfrak{m}$ marker in $\mathcal{L}(\mathbb{B}_\Phi)$ and a θ_0 marker in $\mathcal{L}(\mathbb{B}_\Theta)$, since it is not a mandatory part of the structure.

Note that this handling of the θ_E assignment also provides a reason, in terms of compatibility of coloring algorithms, for why v (unlike v^*) should not be a phase head, otherwise its specifier position would be a receiver of IM movement, hence necessarily a θ_0 -colored position and would not possibly act as discharger for the $\theta_E^\uparrow$. The specifier position for INFL is also not Spec-of-phase, since the INFL head is not a phase head, but it is one of the positions in the interior of the phase

that IM can move to, in which case they carry a θ_0 position, as illustrated here. This proposal of how to handle the external theta role assignment in passive forms, which is here forced by the coloring rules, was previously proposed in [1]. It differs from the proposal of disentangling θ_E from the injected $\theta^\dagger$ and assigning $\theta_E^\dagger$ at the $\mathfrak{h}^{v^*}$ position as in [6], [10], [8], [9]. A discussion of an interpretation of θ_E as associated to the v^* head in this colored operad is included in [20].

8.2. Exceptional case-marking. As another example, to see how to accommodate exceptional case-marking (ECM) in our formulation, consider, for example, the two sentences “*John expected her to win*” and “*John persuaded her to win*”. In the first case, one can argue that IM raises *her* to a position in the phase interior of v^* . In so doing, it moves it to a non-theta position, in ECM, θ_0 -labelled. This is not at the edge of the phase, but it has a “subject-of predication” interpretation. On the other hand if it were inserted in that same position through an EM operation, then it would have to receive a theta role assignment from “expect” but it does not: in the ECM construction the raised pronoun, although it is marked as object case morphology, it semantically received its theta role from the lower verb “win” not from “expect”. In the second sentence, instead, it is indeed the recipient of a theta role assignment from “persuade”, hence it is obtained via an EM operation. Both cases are captured by the different local coloring move involved, namely two different generators of the colored operad of theta roles. Similar examples are discussed in the context of raising vs. control in [6] and in [7].

One can see in examples like this the combined rules of movement for the Φ and the Θ colored Merge operations, with the rule for IM movement and phases marking the position with an $\mathfrak{z}_{v^*}^\downarrow$ and the IM realized by a generator

$$\begin{array}{c} \mathfrak{s}_{\omega'}^\downarrow \\ | \\ \mathfrak{s}_{\omega}^\downarrow \quad (1, \mathfrak{m}) \end{array}$$

that moves to a $\mathfrak{s}_{\omega}^\downarrow$ position located in the interior of the v^* phase, hence below a $\mathfrak{z}_{v^*}^\downarrow$ colored operad insertion. The fact that, in the Merge colored for phases, this movement is to a position inside the $\mathfrak{s}_{v^*}^\downarrow$ -marked interior of the v^* phase is consistent with the object case morphology. On the other hand, at the level of the Merge colored for theta role assignments, this movement by IM is implemented by a generator of the form

$$\begin{array}{c} \theta_0 \\ | \\ \theta_E^\dagger \quad (1, \theta_0) \end{array}$$

which marks *her* with a θ_0 position, while the $\theta_E^\dagger$ marker remains at the trace of movement, hence as a theta role assigned by *win*, not by *expect*.

8.3. Double object constructions and theta hierarchies. We now analyze again some double object constructions discussed in [20], where the coloring by $\mathbb{B}_\Theta$ for theta role assignment would support a IM movement interpretation of certain configurations of theta roles. In the analysis given in [20], we only considered the compatibility of movement with the bud generating system of the colored operad of theta theory. However, as we have been discussing in this paper, movement is also filtered by the colored operad of phases. Thus, we can analyze again the possible syntactic objects that account for double object constructions, and check whether movement is needed and whether it is compatible with the bud generating systems of both operads, the colored operad of phases $\mathbb{B}_\Phi$ and the colored operad of theta roles $\mathbb{B}_\Theta$. This will lead to favoring certain possibilities. It will also serve as another good test case for how to formulate the simultaneous effect of both filters.

8.3.1. *Double object and coloring by phases.* Consider a double object sentence like “*Mary gave a book to John*”. From the point of view of phases, we have a structure of the form

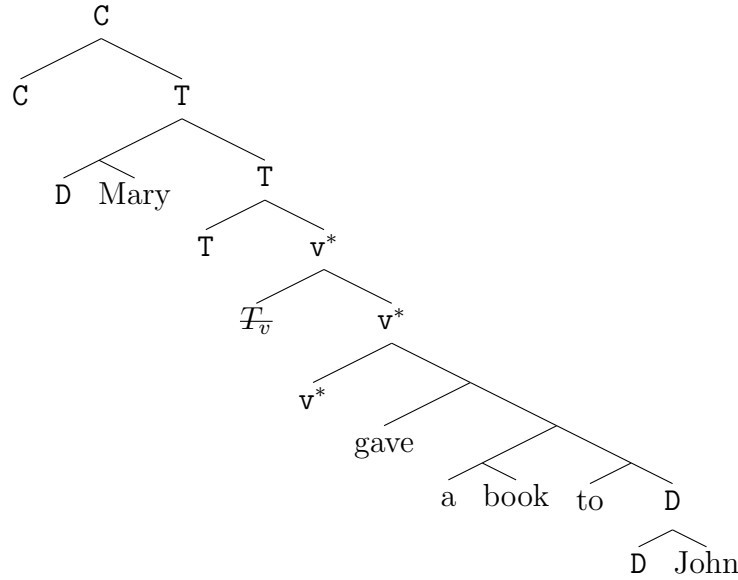
$$\{v^*\{NP, \{\text{Root}, XP\}\}\}.$$

In general, such structures can occur in two different ways: NP can be put into its position by EM (in the case of Control) or by IM (in the case of ECM discussed above). These possibilities clearly differ when it comes to the structure of theta roles. This fact shows that we do not expect the implementation of both filters, for phases and theta roles, to be realized as a matching of conditions via a morphism of colored operads in the sense of Definition 7.1, but it requires the more general form of correspondences as in Definition 7.2. We will return to discuss this in §9.

More specifically in the example of “*Mary gave a book to John*”, we can discuss all the possible candidates for syntactic objects representing this double object construction, with their full Extended Projection, and check what their coloring looks like, according to the bud generating systems for phases and for theta roles. We will see that the interaction between the coloring by phases and by theta roles selects for some of them.

(1) Consider first a candidate form of a syntactic object of the form

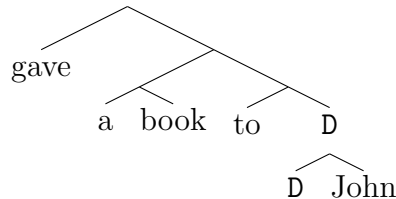
(8.9)



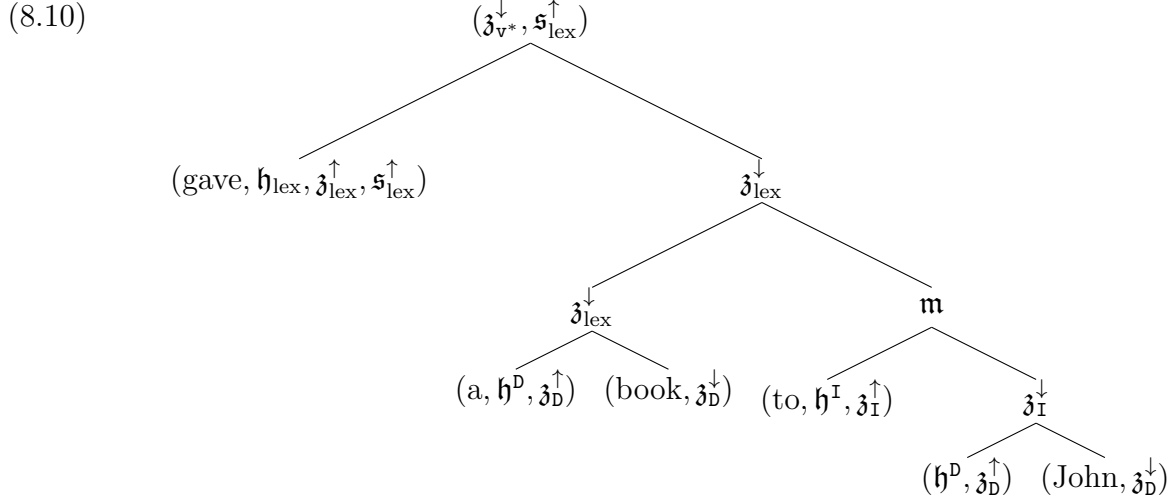
with the IM movement of the term

$$T_v = \begin{array}{c} D \\ \swarrow \searrow \\ D \quad \text{Mary} \end{array}.$$

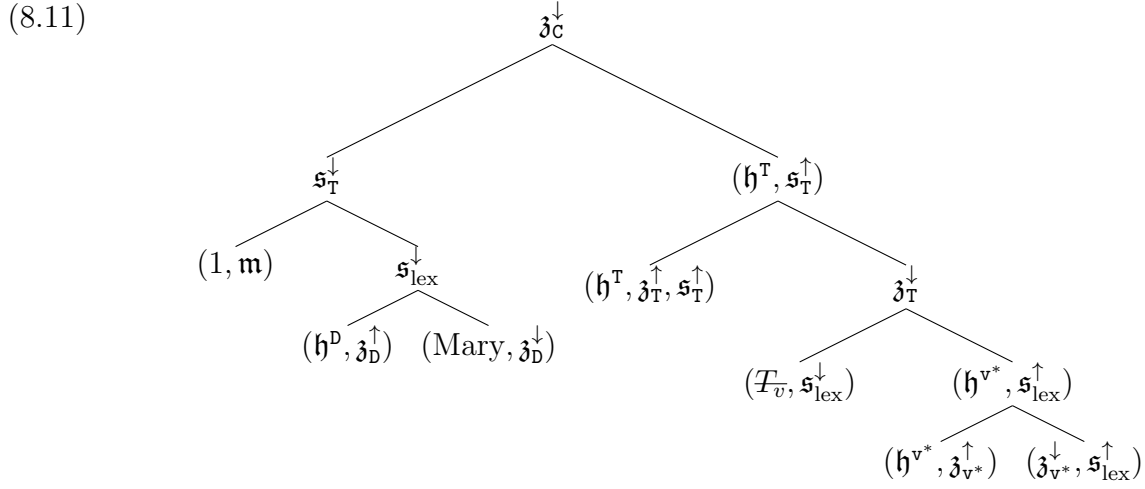
In terms of coloring according to $\mathbb{B}_\Phi$ for the structure of phases, the part



can be colored by operad compositions of generators

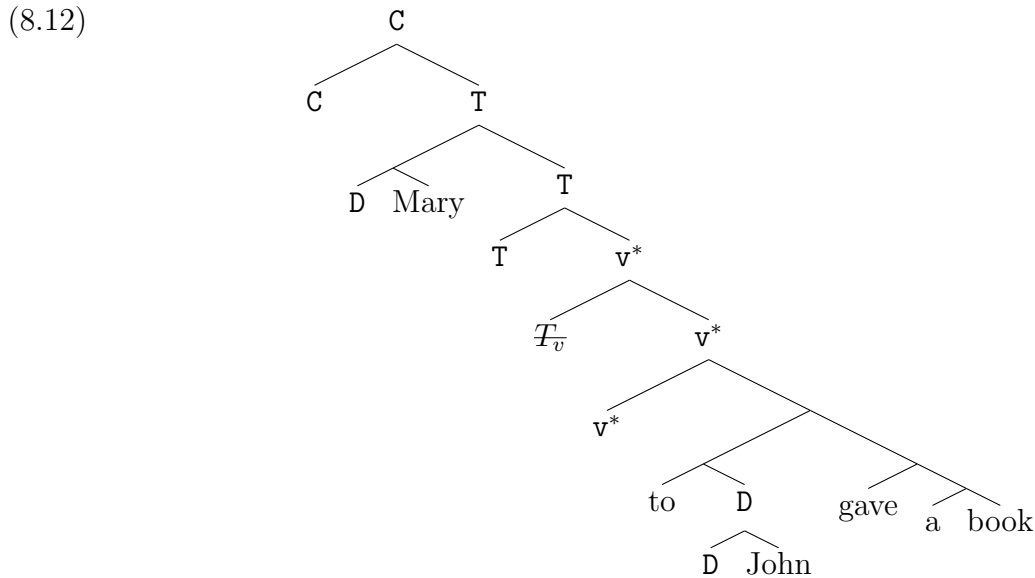


Then, in the rest of the Extended Projection we have further generators

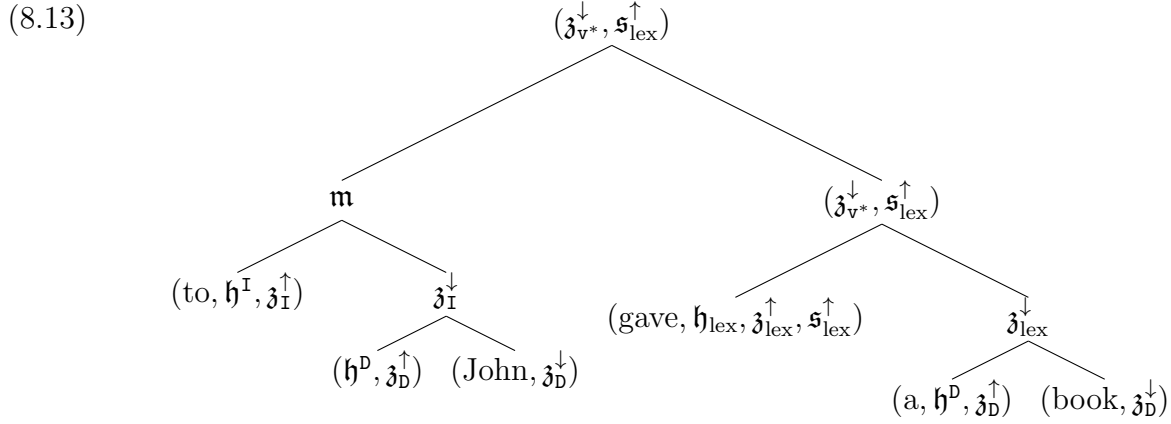


with the last leaf input of (8.11) matched with the root output of (8.10).

(2) We then consider another possible candidate syntactic object of the form

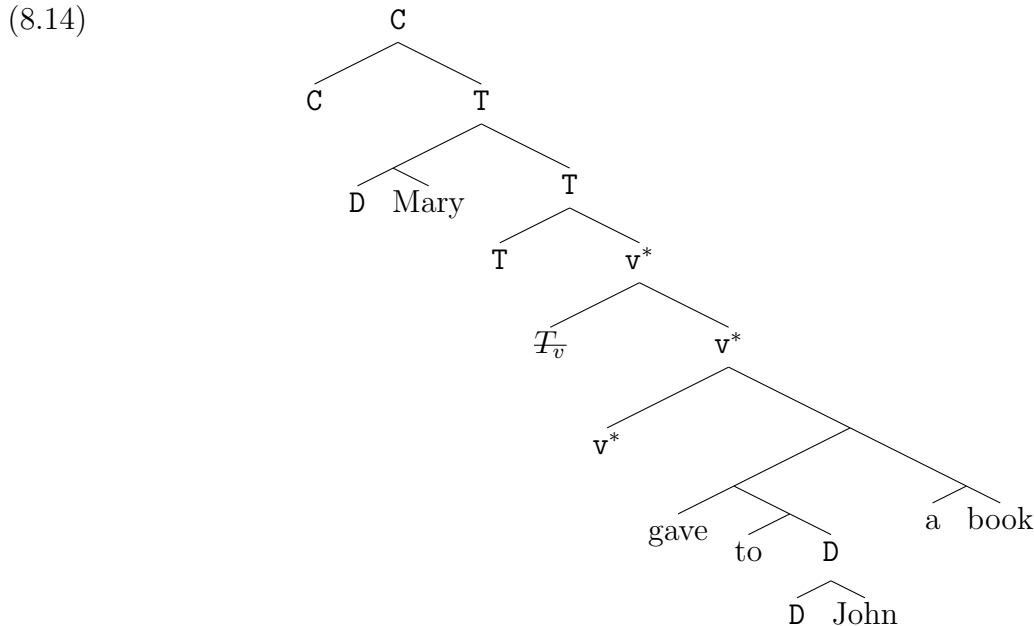


In this case the upper part of the coloring in (8.11) is unchanged but in the lower part, instead of (8.10) we now have



Since all the trees are non-planar, both structures (8.9) and (8.12) can be planarized in Externalization to produce the same string “*Mary gave a book to John*”. Note that (8.9) can also be planarized in the form “*Mary gave (to) John a book*” as well as two verb-last forms that would be filtered out as Externalizations in English. It does not, however, planarize as “*Mary to John gave a book*” (as in the Mandarin “*Tā gěile tā yī běn shū*”). On the other hand, (8.12) can be planarized as “*Mary to John gave a book*” as in Mandarin but it cannot be planarized in the form “*Mary gave (to) John a book*”.

- (3) There is a further possibility that is compatible with coloring with respect to the generators in $\mathcal{R}_{\Phi}$, namely a syntactic object of the form



Here again the coloring of (8.11) is unchanged but instead of (8.10) or (8.13) we have

$$(8.15) \quad \begin{array}{c} (\mathfrak{z}_{v^*}^\downarrow, \mathfrak{s}_{\text{lex}}^\uparrow) \\ \swarrow \quad \searrow \\ (\mathfrak{h}_{\text{lex}}, \mathfrak{z}_{\text{lex}}^\uparrow, \mathfrak{s}_{\text{lex}}^\uparrow) \quad \mathfrak{z}_{\text{lex}}^\downarrow \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ (\text{gave}, \mathfrak{h}_{\text{lex}}, \mathfrak{z}_{\text{lex}}^\uparrow, \mathfrak{s}_{\text{lex}}^\uparrow) \quad \mathfrak{m} \quad (\text{a}, \mathfrak{h}^\text{D}, \mathfrak{z}_\text{D}^\uparrow) \quad (\text{book}, \mathfrak{z}_\text{D}^\downarrow) \\ \swarrow \quad \searrow \\ (\text{to}, \mathfrak{h}^\text{I}, \mathfrak{z}_\text{I}^\uparrow) \quad \mathfrak{z}_\text{I}^\downarrow \\ \swarrow \quad \searrow \\ (\mathfrak{h}^\text{D}, \mathfrak{z}_\text{D}^\uparrow) \quad (\text{John}, \mathfrak{z}_\text{D}^\downarrow) \end{array}] .$$

This can be planarized as ‘*Mary gave (to) John a book*’ or as ‘*Mary to John gave a book*’ as in Mandarin, but not as ‘*Mary gave a book to John*’. (Note that other planarizations are also possible: for example ‘*Mary a book gave to John*’, which will be filtered out in Externalization in both English and Mandarin. We focus here on these three planarizations, as these provide enough cases realized in actual languages, to allow us to distinguish between the different candidate syntactic objects.)

Note that the coloring by $\mathbb{B}_\Phi$ does not rule out any of these three tree structures (8.9), (8.12), and (8.14), as all of them can be colored according to the rules of $\mathcal{R}_\Phi$ and give objects in $\mathcal{L}(\mathbb{B}_\Phi)$, so this coloring by phases alone does not resolve the ambiguity. We will see in §8.4 what happens when one also considers the coloring by theta roles, with the renenerators in $\mathcal{R}_\Theta$.

8.3.2. Movement in double object constructions. In the cases (8.12), and (8.14) consistently colored as objects in $\mathcal{L}(\mathbb{B}_\Phi)$, the remaining planarization ‘*Mary gave (to) John a book*’ for (8.12), and ‘*Mary gave a book to John*’ for (8.14) could in principle be achieved via movement by IM of *gave* to spec-of- v^* position, respectively resulting in $\{\text{gave-}v^*, \{\{\text{gave}, \text{a-book}\}, \text{to-John}\}\}$ and $\{\text{gave-}v^*, \{\{\text{gave}, \text{to-John}\}, \text{a-book}\}\}$. This movement description was used in [20] to formulate the theta roles coloring for the double object constructions. However, when we consider the full Extended Projection and the coloring by phases, for this example we see that the spec-of- v^* position is used in the movement of the T_v accessible term. Moreover, moving a term labelled by $(\mathfrak{h}^\omega, \mathfrak{z}_\omega^\uparrow, \mathfrak{s}_\omega^\uparrow)$ would require introducing additional generators for $\mathcal{R}_\Phi$, in addition to the generators of the form (4.27)

$$T_{\mathfrak{c}, (1, \mathfrak{m})}^{\mathfrak{s}} := \begin{array}{c} \mathfrak{s}_{\omega'}^\downarrow \\ \swarrow \quad \searrow \\ \mathfrak{c}_\omega^\downarrow \quad (1, \mathfrak{m}) \end{array}$$

where we assumed that $\mathfrak{c}$ can only be either a $\mathfrak{s}$ or a $\mathfrak{z}$. However, allowing additional generators of the form

$$(8.16) \quad \begin{array}{c} \mathfrak{s}_{\omega'}^\downarrow \\ \swarrow \quad \searrow \\ (\mathfrak{h}^\omega, \mathfrak{z}_\omega^\uparrow, \mathfrak{s}_\omega^\uparrow) \quad (1, \mathfrak{m}) \end{array} \quad \text{or} \quad \begin{array}{c} \mathfrak{s}_{\omega'}^\downarrow \\ \swarrow \quad \searrow \\ (\mathfrak{h}^\omega, \mathfrak{z}_\omega^\uparrow) \quad (1, \mathfrak{m}) \end{array} \quad \text{or} \quad \begin{array}{c} \mathfrak{s}_{\omega'}^\downarrow \\ \swarrow \quad \searrow \\ (\mathfrak{h}^{\omega'}, \mathfrak{s}_\omega^\uparrow) \quad (1, \mathfrak{m}) \end{array}$$

has some drawbacks. In the first case, for example, if the trace of the movement at the node

$$(8.17) \quad \begin{array}{c} (\mathfrak{s}_\omega^\uparrow, \mathfrak{z}_{\omega'}^\downarrow) \\ | \\ \text{---}(\mathfrak{h}^\omega, \mathfrak{z}_\omega^\uparrow, \mathfrak{s}_\omega^\uparrow)\text{---} \quad \mathfrak{z}_\omega^\downarrow \end{array}$$

remains colored by $(\mathfrak{h}^\omega, \mathfrak{z}_\omega^\uparrow, \mathfrak{s}_\omega^\uparrow)$, which contains a head label $\mathfrak{h}^\omega$ and hence participates in the labeling algorithm, this creates a problem with the ECP, as formulated in Remark 6.5, since the trace should not participate in labeling. In fact, a closer inspection shows that it is possible to include additional generators of the form (8.16) without incurring in a violation of ECP. In fact, for IM movement realized via one of the generator (8.16) to take place, the syntactic object has to include a position with an input marked by $\mathfrak{s}_\omega^\downarrow$, to which the output of a generator as in (8.16) is grafted. This means that both the maximal projection of the head $\mathfrak{h}^\omega$ involved in the generator (8.16) and that of the higher head $\mathfrak{h}^{\omega'}$, whose complement $\mathfrak{z}_{\omega'}^\downarrow$ appears in the output of (8.17), have to be already present in the syntactic object, for a generator of the form (8.16) to be included in the structure. But the presence of these labelled positions implies that all the internal vertices that need to receive a labeling $\mathfrak{h}^\omega$ are already labelled in this syntactic object, hence the trace $\text{---}(\mathfrak{h}^\omega, \mathfrak{z}_\omega^\uparrow, \mathfrak{s}_\omega^\uparrow)\text{---}$ does not participate in any labeling role. In terms of the equivalent colored Merge formulation, this just means that when the colored Merge operation implementing this IM operation takes place, all the labeling in which the head $\mathfrak{h}^\omega$ participates has already been completed, hence the trace has no labeling role, compatibly with ECP.

8.4. Double object and coloring by theta roles. We now compare the proposed forms (8.9), (8.12) and (8.14) of the syntactic object in terms of coloring by theta roles in $\mathcal{L}(\mathbb{B}_\Theta)$. As we will see, this coloring in $\mathcal{L}(\mathbb{B}_\Phi)$ put a strong constraint on the possible coloring in $\mathcal{L}(\mathbb{B}_\Theta)$, including assumptions about theta-hierarchies in the generators in $\mathcal{R}_\Theta$.

In all cases (8.9), (8.12), and (8.14), in the upper part of the structure all nodes from the root until the second T-labelled node correspond to generators

$$\begin{array}{c} \theta_0 \\ | \\ \theta_0 \quad \theta_0 \end{array}$$

while the first $\mathbf{v}^*$ node is a generator

$$\begin{array}{c} \theta_0 \\ | \\ (\mathcal{T}_v, \theta_E^\downarrow) \quad \theta_E^\uparrow \end{array}$$

followed by a generator of the form

$$\begin{array}{c} \theta_E^\uparrow \\ | \\ \theta_0 \quad \theta_E^\uparrow \end{array} \quad .$$

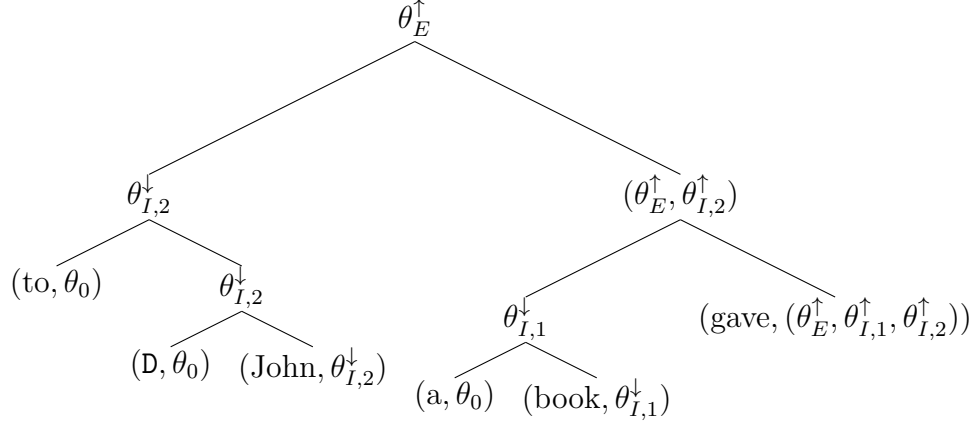
In the lower part of the structure, however, the three cases look different.

Proposition 8.1. *Requiring consistent coloring in both $\mathcal{R}_\Phi$ and $\mathcal{R}_\Theta$ of the candidate syntactic objects (8.9), (8.12), (8.14) and of syntactic objects obtained from (8.12) and (8.14) via IM movement, shows that the possibility favored by the coloring rules is that both (8.12) and (8.14) are available prior to Externalization (hence both hierarchies $\theta_{\text{GO}} \prec \theta_{\text{TH}}$ and $\theta_{\text{GO}} \succ \theta_{\text{TH}}$ are possible). The other possibilities*

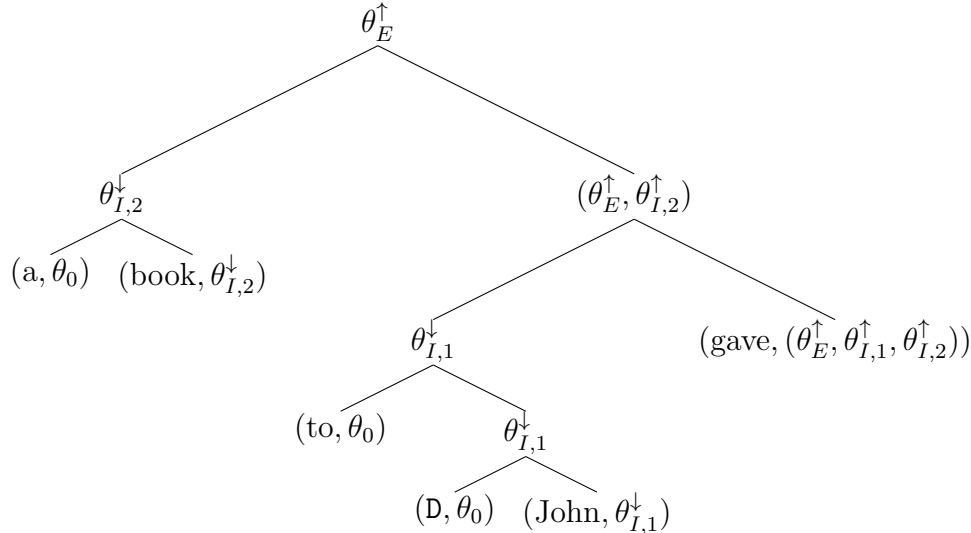
- the syntactic object (8.9),

• *only one of (8.12) or (8.14), together with a syntactic object obtained from it via IM movement that accounts for remaining planarizations that are linguistically attested, are all disfavored by the compatibility of the coloring rules of $\mathcal{R}_\Phi$ and $\mathcal{R}_\Theta$.*

Proof. It is convenient to describe first the case of (8.12) and (8.14). In the case of (8.12) we can give the lower part of the structure a theta coloring



In this case the goal/theme theta-hierarchy of the internal theta roles is $(\theta_{I,1}, \theta_{I,2}) = (\theta_{\text{TH}}, \theta_{\text{GO}})$. In the case of (8.14), similarly, we can color the structure according to the coloring rules of $\mathcal{R}_\Theta$ in the form



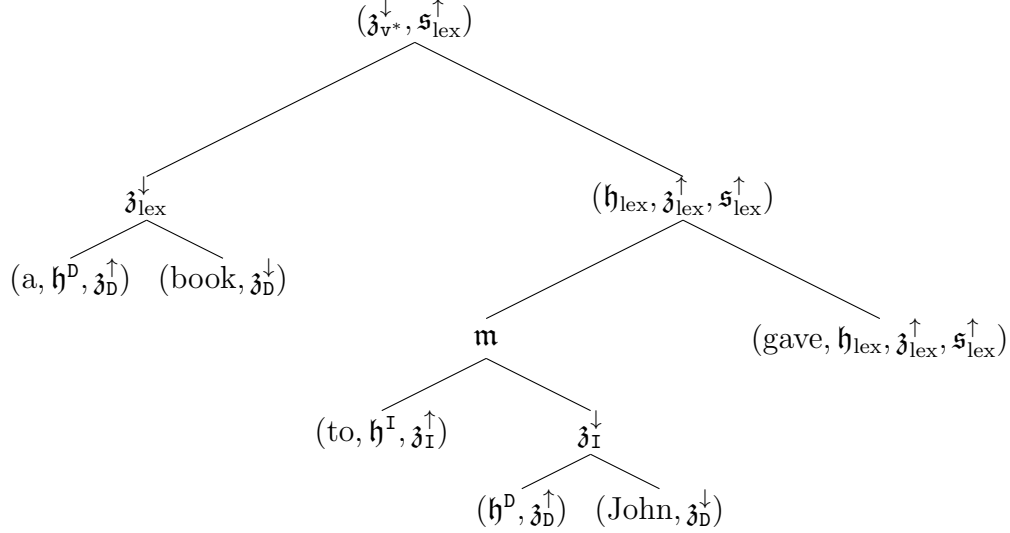
This is also well colored by generators in $\mathcal{R}_\Theta$, this time with a goal/theme theta-hierarchy of the opposite form $(\theta_{I,1}, \theta_{I,2}) = (\theta_{\text{GO}}, \theta_{\text{TH}})$.

If we allow for the possibility that both goal/theme theta-hierarchies $(\theta_{I,1}, \theta_{I,2}) = (\theta_{\text{GO}}, \theta_{\text{TH}})$ and $(\theta_{I,1}, \theta_{I,2}) = (\theta_{\text{TH}}, \theta_{\text{GO}})$ may be possible and that either one or the other is filtered out in Externalization (depending on specific languages), these two possibilities then suffice for all the possible linear orderings, namely all six permutations of the predicate and its two objects (the two internal theta roles).

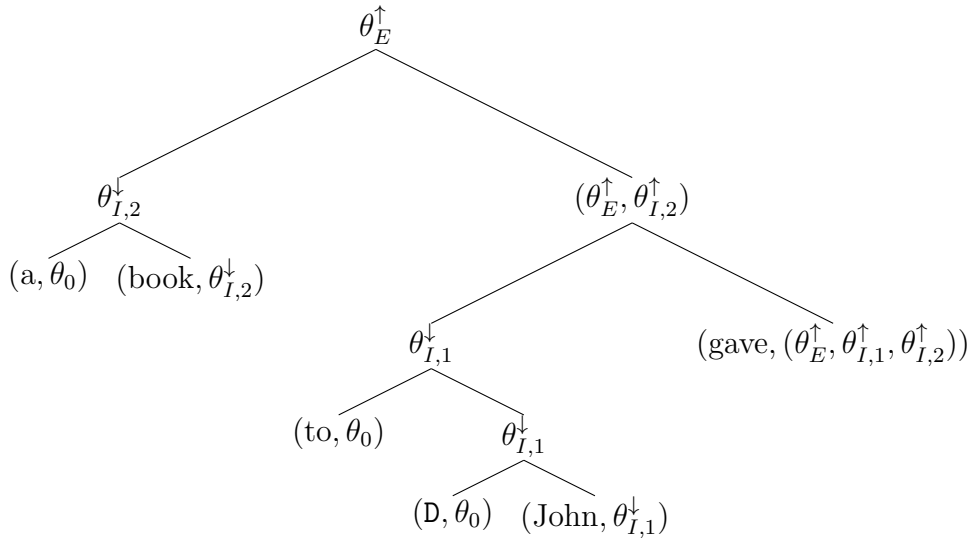
In a stronger formulation of theta-hierarchies, in which only one *or* the other hierarchy $(\theta_{I,1}, \theta_{I,2}) = (\theta_{\text{GO}}, \theta_{\text{TH}})$ or $(\theta_{I,1}, \theta_{I,2}) = (\theta_{\text{TH}}, \theta_{\text{GO}})$ is possible, prior to Externalization, then only one of (8.12) and (8.14) will be accepted by the theta roles coloring. This means that one of the possible linear ordering known to occur will not be obtainable as planar embedding of the accepted theta-colored syntactic object. This would then imply that the remaining linear ordering has to be an effect of movement, which, in this case, would mean movement from a position marked with $(\mathfrak{h}^{\text{lex}}, \mathfrak{z}_{\text{lex}}^\uparrow, \mathfrak{s}_{\text{lex}}^\uparrow)$,

hence this would require the inclusion of additional generators for the form (8.16), as discussed above.

For example, consider again the case of (8.15), with phase coloring

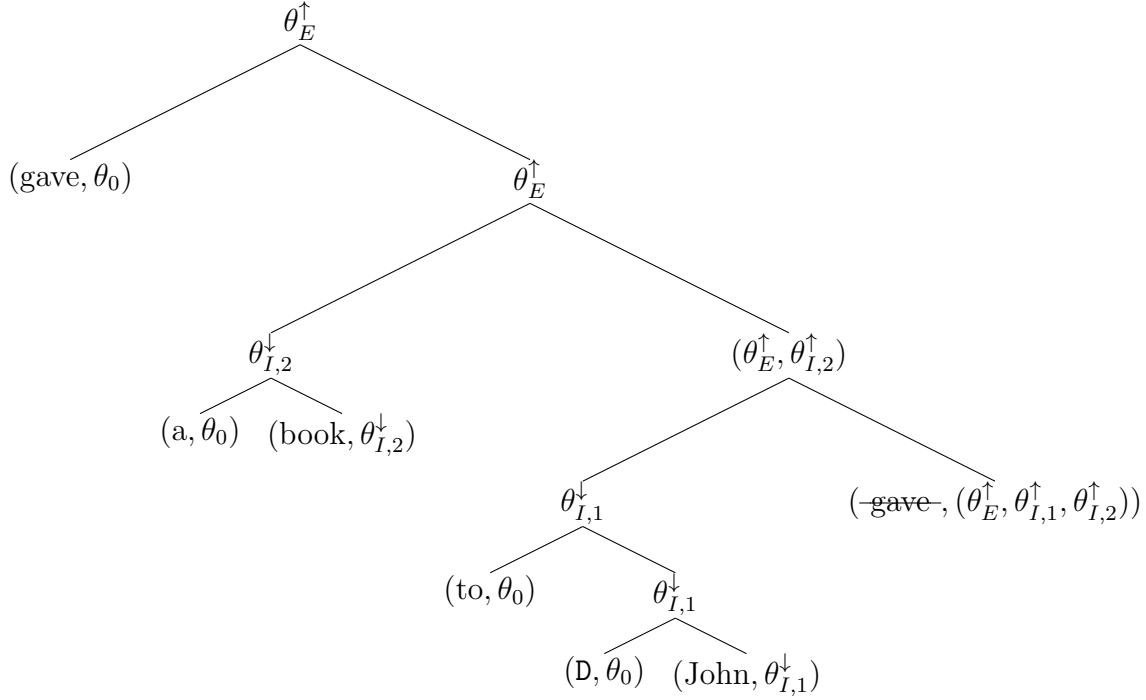


and theta coloring



which can be planarized as “*to John gave a book*” (as in Mandarin) or as ‘*gave (to) John a book*’, but not as “*gave a book to John*”. To obtain the linear ordering “*gave a book to John*” requires movement. In terms of coloring by theta roles, this is compatible with the generators of $\mathcal{R}_\Theta$, as

in [20], and would give a syntactic object of the form



On the other hand, for coloring according to phases, this movement requires adding to $\mathcal{R}_\Phi$ a generator of the form

$$(8.18) \quad \begin{array}{c} \mathfrak{s}_{\omega'}^\downarrow \\ \swarrow \quad \searrow \\ (\mathfrak{h}^\omega, \mathfrak{z}_{\omega'}^\uparrow, \mathfrak{s}_{\omega'}^\uparrow) \quad (1, \mathfrak{m}) \end{array} .$$

This means that the term $(\text{gave}, \mathfrak{h}_{\text{lex}}, \mathfrak{z}_{\text{lex}}^\uparrow, \mathfrak{s}_{\text{lex}}^\uparrow)$ is moved by IM to a specifier position $\mathfrak{s}_{\omega'}^\downarrow$, which also needs to be labelled θ_0 in the theta roles coloring. Because the root output of (8.15) is colored $(\mathfrak{z}_{v^*}^\downarrow, \mathfrak{s}_{\text{lex}}^\uparrow)$, the next generator above can only be of the form

$$\begin{array}{c} (\mathfrak{h}_{v^*}, \mathfrak{s}_{\text{lex}}^\uparrow) \\ \swarrow \quad \searrow \\ (\mathfrak{h}_{v^*}, \mathfrak{z}_{v^*}^\uparrow) \quad (\mathfrak{z}_{v^*}^\downarrow, \mathfrak{s}_{\text{lex}}^\uparrow) \end{array} .$$

However, this creates a compatibility problem because the output coloring $\mathfrak{s}_{\omega'}^\downarrow$ cannot be equal to $\mathfrak{s}_{\text{lex}}^\downarrow$, since as shown in (8.11) that input already receives the root output of the accessible term

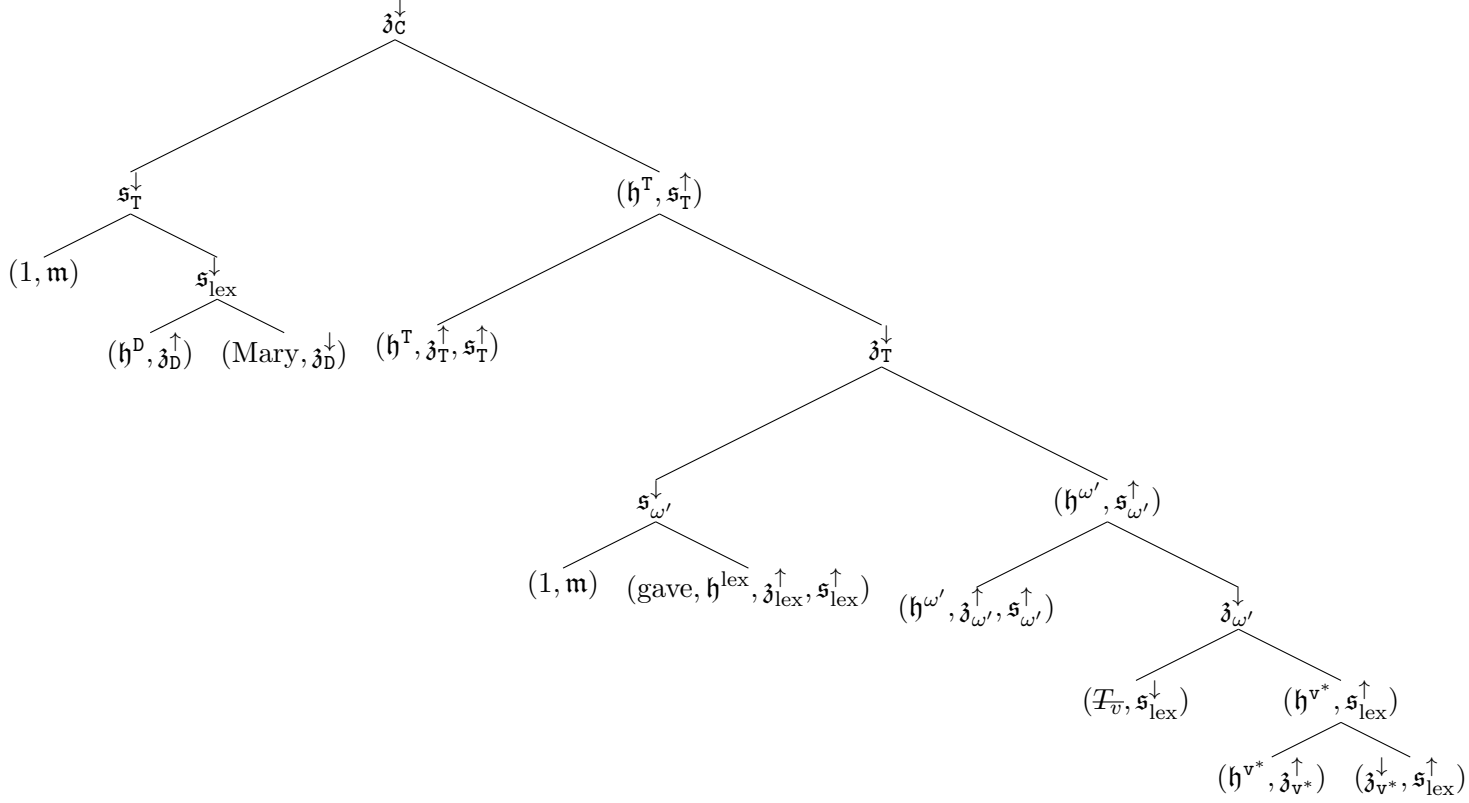
$$T_v = \begin{array}{c} \text{D} \\ \swarrow \quad \searrow \\ \text{D} \quad \text{Mary} \end{array}$$

as shown in (8.14), which carried the external theta role $\theta_E^\downarrow$. The $\mathfrak{s}_{\omega'}^\downarrow$ also cannot be $\mathfrak{s}_{\text{T}}^\downarrow$ as that receives the output of the generator

$$\begin{array}{c} \mathfrak{s}_{\text{T}}^\downarrow \\ \swarrow \quad \searrow \\ \mathfrak{s}_{\text{lex}}^\downarrow \quad (1, \mathfrak{m}) \end{array}$$

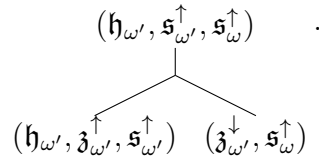
in the movement of the accessible term T_v , as shown in (8.11). Also ω' cannot be a higher head above $\mathfrak{s}_{\text{T}}^\downarrow$, since that would not result in a planarization of the form “*Mary gave a book to John*”

(though it could planarize to “*gave Mary a book to John*”). Moreover, movement to the specifier position $\mathfrak{s}_{\omega'}^{\downarrow}$ of a higher head ω' may also violate the Phase Impenetrability Condition if the head $\mathfrak{h}^{\omega'}$ belongs to the interior of a higher phase. Thus, the position $\mathfrak{s}_{\omega'}^{\downarrow}$ would have to be located between the maximal projection of the head $\mathfrak{h}^{v^*}$ and the head $\mathfrak{h}^T$. This would have the effect of replacing (8.11) with a structure of the form (8.19)



However, this now also creates a problem, because if $\omega' \in \Omega_{\mathfrak{h},\phi}$ is a phase-head, the IM movement of T_v would be from the interior of a lower phase and violate the Phase Impenetrability Condition, and if ω' is not a phase head, then the IM movement of the term $(\text{gave}, \mathfrak{h}^{lex}, \mathfrak{z}_{lex}^{\uparrow}, \mathfrak{s}_{lex}^{\uparrow})$ performed by the added generator would not be movement to Spec-of-Phase, hence by our coloring rules of Proposition would have to be either $\omega' = \text{INFL}$ (when it is not a phase-head) or $\omega' = \text{Root}$. It would have to be $\omega' = \text{INFL}$ given the position with respect to v^* in (8.19), but this violates the EPP requirement of having a noun or determiner phrase in its specifier position, as is the case for the T_v term. Thus, the possibility $\omega' = \text{INFL}$ is also not compatible with the structure of the Extended Projection and the movement of T_v .

One could also consider changing the generators of the colored operad $\mathcal{R}_{\Phi}$ of phases, adding the possibility of generators where more than one $\mathfrak{s}^{\uparrow}$ is flowing upward through the graph, something like



However, this in turn would require a cascade of other changes and additions to the generator to still reflect the correct phase structure and allow for the discharge of a $\mathfrak{s}_{\omega'}^{\downarrow}$ for a higher head ω' , with θ_0 position, before the $\mathfrak{s}_{\omega'}^{\downarrow} = \mathfrak{s}_T^{\downarrow}$ that receives the IM movement originating from the external theta role position $\theta_E^{\downarrow}$. It seems difficult to obtain a system of bud generator that would have

these properties and would still be compatible with the structure of the Extended Projection. So this does not look like a viable option.

We can then consider the remaining possibility, namely the case of a syntactic object of the form (8.9). According to the choice of generators for $\mathcal{R}_\Theta$ in [20], the internal theta roles are discharged above the Root position where they are injected, so that we would again expect, as in the two other cases (8.12) and (8.14), a structure of the form

$$(8.20) \quad \begin{array}{c} \theta_E^\uparrow \\ | \\ \theta_{I,2}^\downarrow \quad (\theta_E^\uparrow, \theta_{I,2}^\uparrow) \\ | \quad \swarrow \quad \searrow \\ \theta_{I,1}^\downarrow \quad (\theta_E^\uparrow, \theta_{I,1}^\uparrow, \theta_{I,2}^\uparrow) \end{array} \quad , \cdot$$

However, what we have in (8.9) would instead require a different coloring with generators that discharge the internal theta roles below the point of insertion, resulting in a composition of two generators of the form

$$(8.21) \quad \begin{array}{c} \theta_E^\uparrow \\ | \\ (\theta_E^\uparrow, \theta_{I,1}^\uparrow, \theta_{I,2}^\uparrow) \quad (\theta_{I,1}^\downarrow, \theta_{I,2}^\downarrow) \\ \swarrow \quad \searrow \\ \theta_{I,1}^\downarrow \quad \theta_{I,2}^\downarrow \end{array}$$

Thus, a coloring by theta roles of the syntactic object (8.9) would require the introduction of additional generators in the bud generating system $\mathcal{R}_\Theta$ that would allow for this type of coloring as in (8.21). This would disfavor (8.9), with respect to (8.12) and (8.14), as accounting for (8.9) would necessarily complicate the structure of the operad of theta roles. Moreover, among the three attested planarizations we considered, the syntactic object (8.9) can only give “*Mary gave a book to John*” and “*Mary gave (to) John a book*” but not the planarization as in Mandarin, “*Mary to John gave a book*”. Thus, once again, if only (8.9) is viable before Externalization, we would have to obtain the remaining from from another syntactic object obtained from (8.9) via IM movement, and this reproduces the same problems that we discussed in the case where only one of (8.12) and (8.14) is viable prior to Externalization. Thus, this also disfavors the possibility of (8.9).

Thus, both the structure (8.9) and the structures obtained by movement in (8.9), (8.12), or (8.14) (required to achieve the remaining planar embeddings known to occur if only one of (8.9), (8.12), (8.14) is available) are disfavored with respect to the possibility of having *both* (8.12) and (8.14) prior to Externalization, hence having both possible hierarchies $(\theta_{I,1}, \theta_{I,2}) = (\theta_{\text{GO}}, \theta_{\text{TH}})$ and $(\theta_{I,1}, \theta_{I,2}) = (\theta_{\text{TH}}, \theta_{\text{GO}})$ in the $\succeq$ preorder of theta-hierarchies. Indeed, in this case the simple generators for $\mathcal{R}_\Theta$ and $\mathcal{R}_\Phi$ suffice to account for both (8.12) and (8.14), with further selection between them occurring in Externalization. $\square$

Thus, we see that considering both coloring by phases and by theta roles significantly restricts the possibilities, compared to only considering one of the two colorings, as in [20].

9. COMBINED FILTERING FOR PHASES AND THETA ROLES

The specific cases discussed in the previous section suggest how one can combine the two colored operads of phases and theta roles into a single filtering system, in the form of a transduction of colored operads as introduced in Definition 7.2.

As in [20], for a theta-hierarchy $\theta_{I,1} > \cdots > \theta_{I,n}$, and for $k \leq n$, we use the notation

$$\underline{\theta}_k^\uparrow := (\theta_E^\uparrow, \theta_{I,1}^\uparrow, \dots, \theta_{I,k}^\uparrow),$$

with $\theta_{I,i}^\uparrow$ are the internal theta roles and θ_E is the external theta role. Thus,

$$\underline{\theta}^\uparrow = \underline{\theta}_n^\uparrow = (\theta_E^\uparrow, \theta_{I,1}^\uparrow, \dots, \theta_{I,n}^\uparrow) = (\theta_E^\uparrow, \underline{\theta}_I^\uparrow)$$

is the theta-grid assigned by a predicate. We also write $\theta_{I,j}^\downarrow$ for one of the theta roles assigned by the predicate that is received by one of its arguments.

Consider the set $\hat{\Omega}$ consisting of the colors

$$(9.1) \quad \begin{array}{cccc} (\mathfrak{h}^\omega, \mathfrak{z}_\omega^\uparrow, \mathfrak{s}_\omega^\uparrow, \underline{\theta}_k^\uparrow) & (\mathfrak{h}^\omega, \mathfrak{z}_\omega^\uparrow, \mathfrak{s}_\omega^\uparrow, \theta_0) & (\mathfrak{h}^\omega, \mathfrak{z}_\omega^\uparrow, \underline{\theta}_k^\uparrow) & (\mathfrak{h}^\omega, \mathfrak{z}_\omega^\uparrow, \theta_0) \\ (\mathfrak{s}_\omega^\uparrow, \mathfrak{z}_{\omega'}^\downarrow, \underline{\theta}_k^\uparrow) & (\mathfrak{s}_\omega^\uparrow, \mathfrak{z}_{\omega'}^\downarrow, \theta_0) & (\mathfrak{h}^\omega, \mathfrak{s}_\omega^\uparrow, \theta_E^\uparrow) & (\mathfrak{h}^\omega, \mathfrak{s}_\omega^\uparrow, \theta_0) \\ (\mathfrak{h}^{\omega'}, \mathfrak{s}_\omega^\uparrow, \underline{\theta}_k^\uparrow) & (\mathfrak{h}^{\omega'}, \mathfrak{s}_\omega^\uparrow, \theta_0) & (\mathfrak{z}_\omega^\downarrow, \theta_{I,j}^\downarrow) & (\mathfrak{z}_\omega^\downarrow, \underline{\theta}_k^\uparrow) \\ (\mathfrak{z}_\omega^\downarrow, \theta_0) & (\mathfrak{s}_\omega^\downarrow, \theta_0) & (\mathfrak{s}_\omega^\downarrow, \theta_E^\downarrow) & (\mathfrak{m}, \theta_0) \\ (\mathfrak{m}, \underline{\theta}_k^\uparrow) & (\mathfrak{m}, \theta_{I,j}^\downarrow) & (1, \mathfrak{m}, \theta_0) & \end{array}$$

with $\hat{\Omega} = \hat{\Omega}_f \sqcup \hat{\Omega}_{\text{lex}}$, where $\hat{\Omega}_f$ consists of colors as above with heads $\omega, \omega' \in \Omega_{\mathfrak{h},f}$ and $\hat{\Omega}_{\text{lex}}$ consists of the element $(1, \mathfrak{m}, \theta_0)$ and pairs of the form $(\alpha, \mathfrak{m})$ of $(\alpha, \mathfrak{c})$ with $\alpha \in \mathcal{SO}_0$ and $\mathfrak{c}$ a color as above with $\omega \in \Omega_{\mathfrak{h},\text{lex}}$. There are two projection maps

$$\pi : \hat{\Omega} \rightarrow \Omega_\Phi \quad \text{and} \quad \pi' : \hat{\Omega} \rightarrow \Omega_\Theta$$

where Ω_Φ is the set of colors Ω_{ext} of Definition 4.14 of the bud generating system $\mathbb{B}_\Phi$ of the colored operad of phases, and Ω_Θ is the set of colors denoted by Θ in Definition 3.12 of [20], of the bud generating system $\mathbb{B}_\Theta$ of the colored operad of theta roles. The projections π, π' consist, respectively, of dropping the θ coordinates in the colors of (9.1) or dropping the phase coordinates $\mathfrak{h}, \mathfrak{s}, \mathfrak{z}, \mathfrak{m}$. We set $\mathcal{T} := \pi^{-1}(\mathcal{T}_\Phi) \cap \pi'^{-1}(\mathcal{T}_\Theta)$ where we write $\mathcal{T}_\Phi$ and $\mathcal{T}_\Theta$ for the terminal colors of the bud generating systems $\mathbb{B}_\Phi$ and $\mathbb{B}_\Theta$, respectively. We take $\mathcal{I} := \hat{\Omega} \setminus \mathcal{T}$.

Definition 9.1. Consider the colored operad $\mathbb{B}_{\hat{\Omega}}(\mathcal{M})(n) = \hat{\Omega} \times \mathcal{M}(n) \times \hat{\Omega}^n$ and all the elements of $\mathbb{B}_{\hat{\Omega}}(\mathcal{M})(2)$ of the form

$$T_{\mathfrak{c}', \mathfrak{c}''}^\mathfrak{c} := \begin{array}{c} \mathfrak{c} \\ \swarrow \quad \searrow \\ \mathfrak{c}' \quad \mathfrak{c}'' \end{array}$$

with $\mathfrak{c}, \mathfrak{c}', \mathfrak{c}''$ in $\hat{\Omega}$. We extend the projection maps to

$$\pi : \mathbb{B}_{\hat{\Omega}}(\mathcal{M})(2) \rightarrow \mathbb{B}_{\Omega_\Phi}(\mathcal{M})(2) \quad \text{and} \quad \pi' : \mathbb{B}_{\hat{\Omega}}(\mathcal{M})(2) \rightarrow \mathbb{B}_{\Omega_\Theta}(\mathcal{M})(2)$$

by setting

$$\begin{aligned} \pi(T_{\mathfrak{c}', \mathfrak{c}''}^\mathfrak{c}) &:= T_{\pi(\mathfrak{c}'), \pi(\mathfrak{c}'')}^{\pi(\mathfrak{c})} = \begin{array}{c} \pi(\mathfrak{c}) \\ \swarrow \quad \searrow \\ \pi(\mathfrak{c}') \quad \pi(\mathfrak{c}'') \end{array} \\ \pi'(T_{\mathfrak{c}', \mathfrak{c}''}^\mathfrak{c}) &:= T_{\pi'(\mathfrak{c}'), \pi'(\mathfrak{c}'')}^{\pi'(\mathfrak{c})} = \begin{array}{c} \pi'(\mathfrak{c}) \\ \swarrow \quad \searrow \\ \pi'(\mathfrak{c}') \quad \pi'(\mathfrak{c}'') \end{array} \end{aligned}$$

We define the subset $\mathcal{R}_{\Phi} \subset \mathbb{B}_{\hat{\Omega}}(\mathcal{M})(2)$ as the subset of those generators $T_{\mathfrak{c}', \mathfrak{c}''}^\mathfrak{c}$ with the property that

$$(9.2) \quad \pi(T_{\mathfrak{c}', \mathfrak{c}''}^\mathfrak{c}) \in \mathcal{R}_\Phi \quad \text{and} \quad \pi'(T_{\mathfrak{c}', \mathfrak{c}''}^\mathfrak{c}) \in \mathcal{R}_\Theta.$$

We then obtain a proposal for a single filtering system that simultaneously implements filtering by phases and by theta roles.

Proposition 9.2. *The bud generating system $\mathbb{B}_{\Phi\Theta} = (\mathcal{M}_h, \hat{\Omega}, \mathcal{R}_{\Phi\Theta}, \mathcal{I}, \mathcal{T})$ determines a transduction of colored operads, as in Definition 7.2 relating the colored operads of phases and or theta roles. Let $\mathcal{L}(\mathbb{B}_{\Phi\Theta}) \subset \mathbb{B}_{\hat{\Omega}}(\mathcal{M})$ be the language of the bud generating system $\mathbb{B}_{\Phi\Theta}$. We can also consider, under the projections π, π' , the subsets $\pi^{-1}(\mathcal{L}(\mathbb{B}_{\Phi})) \subset \mathbb{B}_{\hat{\Omega}}(\mathcal{M})$ and $\pi'^{-1}(\mathcal{L}(\mathbb{B}_{\Theta})) \subset \mathbb{B}_{\hat{\Omega}}(\mathcal{M})$. The syntactic objects that are compatibly well formed with respect to both the filtering by phases and by theta roles are the elements of*

$$\pi^{-1}(\mathcal{L}(\mathbb{B}_{\Phi})) \cap \pi'^{-1}(\mathcal{L}(\mathbb{B}_{\Theta})),$$

which in this case is the same as $\mathcal{L}(\mathbb{B}_{\Phi\Theta})$. All the cases discussed in §8 of compatible coloring by phases and theta roles are elements in this set.

Proof. It suffices to check that we can indeed take $\pi^{-1}(\mathcal{L}(\mathbb{B}_{\Phi})) \cap \pi'^{-1}(\mathcal{L}(\mathbb{B}_{\Theta}))$ instead of $\pi^{-1}(\mathcal{L}(\mathbb{B}_{\Phi})) \cap \pi'^{-1}(\mathcal{L}(\mathbb{B}_{\Theta})) \cap \mathcal{L}(\mathbb{B}_{\Phi\Theta})$. The syntactic objects in $\mathcal{L}(\mathbb{B}_{\Phi\Theta})$ are composed of generators in $\mathcal{R}_{\Phi\Theta}$. Under the projections π, π' these generators to generators in $\mathcal{R}_{\Phi}$, respectively $\mathcal{R}_{\Theta}$, so that the image of $\mathcal{L}(\mathbb{B}_{\Phi\Theta})$ is contained in $\mathcal{L}(\mathbb{B}_{\Phi})$, respectively $\mathcal{L}(\mathbb{B}_{\Theta})$ and $\mathcal{L}(\mathbb{B}_{\Phi\Theta}) \subset \pi^{-1}(\mathcal{L}(\mathbb{B}_{\Phi})) \cap \pi'^{-1}(\mathcal{L}(\mathbb{B}_{\Theta}))$. If a syntactic object is in $\pi^{-1}(\mathcal{L}(\mathbb{B}_{\Phi}))$, then it is obtained from generators of the form $T_{\mathfrak{c}', \mathfrak{c}''}^{\pi(\mathfrak{c})}$ with $T_{\pi(\mathfrak{c}'), \pi(\mathfrak{c}'')}^{\pi(\mathfrak{c})} \in \mathcal{R}_{\Phi}$ and $\mathfrak{c} = (\pi(\mathfrak{c}), \mathfrak{c}_{\Theta})$, $\mathfrak{c}' = (\pi(\mathfrak{c}'), \mathfrak{c}'_{\Theta})$, $\mathfrak{c}'' = (\pi(\mathfrak{c}''), \mathfrak{c}''_{\Theta})$ with $\mathfrak{c}_{\Theta}, \mathfrak{c}'_{\Theta}, \mathfrak{c}''_{\Theta}$ colors in Ω_{Θ} , and conversely for syntactic objects in $\pi'^{-1}(\mathcal{L}(\mathbb{B}_{\Theta}))$, so that, if a syntactic object is in $\pi^{-1}(\mathcal{L}(\mathbb{B}_{\Phi})) \cap \pi'^{-1}(\mathcal{L}(\mathbb{B}_{\Theta}))$ it is in fact in $\mathcal{L}(\mathbb{B}_{\Phi\Theta})$. Thus, the filtering by both phases and theta roles is achieved by intersecting the preimages of the two filtering systems, namely the preimages of $\mathcal{L}(\mathbb{B}_{\Phi})$ and $\mathcal{L}(\mathbb{B}_{\Theta})$ in the common space $\mathbb{B}_{\hat{\Omega}}(\mathcal{M})$, or equivalently the result of this symultaneous filtering is given by $\mathcal{L}(\mathbb{B}_{\Phi\Theta})$. One can check by direct inspection that all the decompositions into generators in $\mathcal{R}_{\Phi}$ and $\mathcal{R}_{\Theta}$ for the cases discussed in §8 are indeed in $\mathcal{L}(\mathbb{B}_{\Phi\Theta})$. $\square$

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